

嵌套采样方法

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摘要: 嵌套采样 (NS) 计算参数后验分布, 并使贝叶斯模型比较在计算上可行。其优势在于能够无监督地导航复杂、潜在的 multimodal 后验分布, 直到达到一个明确的终止点。本文对嵌套采样算法及其变体进行了系统性文献综述。我们重点关注完整算法, 包括对似然受限先验采样的解决方案、并行化、终止和诊断。研究了两种完整算法中活点数量、维度与计算成本之间的关系。提出了一种新的NS公式, 将参数空间探索表述为在树形数据结构上的搜索。之前发表过的稳健误差估计方法以及活点数量动态变化的各种方法被作为该公式的特例提出。基于之前插入排序顺序工作的在线诊断测试被提出。嵌套采样方法的综述以未来研究展望结束。

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Nested Sampling Methods

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Abstract: Nested sampling (NS) computes parameter posterior distributions and makes Bayesian model comparison computationally feasible. Its strengths are the unsupervised navigation of complex, potentially multi-modal posteriors until a well-defined termination point. A systematic literature review of nested sampling algorithms and variants is presented. We focus on complete algorithms, including solutions to likelihood-restricted prior sampling, parallelisation, termination and diagnostics. The relation between number of live points, dimensionality and computational cost is studied for two complete algorithms. A new formulation of NS is presented, which casts the parameter space exploration as a search on a tree data structure. Previously published ways of obtaining robust error estimates and dynamic variations of the number of live points are presented as special cases of this formulation. A new online diagnostic test is presented based on previous insertion rank order work. The survey of nested sampling methods concludes with outlooks for future research.

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1. 背景

嵌套采样 (NS, Skilling, 2004) 是一种蒙特卡洛算法, 用于计算模型参数空间上的积分。在贝叶斯推断分析某些数据 D 的背景下, 被积函数是似然函数 $L(D|\theta)$, 它根据先验概率密度 $\pi(\theta) d\theta$ 对参数 θ 进行边缘化, 从而给出参数空间的度量。对后验密度 $L(D|\theta) \times \pi(\theta) d\theta$ 的积分可以提供关于哪些模型参数区域是可能或不可能的深刻见解。整个参数空间上的积分, $Z = \int L(D|\theta) \pi(\theta) d\theta$, 被称为边缘似然或贝叶斯证据。贝叶斯因子是两个不同模型的边缘似然之比, Z_{M1}/Z_{M2} 。乘以模型先验概率比, 得到的结果后验概率比可以解释为在给定观测数据的情况下这两个模型之间的相对证据。使用贝叶斯因子选择模型可以根据经验尺度 (例如, Jeffreys, 1998; Kass 和 Raftery, 1995) 或校准错误决策率 (例如, Veitch 和 Vecchio, 2008; Buchner, 2019) 进行。因此, 边缘似然的计算对于参数概率模型的科学应用通常非常重要 (Evans, 2007)。在实践中, 后验概率和积分的计算是通过蒙特卡洛算法实现的。

探索、导航和整合这些参数空间可能表现出以下挑战 (与 F. Beaujeau 私下通信中的分类相关):

- (P) 特殊形状, 如非椭球体 (例如, 来自非高斯分布) 和非凸后验轮廓 (例如, 香蕉状)。
- (M) 多个、分离良好的模式, 当后验分布中存在多个概率相当的峰值时。可以通过形成非连通集的轮廓来定义这些模式。
- (D) 高维性 (此处: 中等维度: $d \sim 10$, 高维: $d > 30$)。高维空间会导致维数灾难, 几何直观性会失效。
- (I) 高信息增益使得后验分布成为先验体积中非常小的一部分 (例如, e^{-10000}), 需要被识别。衡量似然更新先验的多少的一个有用指标是信息增益:

$$H = \int \log \frac{\pi(\theta)}{L(D|\theta) \times \pi(\theta)} \pi d\theta.$$

- (T) 相变是可访问参数空间 X 的令人惊讶的、突发的、似然增加 L 的变化, 即当 $g = \frac{d \log L}{d \log X}$ 不是上凹函数时 (Skilling 等人, 2006 年)。一个说明性的例子是尖峰-板条似然, 其中高斯与另一个更宽、同心的高斯 ($\sigma_2^2 \gg \sigma_1^2$) 相加。在中心与量级为 σ_1 的范围内, g 大约以 $-(x-\mu)^2/\sigma_1^2$ 的平方形式减少, 但随后减慢减少速度, 并在 σ_1 与 σ_2 之间几乎保持恒定, 其中“尖峰” (σ_1) 占据了宽板条 (σ_2) 顶部的微小区域。因此, 在给定 L 时可访问的参数空间首先随着对数 L 的减小而非常缓慢地增加, 然后极其迅速地增加, 类似于水被加热成水蒸气时的体积膨胀 (Skilling 等人, 2006 年)。这种相变通常发生在次主导模型组件在主导组件被约束后变得相关时, 例如在混合模型中。似然平台可以被认为是极端的相变。

当然, 一个特定的问题可能表现出这些挑战的任何组合。

嵌套采样 (在第3节中介绍) 解决了这些挑战。它使计算 Z 对各种问题都变得实用。后验样本由NS同时计算。除了应用于贝叶斯推理之外, NS已被用作通用积分算法 (例如, Murray等人, 2006年; Pártay、Bartók和Csányi, 2010年; Malakar和Knuth, 2011年; Goggans、Henderson和Cao, 2014年; Birge、Chang和Polson, 2012年), 以及用于计算熵 (Malakar和Knuth, 2011年; Brewer, 2017年)。

2. 审查方法

本综述介绍了过去15年间发展起来的NS方法。我们进行了系统文献综述以查找关于嵌套采样的研究。我们使用了四个来源: (1) 2017年9月使用引号内的“nested sampling”术语在Google Scholar中进行搜索。在6080个搜索结果中, 我们考虑了排名靠前的260个结果 (由Google Scholar按相关性排序)。我们通过从查询中移除一位作者 (“PC Mahalanobis”) 的出版物并进一步手动移除结果, 排除了与土壤测量无关的嵌套采样技术相关结果。(2) 2017年9月使用搜索查询“nested sampling”¹ 搜索Skilling原始的2004年嵌套采样论文的Google Scholar引用。这产生了420个搜索结果, 我们全部考虑了。(3) 首先使用查询“nested sampling”在NASA摘要数据库系统中进行搜索。这给出了1215个结果, 其中许多是嵌套采样的简单应用而没有方法论贡献, 大多来自天体物理学。因此, 我们将搜索限制在arXiv类别 (arxiv_class:"stat.*" OR arxiv_class:"math.*" OR

¹精确查询链接

1. Context

Nested Sampling (NS, [Skilling, 2004](#)) is a Monte Carlo algorithm for computing an integral over a model parameter space. In the context of Bayesian inference of analysing some data D , the integrand is the likelihood function $L(D|\theta)$, which is marginalised over the parameters θ according to the prior probability density $\pi(\theta) d\theta$, which gives a measure of the parameter space. Integrals over the posterior density $L(D|\theta) \times \pi(\theta) d\theta$ allow insightful statements about what model parameters regions are probable or improbable. The integral over the entire parameter space, $Z = \int L(D|\theta) \pi(\theta) d\theta$, is known as the marginal likelihood or Bayesian evidence. The Bayes factor is the ratio of marginalised likelihoods of two different models, Z_{M1}/Z_{M2} . Multiplied by the model prior odds, the resulting posterior odds can be interpreted as the relative evidence among these two models, given the observed data. Selecting models using the Bayes factor can be performed based on empirical scales (e.g., [Jeffreys, 1998](#); [Kass and Raftery, 1995](#)) or calibrated on false decision rates (e.g., [Veitch and Vecchio, 2008](#); [Buchner, 2019](#)). The computation of marginal likelihoods is thus generally important for science with parametric probabilistic models ([Evans, 2007](#)). In practice, the computation of posteriors and the integral is achieved with Monte Carlo algorithms.

Exploring, navigating and integrating these parameter spaces can exhibit the following challenges (classification in priv. comm. with F. Beaujeau):

- **(P) Peculiar shapes** such as non-ellipsoidal (e.g., from non-Gaussian profiles) and non-convex posterior contours (e.g., bananas).
- **(M) Multiple, well-separated modes**, when several peaks in the posterior with comparable probability exist. One can define these by contours forming non-connected sets.
- **(D) High dimensionality** (here: intermediate: $d \sim 10$, high: $d > 30$). High-dimensional spaces incur the curse of dimensionality and geometric intuition breaks down.
- **(I) High information gain** makes the posterior a very small portion (e.g., e^{-10000}) of the prior volume, that needs to be identified. A useful measure of how much the likelihood updates the prior is the information gain:

$$H = \int \log \frac{\pi(\theta)}{L(D|\theta) \times \pi(\theta)} \pi d\theta.$$
- **(T) Phase transitions** are surprising and abrupt changes of the accessible parameter space X with increasing likelihood L , i.e., when $g = \frac{d \log L}{d \log X}$ is not an up-concave function ([Skilling et al., 2006](#)). A illustrating example is the spike-and-slab likelihood, where a Gaussian is summed with another, much wider, co-centred Gaussian ($\sigma_2^2 \gg \sigma_1^2$). Between the centre and scales of order σ_1 , g decreases quadratically approximately as $-(x-\mu)^2/\sigma_1^2$, but then slows its decrease and becomes almost constant between σ_1 and σ_2 , where the “spike” (σ_1) occupies a tiny region on top of the wide slab (σ_2). Therefore, the parameter space accessible at a given L increases first very slowly with decreasing $\log L$, then extremely rapidly, analogous to the volume expansion of water as it is heated to water vapor ([Skilling et al., 2006](#)). Such phase transitions commonly occur when a subdominant model component becomes relevant after a dominant component is constrained, for example in mixture models. Likelihood plateaus can be considered extreme phase transitions.

Naturally, a given problem can exhibit any combination of these challenges.

Nested sampling (introduced in §3) addresses these challenges. It makes computing Z practical for a wide variety of problems. Posterior samples are simultaneously computed by NS. Beyond the application to Bayesian inference, NS has been applied as a general integration algorithm (e.g., [Murray et al., 2006](#); [Pártay, Bartók and Csányi, 2010](#); [Malakar and Knuth, 2011](#); [Goggans, Henderson and Cao, 2014](#); [Birge, Chang and Polson, 2012](#)) and to compute entropies ([Malakar and Knuth, 2011](#); [Brewer, 2017](#)).

2. Review methodology

This review presents NS methods developed over the last 15 years. We conducted a systematic literature review to find works on nested sampling. We used four sources: (1) Google Scholar was searched with the terms “nested sampling” (including quotes) in Sep 2017. Of the 6080 search results of which we consider the first 260 (ranked by relevance by Google Scholar). We excluded results on the unrelated nested sampling technique for soil measurements by removing publications by one author from the query “PC Mahalanobis” and further manually removed results. (2) Google Scholar citations of Skilling’s original 2004 Nested Sampling paper were searched with the search query “nested sampling”¹ in September 2017. This yielded 420 search results which we all considered. (3) The NASA Abstract Database System was initially searched with the query “nested sampling”. This gave 1215 results, many of which are simple applications of nested sampling without methodological contributions, mostly from astrophysics. We thus limit our search to the arXiv classes (arxiv_class:“stat.*” OR arxiv_class:“math.*” OR

¹exact query link

arxiv_class:"physics.*"), 其中发布统计方法和算法的论文被 (交叉) 发布。这产生了78个结果, 所有结果都被考虑在内。(4) 之前已知的通讯作者是第一作者的作品也被包括在内。在那些四个查询中, 我们考虑对嵌套抽样方法的任何方面有新贡献的作品。

需要一些限制来聚焦内容。首先, 我们将自己限制在连续参数空间上的推理问题。NS 不要求空间必须是连续的, 只需要定义一个从中可以采样的先验即可。“考虑的对象”可以是分类性质的, 也可以是不同维度的 (例如, Brewer, 2014)。此外, 本综述并未深入探讨 NS 的概率理论分析。这在数学上是复杂的, 并且已在其他地方涵盖, 我们将在第 3.2 节和第 4 节中向感兴趣的读者进行参考。我们排除那些仅将嵌套采样以先前发表的形式应用于新问题, 而没有对现有算法的任何方面进行修改的工作, 并且进一步排除那些没有描述其使用特定方法的工作。

本综述表明 NS 是在沟通有限的社区中开发的。前几节从贝叶斯实践者 (3.1)、统计学家 (3.2)、物理学家 (3.3) 和计算机科学家 (3.4) 的角度介绍了 NS。这些部分使用每个群体的语言, 试图让专家更好地交流他们的想法。本综述的重点是 NS 组件的实现技术, 这些技术列在第 3.5 节中, 以有效应对 PMDIT 挑战。第 4 节介绍了积分过程, 并引用了收敛性证明。终止标准在 § 4.3 中讨论, 随后在 § 4.4 中讨论了计算复杂性。为了确定 NS 运行的正确性和质量, § 4.5 中提供了诊断方法, 包括一个新的测试。广泛回顾了用于 NS 内部的采样方法, 包括基于随机游走 (§ 5.1)、拒绝采样 (§ 5.2) 和混合方法 (§ 5.3)。第 6 节讨论了 NS 的变体, 这些变体缓和了硬似然约束, 改变了活点的数量, 并并行化了算法。在 § 7 中, 一个简单的数值实验演示了 NS 实现的一些行为。

本次调查中介绍的这些技术帮助了我们对开源 NS 实现的设计决策。UltraNest² 是一个高性能通用嵌套抽样库, 适用于用 Python、C、C++、Fortran、Julia 或 R 编程语言编写的模型, 重点关注可靠性。我们在相关章节中记录了 UltraNest 的设计决策。

3. 嵌套抽样简介

我们提供了面向不同受众的介绍, 旨在让不同群体能够翻译他们使用的语言。这些章节从贝叶斯从业者 (§ 3.1)、理论统计学家 (§ 3.2)、物理学家 (§ 3.3) 和计算机科学家的角度介绍了 NS (§ 3.4)。然后, § 3.5 中确定了 NS 的组成部分。

3.1. 概念性介绍

首先, 介绍算法的参考版本。理论背景、理由和变体将在后续章节中讨论。虽然该算法不限于连续先验, 但要理解这些概念, 首先考虑具有均匀先验的参数空间可能会有所帮助。例如, $V = \int \pi(\theta) d\theta$ 可以直观地与体积相关联。一些实现也倾向于这种方法, 并通过使用累积先验分布进行逆变换来支持非均匀 (包括相关的) 先验 (参见附录 A)。

NS 是一种集成算法, 可提供后验样本和边缘似然 Z 。该方法类似于勒贝格积分, 需要跟踪高度 (似然) 和体积。假设我们想要在 d 维连续参数空间上计算边缘似然。图 1 展示了下述流程。

初始化: 从先验 N 随机采样若干点 (例如 $N = 400$), 并在每个点上评估似然函数。

收缩移除可能性最低的活点, L_1 (最差拟合), 它成为第一个死点。考虑到每个点代表 $1/N$ 总体积, 这将使体积减少约 $\delta V = 1/N$ 倍。移除的体积还有三个常用估计器: 考虑到样本通过采样的 L 阈值均匀分割体积, 移除的壳体积 δV , $L > L_{\min}$ 是从 Beta(1, N) 分布中抽取的随机变量。这个分布可以随机采样, 或几何

²<https://johannesbuchner.github.io/UltraNest/>

`arxiv_class:"physics.*")`, where papers developing statistical methods and algorithms are (cross-)posted. This yields 78 results, all of which were considered. (4) Works previously known to the first author were also included. Out of those four queries we consider works with novel contributions to any aspect of nested sampling methods.

Some restrictions are necessary to focus the content. Firstly, we limit ourselves to inference problems over continuous parameter spaces. NS does not require the space to be continuous, only that a prior is defined from which can be sampled. The “objects” considered can be of categorical nature or of varying dimensionality (e.g., [Brewer, 2014](#)). Furthermore, the review does not go into depth on probability theoretical analyses of NS. This is mathematically involved and already covered elsewhere, to which we refer the interested reader in sections 3.2 and 4. We exclude works which merely apply nested sampling in a previously published form to a new problem without modifications of any aspect of an existing algorithm, and further exclude works which do not describe the specific method they use.

The review made clear that NS is developed in communities with limited communication. The first sections introduce NS from the view points of Bayesian practitioners (3.1), statisticians (3.2), physicists (3.3) and computer scientists (3.4). Using the language of each group these sections attempt to allow experts to exchange their ideas better. The focus of this review are techniques for implementing the components of NS, enumerated in section 3.5, in such a way that they efficiently address the PMDIT challenges. Section 4 gives an introduction to the integration procedure, and references convergence proofs. Termination criteria are discussed in §4.3, followed by a discussion of the computational complexity in §4.4. Diagnostics to determine the correctness and quality of a NS run are presented in §4.5, including a new test. Sampling methods for use inside NS are extensively reviewed, including methods based on random walks (§5.1), rejection sampling (§5.2), and hybrids (§5.3). Section 6 then discusses variations of NS, which soften the hard likelihood constraint, vary the number of live points and parallelise the algorithm. In §7, a simple numerical experiment demonstrates some of the behaviours of NS implementations.

The survey of techniques presented helped inform design decisions for our own open-source NS implementation. *UltraNest*² is a high-performance general purpose nested sampling library for models written in the Python, C, C++, Fortran, Julia or R programming languages, with a focus on reliability. We document the design decisions for *UltraNest* in the relevant sections.

3. Introduction to Nested Sampling

We present introductions aimed at different audiences, with the goal of enabling different groups to translate between their languages used. The sections present an introduction NS from the perspective of a Bayesian practitioner (§3.1), theoretical statistician (§3.2), physicist (§3.3) and computer scientist (§3.4). The components of NS are then identified in §3.5.

3.1. Conceptual introduction

To get started, a reference version of the algorithm is introduced. The theoretical background, justifications and variations are discussed in subsequent sections. While the algorithm is not limited to continuous priors, to understand the concepts, it can help to first consider a parameter space with uniform priors. For example, $V = \int \pi(\theta) d\theta$ can intuitively be associated with a volume. Some implementations also prefer this approach, and support nonuniform, including dependent, priors by inverse transforming with the cumulative prior distribution (see Appendix A).

NS is an integration algorithm that provides both the posterior samples and the marginal likelihood Z . The approach is akin to Lebesgue integration, which requires keeping track of the height (the likelihood) and the volume. Lets consider that we want to compute the marginal likelihood over a d -dimensional continuous parameter space. Figure 1 illustrates the procedure described below.

Initialisation Sample randomly from the prior N live points (e.g., $N = 400$) and evaluate the likelihood function at each point.

Shrinkage Remove the live point with the lowest likelihood, L_1 (the worst fit), which becomes the first dead point. Considering that each point represents $1/N$ of the total volume, this reduces the volume by a factor of approximately $\delta V = 1/N$. Three more estimators of the removed volume are common: Considering that the samples split the volume in a uniformly sampled fashion by the sampled L thresholds, the volume δV of the removed shell, $L > L_{\min}$, is a random variable drawn from a Beta(1, N) distribution. This distribution can be randomly sampled, or the geometric

²<https://johannesbuchner.github.io/UltraNest/>

均值 $\delta V = 1 - \exp(-\frac{1}{N})$ 和算术均值 $\delta V = \frac{1}{N+1}$ 被视为估计器。这将在 § 4.2 中进一步讨论。对于最小的 N 之外的所有情况，这些估计器之间的差异在实践中可以忽略不计。为简化教学， $\delta V = \frac{1}{N}$ 在本工作中被采用。移除一个点后，剩余体积为 $V = 1 - \frac{1}{N}$ ，即体积缩小了 $(N-1)/N$ 倍。

似然约束先验采样 (LRPS) 随机从先验中采样一个新的独立活点，但要求其似然性必须超过 L_1 。这一步称为似然约束先验采样，LRPS，也称为约束采样或约束模拟。第5节详细讨论了LRPS方法。任何似然性低于 L_1 的区域都不会再考虑，我们在体积内再次有 N 个活点。

迭代我们重复替换活点（收缩和LRPS步骤），这会持续提高似然性阈值并将体积缩小约一个常数因子。换句话说，移除 N 的最低点会自动选择似然性阈值，使得体积至少平均缩小一个常数因子。NS从最差到最好似然性扫描先验。通过常数收缩因子推进会指数级减少剩余体积。

终止经过 i 次迭代后，剩余体积呈指数级缩小， $V_i = (1 - \frac{1}{N})^i$ ，高似然性阈值会选择接近最佳参数峰值的活点，即剩余点的似然性趋于平坦。因此对 Z 的进一步 $\Delta V_i \times L_i$ 贡献可以忽略不计，积分可以停止。算法已收敛，即进一步迭代不会显著改变结果。其他终止标准在 § 4.3 中讨论。

积分在迭代 i 移除一个点后减少了体积 $V_i = (1 - \frac{1}{N})^i$ by $\Delta V_i = V_i - V_{i-1} = (1 - \frac{1}{N})^i \times \frac{1}{N}$ 。这可以想象成先前体积的壳被剥掉。这个积分贡献的“层高度”就是死点的似然性 L_{i0} 。因此，每个死点被分配未归一化的权重 $\Delta V_i \times L_i$ ，积分 Z 简单地是： $Z \approx \sum_i \Delta V_i \times L_i$ ，并提供误差估计。终止时剩余的活点也可以被包含，它们的似然性乘以剩余体积并平均分配给他们， V_i/N 。加权死点是后验分布的近似样本，为了方便可以按 $\Delta V_i \times L_i$ 比例重新采样为未加权后验样本。

特性图 1 说明了此过程的一些重要方面。首先，NS 定义了初始化和终止点，因此可以在没有用户参与的情况下运行。它对参数空间进行全局探索。这使得 NS 能够鲁棒地识别和表征多个模式。从整个体积到逐渐更小的区域的扫描是通过不断增加的 L 阈值完成的。收缩是通过常数因子进行的，这意味着 L 层动态增加。这能同样很好地处理重尾和轻尾分布。此外，它还遍历相变，例如图 1 中从黄色平台到模式橙色基底的相变。这个相变在图 1 的底部面板中突出显示，其中首先随着体积缩小，似然性保持近似恒定，然后迅速增加，使 $\log X - \log L$ 曲线变成非凹形。

图1中第二靠下的面板中的体积-似然图显示，在这个简单示例中，大部分空间具有非常低的似然值，对积分的贡献很小。NS需要遍历这个空间，有时需要很长时间，直到大部分后验分布被达到。

新点的几何发现是基于先验分布进行的。似然函数被用作一个预言机，用于二元决策，即建议的新点是在内部还是外部。在这个先验空间的几何采样中，NS不需要似然函数梯度，这使得它易于与来自传统代码的复杂、数值似然函数集成。

最后，当剩余的活点占据极小的先验体积时，NS会终止无监督过程，这个体积对积分的贡献微乎其微。

总结来说，NS是一种吸引人的贝叶斯推理算法框架，因为

1. 它全局探索参数空间，
2. 它很好地处理多模态分布和相变，
3. 它在定义明确的位置初始化和终止，无需繁琐的监督，并且
4. 它同时提供边际似然和后验样本。

mean $\delta V = 1 - \exp(-\frac{1}{N})$ and arithmetic mean $\delta V = \frac{1}{N+1}$ considered as estimators. This is discussed further in §4.2. For all but the smallest N , the discrepancy between these estimators is negligible in practice. For pedagogical simplicity, $\delta V = \frac{1}{N}$ is adopted in this work. With one point removed, the remaining volume is $V = 1 - \frac{1}{N}$, i.e., the volume shrank by the factor $(N-1)/N$.

Likelihood-restricted prior sampling (LRPS) A new, independent live point is sampled randomly from the prior, but it is required that its likelihood exceeds L_1 . This step is called likelihood-restricted prior sampling, LRPS, also known as constrained sampling or constrained simulation. Section 5 extensively discusses LRPS methods. Any region with likelihoods below L_1 is not considered any further, and we have again N live points within a volume.

Iterations We repeat replacing live points (shrinking and LRPS steps), which continuously increases the likelihood threshold and shrinks the volume by approximately a constant factor. Put in another way, removing the lowest point of N automatically chooses likelihood thresholds such that the volume decreases by a constant factor, at least on average. NS scans the prior from the worst to best likelihood. The progression by constant shrinkage factors reduces the remaining volume exponentially.

Termination After i iterations the remaining volume is exponentially small, $V_i = (1 - \frac{1}{N})^i$, with a high likelihood threshold selecting live points close to the best-fit parameter peak(s), i.e., the likelihood of the remaining points is flat. Further $\Delta V_i \times L_i$ contributions to Z are thus negligible and the integration can be stopped. The algorithm has converged in the sense that iterating further would not significantly alter the result. Other termination criteria are discussed in §4.3.

Integration Removing a point at iteration i reduced the volume $V_i = (1 - \frac{1}{N})^i$ by $\Delta V_i = V_i - V_{i-1} = (1 - \frac{1}{N})^i \times \frac{1}{N}$. This can be envisioned as a shell of prior volume being peeled off. The “level height” for this integration contribution is just the likelihood of the dead point, L_i . Accordingly, each dead point is assigned the unnormalised weight $\Delta V_i \times L_i$, and the integral Z is simply: $Z \approx \sum_i \Delta V_i \times L_i$, with an error estimate available. The remaining live points at termination can also be included, with their likelihoods multiplied by the remaining volume distributed equally among them, V_i/N . The weighted dead points are approximate samples from the posterior, and can for convenience be resampled proportional to $\Delta V_i \times L_i$ into unweighted posterior samples.

Properties Figure 1 illustrates some important aspects of this procedure. Firstly, NS defines an initialisation and termination point, and can thus run without the user in the loop. It performs a global exploration of the parameter space. This makes NS robust to identify and characterise multiple modes. The scanning from the entire volume to progressively smaller regions is done by an ever-increasing L threshold. The shrinkage is done by constant factors, which means the L level is increased dynamically. This handles heavy and light tailed distributions equally well. Additionally, it traverses phase transitions, such as from the yellow plateau to the orange base of the modes in Figure 1. This phase transition is highlighted in the bottom panel of Figure 1, where first the likelihood stays approximately constant as the volume shrinks, then increases rapidly, making the $\log X - \log L$ curve non-concave.

The volume-likelihood plot in the second-to-bottom panel of Figure 1 shows that much of the space in this simple example has a very low likelihood, which contributes little to the integral. NS needs to traverse this space, sometimes for a long time, until the bulk of the posterior is reached.

The geometric discovery of new points is done based on the prior. The likelihood function is queried as an oracle for a binary decision, namely, whether a suggested new point is inside or outside. In this geometric sampling of the prior space, NS does not require likelihood function gradients, making it easy to integrate with complex, numerical likelihoods from legacy codes.

Finally, NS terminates unsupervised, when the remaining live points occupy a tiny prior volume, which contributes vanishingly little to the integral.

To summarise, NS is an attractive algorithm framework for Bayesian inference because

1. it explores the parameter space globally,
2. it handles multi-modal distributions and phase transitions well,
3. it initialises and terminates at a well-defined point without cumbersome supervision, and
4. it provides both the marginal likelihood and posterior samples.

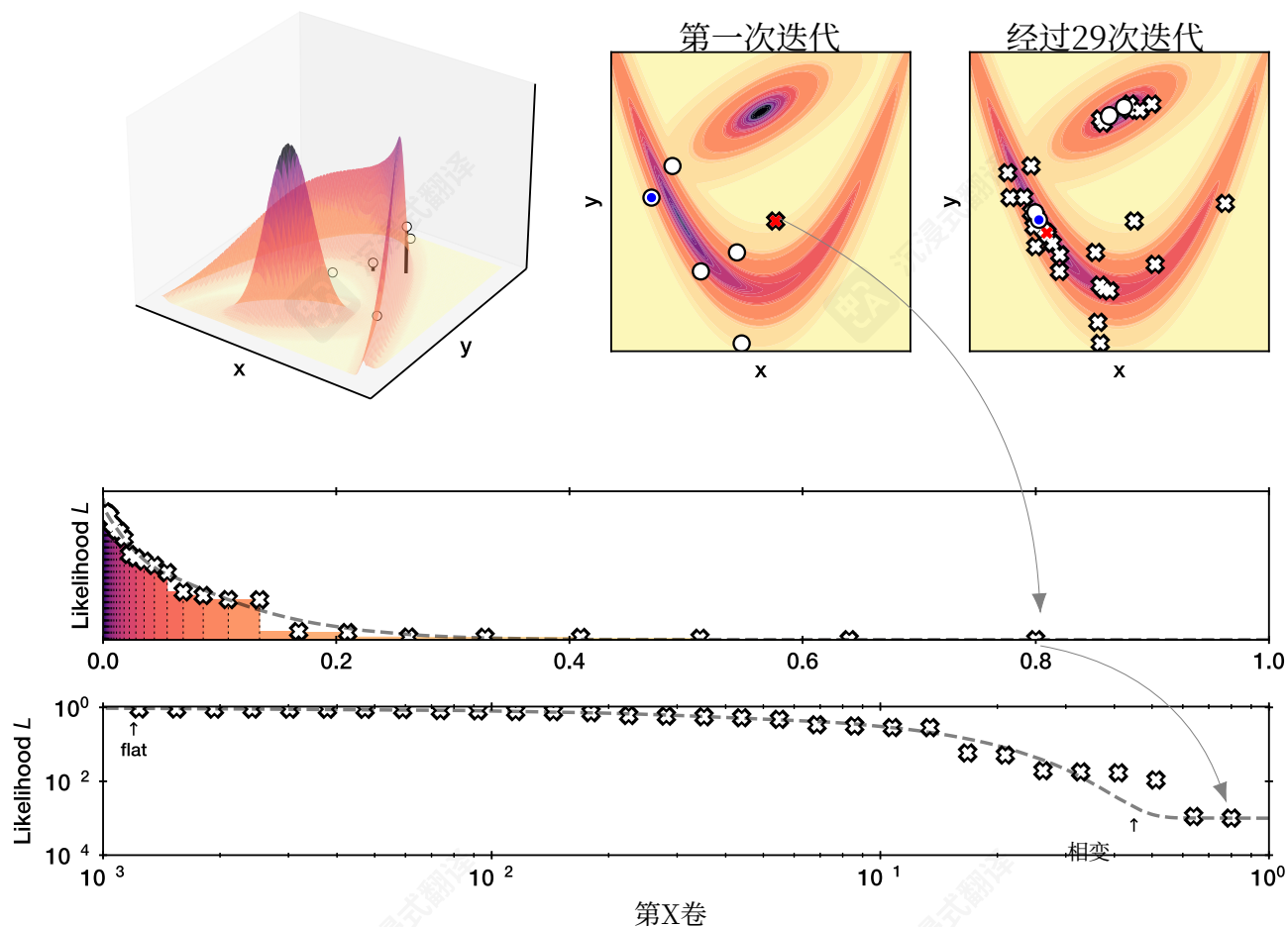


图1。左上：在一个二维参数空间中定义了一个复杂的似然函数。NS从评估 $N = 5$ 个随机点开始。上中：每个活点（圆圈）定义一个似然等高线。最低似然点（红色十字）成为死点。它被一个新活点（蓝色圆圈）替换，该点从先验分布中随机采样，但位于由死点定义的等高线上方。右上：经过几次迭代后，活点集中在似然峰值处的小体积内。底部面板（上部是线性，下部是对数）：每次迭代都会放置一个死点，并标出其似然值，并通过几何收缩估计先验体积。该序列的先验体积从右到左呈指数缩小。垂直条表示被移除的似然壳，颜色与其它面板显示的等高线一致。条形面积是后验权重，条形总和给出边际似然 Z 。灰色虚线表示该函数的真实体积-似然关系。在下部的对数-对数图中，标记了相变，它对应于上部面板中从宽而浅的黄色区域到高而陡峭的橙色区域的过渡。

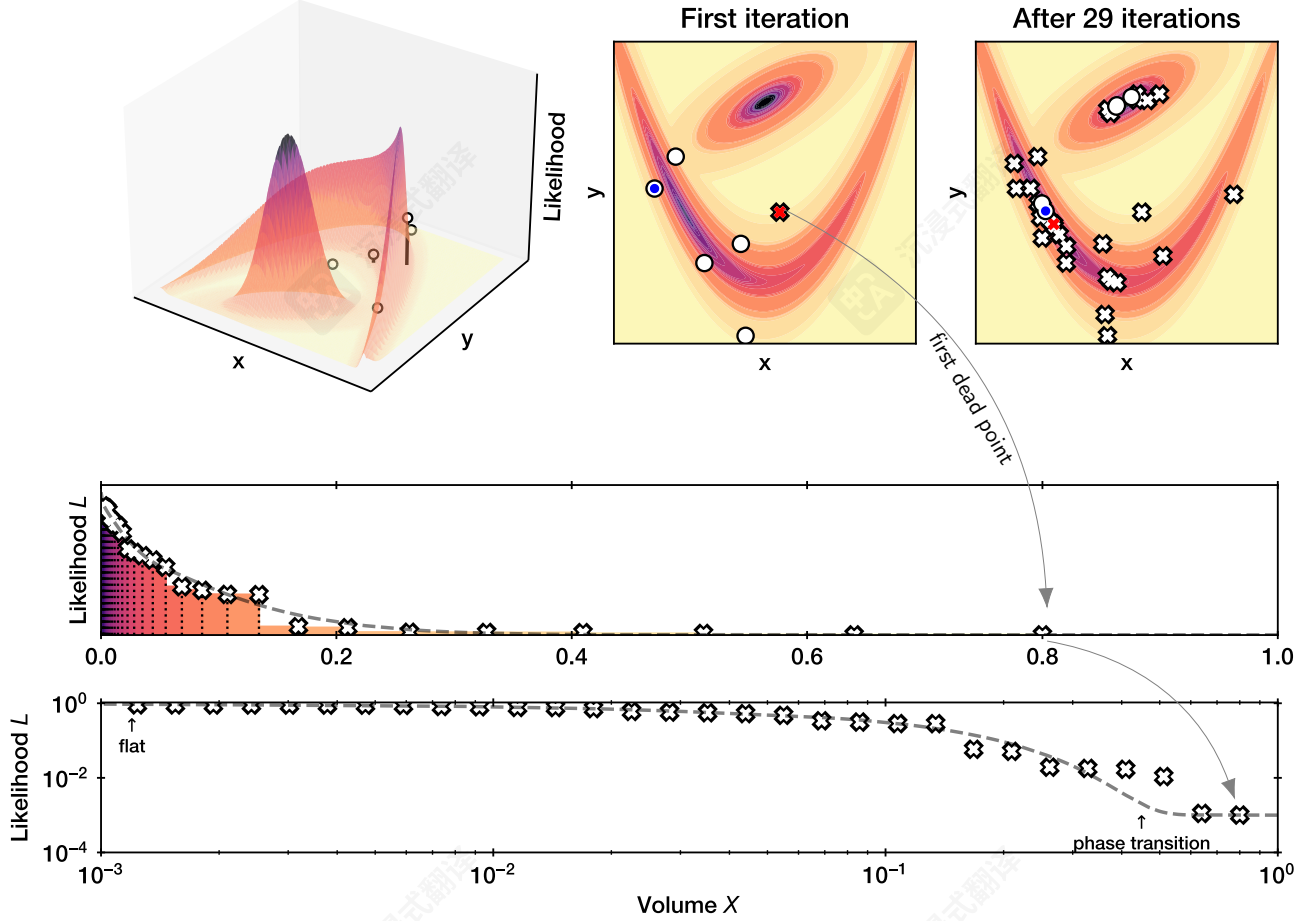


FIGURE 1. Top left: A complicated likelihood function is defined over a two-dimensional parameter space. NS begins by evaluating $N = 5$ random points. Top center: Each live point (circle) defines a likelihood contour. The lowest likelihood point (red cross) becomes a dead point. It is replaced by a new, live point (blue circle), sampled randomly from the prior but above the contour defined by the dead point. Top right: After a few iterations, the live points concentrate in a small volume at the likelihood peak. Bottom panels (top is linear, bottom is logarithmic): For each iteration, a dead point is placed with its likelihood and the prior volume estimated by geometric shrinkage. The prior volume of the sequence shrinks exponentially from right to left. Vertical bars represent the likelihood shell removed, and are coloured consistent with the contours shown in the other panels. The bar area is the posterior weight, and the sum of the bars gives the marginal likelihood Z . The gray dashed curve indicates the true volume-likelihood relation for this function. In the bottom log-log plot, the phase transition is marked, which corresponds to the transition from the wide, shallow yellow regions to the high and steep orange regions in the upper panels.

3.2. 理论统计学家视角

为求解 d 维的 θ 和 D 联合分布积分,

$$Z = \int \cdots \int P(D|\theta) \times \pi(\theta) d\theta_1 \cdots d\theta_d, \quad (1)$$

NS 将其转化为一个一维积分。似然约束先验的生存函数为 (Chopin and Robert, 2007a, 2010):

$$\begin{aligned} X(L_{\min}) &= \text{pr}\{L(D|\theta) > L_{\min}\} \\ &= \int_{L(D|\theta) > L_{\min}} \pi(\theta) d\theta \end{aligned}$$

然后通过逆变换, 利用似然函数实现了对先验空间的“排序”:

$$Z = \int_0^1 L_{\min}(X) dX \quad (2)$$

乍一看, 这种概念上的转换似乎并未带来任何进展, 因为在大多数非平凡函数中, 相关的多维空间无法与 $L_{\min}$ 直接对应。

相反, NS 选择 L 层级, 以便对应的 X 可以估计。请注意, X 、 $L_{\min}(X)$ 的逆是一个单调递增函数。它在图 1 的底部面板中可视化。假设 $\theta_1 \dots \theta_N, \dots$ 是来自先验分布的独立同分布样本, 它们的似然值为 $L_1, \dots, L_N$ 。根据定义, 这些似然样本的生存函数是 X 。因此, 样本的概率积分变换 $X(L_1) \dots X(L_N), \dots$ 是独立同分布的标准均匀分布。这意味着从先验分布中采样的点在先验体积 X 中均匀分布 L 层级。假设样本按 $L_1 = \min\{L_1, \dots, L_N\}$ 索引。那么根据均匀随机变量集合的顺序统计量的性质, 相应的 X_1 遵循 $X(L_1) \sim \text{Beta}(N, 1)$ (Skilling, 2004)。第 4.2 节将讨论具有不同假设的此步骤的替代方案。

设置 $L_{\min} = L_1$ 并重复采样程序, 其中先验范围限制为 $L(D|\theta) > L_{\min}$, 从而诱导出嵌套采样。递归过程追踪一个不断缩小的 X , 同时其似然阈值不断增大 $L_{\min}$ 。在受限的先验空间内, $X(L_2) \dots X(L_N), \dots$ 等也是均匀分布的, 具体范围从 $X(L_{\min})$ 到 1。因此, 采样的 X_i 序列具有 $X_{i+1}/X_i \sim \text{Beta}(N, 1)$ 的性质, 其中 $X_1 = 1$ 。这使得诸如 $Z = \sum_i (X_{i-1} - X_i) \times L_i$ 的估计量可计算。第 4 节讨论了 X 、 Z 和 $\log Z$ 的收敛性证明以及无偏积分估计量的构造。

上文仅简要介绍了 NS 涉及的思想。我们建议感兴趣的读者参考 Chopin 和 Robert (2010) 以及 Schittenhelm 和 Wacker (2020) 以获取更正式的介绍, 参考 Walter (2017) 分析在用于通过从一个似然阈值采样到下一个似然阈值来追踪收缩的有限有序点序列中出现的蒙特卡洛点过程, 以及参考 Salomone 等人 (2018) 了解与序贯蒙特卡洛的联系。

马尔可夫链蒙特卡洛 (MCMC) 的流行使其值得比较这两种迭代蒙特卡洛算法。从一个起点开始, 随机游走 Metropolis MCMC 构建一系列点。为了选择下一个点, 需要定义一个提议或转换核, 而 Metropolis 接受规则根据后验概率比选择提议点或当前点作为下一个点。如果链无限运行, 链点的分布将收敛到后验分布。具有有限链的 MCMC 性能关键取决于转换核, 许多方法已被提出。类似地, NS 的性能关键取决于 LRPS, 许多方法已被提出。

MCMC 和 NS 算法也可以通过它们在典型应用中的涌现行为进行定性比较。MCMC 通常表现出一个初始阶段, 试图识别后验主体。在这个阶段, 后验密度通常以数量级快速增加。这个初始阶段后, 开始探索后验分布, 链开始收敛, 有效独立样本数与链长成正比。

我们还可以识别出 NS 算法表现出的三个新兴阶段。最初, 体积很大, 活动点差异很大, 包括许多不良匹配, 因此死点接收的权重 $\Delta V_i L_i$ 最终可以忽略不计。由于活动点在其似然值上差异很大, 如果算法在这些迭代中终止, 所有后验权重都将集中在迄今为止找到的最可能点 $Z \approx L_i V_i / N$ 。当体积仍然很大且似然值很高时, 使得 $L_i \times V_i$

3.2. Viewpoint for theoretical statisticians

To solve the d -dimensional integral of the joint distribution of θ and D ,

$$Z = \int \cdots \int P(D|\theta) \times \pi(\theta) d\theta_1 \cdots d\theta_d, \quad (1)$$

NS transforms it into a one-dimensional integral. The survival function of a likelihood-restricted prior is (Chopin and Robert, 2007a, 2010):

$$\begin{aligned} X(L_{\min}) &= pr\{L(D|\theta) > L_{\min}\} \\ &= \int_{L(D|\theta) > L_{\min}} \pi(\theta) d\theta \end{aligned}$$

Then a “sorting” of the prior space via the likelihood function is achieved by the inverse:

$$Z = \int_0^1 L_{\min}(X) dX \quad (2)$$

At first glance, this conceptual transformation has not achieved anything, because the relevant multi-dimensional spaces cannot be identified with $L_{\min}$ in all but the most trivial functions.

Instead, NS chooses the L levels such that the corresponding X can be estimated. Note that the inverse of X , $L_{\min}(X)$, is a monotonically increasing function. It is visualised in the bottom panel of Figure 1. Suppose $\theta_1, \dots, \theta_N$ are i.i.d. samples from the prior, and their likelihood is $L_1, \dots, L_N$. By definition, the survival function of these likelihood samples is X . Therefore, the probability integral transform of the samples, $X(L_1), \dots, X(L_N)$ is i.i.d. standard uniform distributed. The implication is that points sampled from the prior generate L levels uniform in the prior volume X . Lets assume the samples were indexed so that $L_1 = \min\{L_1, \dots, L_N\}$. Then by the properties of order statistics of a collection of uniform random variables, the corresponding X_1 follows $X(L_1) \sim \text{Beta}(N, 1)$ (Skilling, 2004). Section 4.2 below discusses alternatives for this step with different assumptions.

Setting $L_{\min} = L_1$ and repeating the sampling procedure with the prior restricted to $L(D|\theta) > L_{\min}$ induces nested sampling. The recursion tracks an ever-shrinking X with an ever-increasing likelihood threshold $L_{\min}$. Within the restricted prior space, $X(L_2), \dots, X(L_N)$ are also uniformly distributed, specifically from $X(L_{\min})$ to unity. The sequence of sampled X_i thus has the property $X_{i+1}/X_i \sim \text{Beta}(N, 1)$, with $X_1 = 1$. This makes estimators such as $Z = \sum_i (X_{i-1} - X_i) \times L_i$ computable. Section 4 discusses convergence proofs and construction of unbiased integral estimators for X , Z and $\log Z$.

Above, only a very brief introduction of the ideas involved in NS was given. We refer to interested reader to Chopin and Robert (2010) and Schittenhelm and Wacker (2020), for more formal introductions, to Walter (2017) for an analysis of the Monte Carlo point process occurring in the sequence of finite ordered points used to track shrinkages by sampling from one likelihood threshold to the next, and Salomone et al. (2018) for connections to Sequential Monte Carlo.

The popularity of Markov Chain Monte Carlo (MCMC) makes it worthwhile to draw comparisons between the two iterative Monte Carlo algorithms. From a starting point, Random Walk Metropolis MCMC constructs a sequence of points. For choosing the next point, a proposal or transition kernel needs to be defined, and the Metropolis acceptance rule either chooses the proposed point or the current point as the next point, proportional to the posterior probability ratio. If chains are run infinitely long, the distribution of chain points converges to the posterior distribution. The performance of MCMC with finite chains crucially depends on the transition kernel, and many methods have been proposed. Similarly, the performance of NS crucially depends on the LRPS, and many methods have been proposed.

The MCMC and NS algorithms can also be qualitatively compared by their emergent behaviour in typical applications. MCMC typically exhibits an initial phase where it attempts to identify the posterior bulk. In this phase, the posterior density is typically rapidly increasing by orders of magnitude. This initial phase is followed by exploration of the posterior, where the chain begins to converge, and the number of effectively independent samples is proportional to the length of the chain.

We can also identify three emerging phases the NS algorithm exhibits. Initially, the volume is large and the live points vary by many orders of magnitude, including many bad fits, so that the dead points receive weights $\Delta V_i L_i$ that are ultimately negligible. Because the live points vary in their likelihood value by many orders of magnitude, if the algorithm were terminated in these iterations, all of the posterior weight would be concentrated in the most likely point found so far $Z \approx L_i V_i / N$. Where the volume is still substantial and likelihoods are high, so that $L_i \times V_i$

达到最大值，后验主体被达到，我们可以将其识别为第二阶段。在这里，多个点获得可比权重 $V_i L_i$ ，即后验被解析为多个后验样本。由于NS需要跟踪收缩，它不能像MCMC那样快速跳到这个阶段，因此（但见 § 6.2）识别后验主体的阶段可能需要许多迭代（与信息增益（Skilling, 2004）成正比）。最后，NS表现出一个阶段，其中似然值很高且非常接近最大似然值，但体积已经变得非常小。因此，大多数点接收较小的权重，后验主体已经被通过。在这里，NS与MCMC的不同之处在于，继续运行不会线性增加有效样本量。第 § 3.4和 § 6.2节讨论了使用额外迭代来增加后验样本量的方法。

3.3. 物理学家视角

许多蒙特卡洛算法源于对物理系统的类比。以Skilling (2012) 和Habeck (2015) 的例子为例，考虑一个盒子中的多个气体粒子。所有粒子的位置和速度完全描述了系统的微观状态，或配置 θ 。如果粒子在（可能形状怪异）的盆地中受重力滚动，系统的总势能 $E(\theta)$ 可以被识别。这就是与负对数似然 $E(\theta) = -\log L$ 的类比。NS最初生成随机粒子配置。选择能量最高（即 $E(\theta)$ 最高）的配置，能量为 ϵ 。然后生成具有比当前能量限制 $E(\theta') \leq \epsilon$ 更低的能量状态 θ' 的新配置，例如通过抖动粒子。

迭代地将最热配置替换为较冷的配置对应于冷却方案。蒙特卡洛冷却方案已知于模拟退火（例如，Kirkpatrick、Gelatt和Vecchi, 1983）和并行退火（Swendsen和Wang, 1986）。NS在此处通过自适应选择冷却方案，并通过几何方式探索截断盆地而不是按比例探索平滑盆地而有所不同。在冷却过程中，可能会发生能量变化很小而体积持续减小的情况，随后出现行为突然变化，能量迅速增加。这种相变（例如，Raghavan和Cohen, 1975，为物理背景）在模拟退火中是有问题的，因为它考虑了能量变化的幅度。相比之下，因为NS在体积中以恒定速度推进，并且只考虑活点的顺序，所以它可以毫无问题地穿越相变（Skilling, 2004）。有关平台，参见 § 4.5.2。

NS 认为孤立系统具有最大能量（微正则观点），而不是系综平均（正则观点）。更明确地说，Habeck (2015) 在 NS 程序中识别了统计力学中的几个术语，如下：配置的体积 X ，能量低于阈值 ϵ ，是

$$X(\epsilon) = \int_{-\infty}^{\epsilon} g(E) dE,$$

其中能量 E 的态密度为（另见 Cameron 和 Pettitt, 2014）：

$$g(E) = \int \delta(E - E(\theta)) \pi(\theta) d\theta$$

从概率术语来看， $g(E)$ 描述了在先验上边缘化的负对数似然值的分布，而 $X(\epsilon)$ 是其累积概率分布。然后 X 的对数可以识别为微正则熵 $S_G(E) = \log X(E)$ ，而 g 的对数是表面熵 $S_B(E) = \log g(E)$ 。然后可以定义微正则温度为 $T_G = X(E)/g(E)$ 。所有配置的总能量，即配分函数 Z ，定义为 $Z = \int e^{-\beta E(\theta)} \pi(\theta) d\theta$ ，其中 $\beta = 1$ 由 NS 评估为 $Z = \int e^{-E} g(E) dE$ ，即通过能量级。更多细节，参见 Habeck (2015) 和 Cameron 和 Pettitt (2014)。

新配置的生成也可以类比于物理系统来考虑。在这种情况下，每个配置被视为一个粒子，它占据一个能量势。在能量盆地中，粒子的加速度可以像哈密顿蒙特卡洛（HMC, Neal 等人, 2011）的发展那样来解释。HMC 使用势能构建轨迹，这可以被视为随机方向的类开普勒轨道（Betancourt, 2017）。NS 和 HMC 类比在两个方面存在重要差异。HMC 轨迹守恒总能量，将其划分为势能和动能。这倾向于只探索一个狭窄的势能范围，该范围由维度数决定，并限制了 HMC 对新配置的探索（例如遥远的盆地）。相比之下，NS 从最热到最冷扫描势能，并在所有能量级别生成配置。第二个差异是，NS 无论能量如何都搜索新配置，只要它们满足能量阈值。因此，粒子不会获得加速度，探索完全是几何的。我们建议读者参考 Nielsen (2013)；Martiniani 等人 (2014)；Habeck (2015)；Baldock 等人 (2016)，以了解基于统计力学的 NS 公式，以及关于类似台球轨迹的 § 5.1.3。

is maximal, the posterior bulk is reached, which we can identify as a second phase. Here, multiple points receive comparable weights $V_i L_i$, i.e., the posterior becomes resolved into multiple posterior samples. Because NS needs to track the shrinkage, it cannot rapidly skip ahead to this phase like MCMC, and therefore (but see §6.2) the phase of identifying the posterior bulk can take many iterations (proportional to the information gain (Skilling, 2004)). Finally, NS exhibits a phase where the likelihoods are high and very close to the maximum likelihood, but the volume has become very small. Therefore, most points receive a small weight and the posterior bulk has been passed. Here NS differs from MCMC in that continuing the run does not linearly increase the effective sample size. Section §3.4 and §6.2 discuss methods for bulking the posterior samples with additional iterations.

3.3. Viewpoint for physicists

Many Monte Carlo algorithms stem from analogies to physical systems. To give an example (from Skilling, 2012; Habeck, 2015), consider several gas particles in a box. The position and velocities of all particles then completely describe the microstate, or configuration θ of the system. If the particles are rolling under gravity within a (perhaps strangely shaped) basin, the total potential energy of the system $E(\theta)$ can be identified. This is the analogy to the negative log-likelihood, $E(\theta) = -\log L$. NS initially generates random particle configurations. The hottest configuration (highest $E(\theta)$), with energy ϵ is selected. New configurations θ' accessible with a lower energy state $E(\theta') \leq \epsilon$ than the current energy limit are generated, for example by jittering the particles.

Iteratively replacing the hottest configuration by a cooler configuration corresponds to a cooling schedule. Monte Carlo cooling schemes are known from simulated annealing (e.g., Kirkpatrick, Gelatt and Vecchi, 1983) and parallel tempering (Swendsen and Wang, 1986). NS differs here by choosing the cooling schedule adaptively, and that it explores a truncated basin geometrically rather than a smoothed basin proportionally. During the cooling process, it can occur that the energy changes very little, while the volume keeps decreasing, followed by an abrupt change of behaviour where the energy increases rapidly. Such phase transitions (see e.g., Raghavan and Cohen, 1975, for physics background) are problematic in simulated annealing because it considers the magnitude of energy changes. In contrast, because NS progresses with constant speed in volume and considers only the order of the live points, it traverses phase transitions without issue (Skilling, 2004). For plateaus, see §4.5.2.

NS considers an isolated system with a maximum energy (microcanonical view) rather than the ensemble average (canonical view). More explicitly, Habeck (2015) identified several terms from statistical mechanics in the NS procedure, as follows: The volume of configurations X , with less energy than a threshold ϵ , is

$$X(\epsilon) = \int_{-\infty}^{\epsilon} g(E) dE,$$

where the density of states at energy E is (see also Cameron and Pettitt, 2014):

$$g(E) = \int \delta(E - E(\theta)) \pi(\theta) d\theta$$

In probability terms, $g(E)$ describes the distribution of negative log-likelihood values marginalised over the prior, with $X(\epsilon)$ its cumulative probability distribution. The logarithm of X can then be identified as the microcanonical entropy $S_G(E) = \log X(E)$, while the logarithm of g is the surface entropy $S_B(E) = \log g(E)$. A microcanonical temperature can then be defined as $T_G = X(E)/g(E)$. The total energy of all configurations, the partition function Z , is defined as $Z = \int e^{-\beta E(\theta)} \pi(\theta) d\theta$ with $\beta = 1$, which is evaluated by NS as $Z = \int e^{-E} g(E) dE$, i.e., over the energy levels. For more details, see Habeck (2015) and Cameron and Pettitt (2014).

The generation of new configurations can also be considered in analogy with physical systems. In this case, each configuration is considered a particle, which inhabits an energy potential. The acceleration of a particle in an energy basin can be motivated as in the development of Hamiltonian Monte Carlo (HMC, Neal et al., 2011). HMC constructs trajectories using the potential energy, which can be considered Keplerian orbits (Betancourt, 2017) of random orientation. NS and HMC analogies differ in two important ways. HMC trajectories conserve the total energy, partitioned into potential and kinetic energy. This tends to explore only a narrow range of potential energies, set by the number of dimensions, and limits HMC's exploration of new configurations (such as distant basins). In contrast, NS scans potential energies from hottest to coolest, and generates configurations at all energy levels. The second difference is that NS searches for new configurations regardless of their energy, so long as they fulfil the energy threshold. Thus, the particles receive no acceleration, and the exploration is purely geometric. We refer the reader to Nielsen (2013); Martiniani et al. (2014); Habeck (2015); Baldock et al. (2016) for formulations of NS based on statistical mechanics, and for billard-like walks, to §5.1.3.

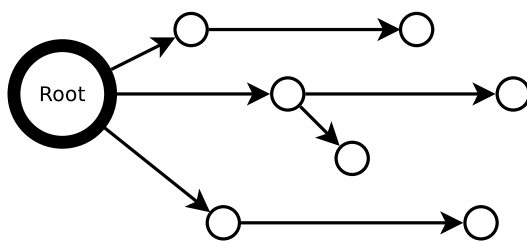


图2。将NS探索呈现为树状结构。根节点表示整个体积，其似然值为零。每个节点代表从先验分布中采样的一个点，且仅限于其父节点的似然阈值之上。在此呈现方式中，节点还按似然值从左到右排序。每条出边将父节点关联的体积分割，并将其子节点分配一个体积为 $\frac{1}{N}$ 的体积，其中 N 是并行边的数量。

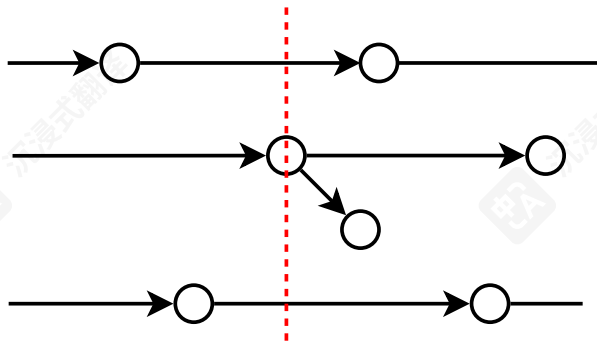


图3。搜索操作：广度优先搜索按似然值顺序（从左到右）迭代。活点的数量 N 是节点列表的大小，对应于并行边的数量（此处：红色虚线前有三个，之后有四个）。体积被缩减，节点被分配后验权重，积分也得到更新。

3.4. 计算机科学家的视角

在计算机科学中，为了快速缩小搜索空间，经常采用二分搜索或 k-d 树等分治算法。通常，算法与特定数据结构紧密相关，该数据结构在任何时候都能完全表示状态。研究此类数据结构的特性可能很有启发性。在本节中，我们识别了这样的数据结构，并将 NS 表述为在该数据结构上运行的算法。这种表示使恢复现有的 NS 运行、并行化和动态更改活点数变得简单，并避免了算法最终阶段的特殊处理。

我们在树结构（见图2）上调整广度优先搜索。搜索从初始节点开始，并保持一个待探索节点的排序列表。最初，NS树是一个单独的根节点，代表完整的先验体积。接下来，我们考虑在该体积内采样 N 个随机点。这通过为根节点添加 N 个子节点来表示，并在图2中说明。我们为子节点分配似然值。一个简单的广度优先搜索算法从根节点开始，保持一个“开放”节点列表，并反复探索具有最低似然值的节点，直到列表为空。剩余体积、积分似然值和后验值也一并计算。这以算法1呈现，并在图3中说明。

算法1中的公式简洁优雅，因为它统一了嵌套采样阶段和剩余积分：在剩余积分阶段，没有任何 N 并行节点有子节点，每个被移除的节点会使该体积缩小 $(N-1)/N$ ，从而在这个阶段实现权重相等。

此过程很快就会用尽要探索的节点，因此需要一个代理来为节点添加子节点。图4说明，向节点添加子节点意味着使用父节点的似然度作为阈值，进行LRPS采样一个新点。具有恒定活点数量的最简单公式为每个被传递的节点添加一个子节点，并在剩余体积可以忽略时停止。这个代理显示在算法2中。通常，任意数量的代理可以任意地为树中的节点添加子节点。

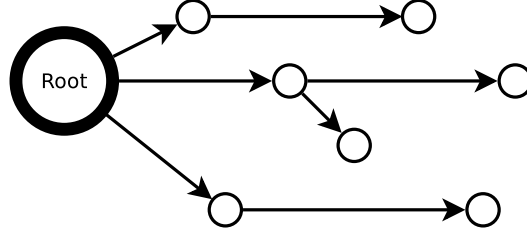


FIGURE 2. NS exploration presented as a tree. The root node presents the entire volume, with likelihood zero. Each node represents a point sampled from the prior, restricted to be above the likelihood threshold of its parent node. In this presentation, the nodes are also ordered from left to right by likelihood value. Each outgoing edge splits the volume associated with the parent and donates its child a volume of $\frac{1}{N}$ where N is the number of parallel edges.

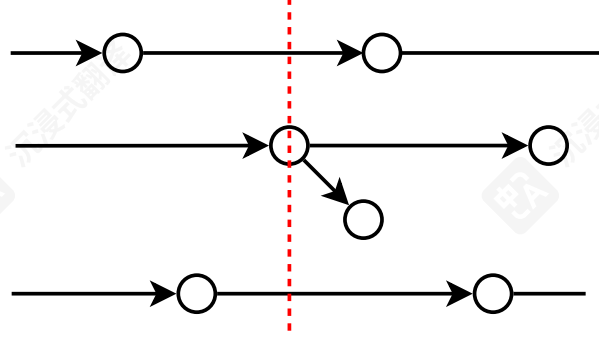


FIGURE 3. Search operation: The breadth-first search iterates in order of likelihood (from left to right). The number of live points N is the size of the node list, and corresponds to the number of parallel edges (here: three before the red dashed line, four after). The volume is reduced, the node assigned a posterior weight, and the integral updated.

3.4. Viewpoint for computer scientists

In computer science, to quickly narrow down a search space, divide-and-conquer algorithms such as binary search or k-d trees are frequently employed. Often, algorithms are closely identified with a specific data structure that fully represents the state at any time. It can be insightful to investigate the properties of such data structures. In this section, we identify such a data structure, and phrase NS as an algorithm operating on it. The representation makes resuming an existing NS run, parallelisation and dynamically varying the number of live points trivial, and avoids a special treatment of the final phase of the algorithm.

We adapt breadth-first search on a tree structure (see Figure 2). The search starts from an initial node and keeps a sorted list of nodes yet to be explored. Initially, the NS tree is a lone root node representing the full prior volume. Next, we consider sampling N random points inside that volume. This is represented by adding N child nodes of the root node, and illustrated in Figure 2. We assign the child nodes their likelihood value. A simple breadth-first search algorithm starts from the root node, keeps a list of “open” nodes, and repeatedly explores the one with the lowest likelihood value until the list is empty. The remaining volume, integrated likelihood and posteriors are also computed. This is presented in Algorithm 1, and illustrated in Figure 3.

The formulation in Algorithm 1 is elegant, because it unifies the nested sampling phase and the remainder integration: During the remainder integration, none of N parallel nodes have children, each removed node shrinks that volume by $(N - 1)/N$, leading to equal weights in this phase.

This procedure would soon run out of nodes to explore, so an agent is needed which adds children to nodes. Figure 4 illustrates that adding a child to a node means LRPS sampling a new point using the parent’s likelihood as the threshold. The simplest formulation with a constant number of live points adds a single child node to each node being passed, and stops when the remaining volume is negligible. This agent is shown in Algorithm 2. In general, an arbitrary number of agents can add children to arbitrary nodes in the tree.

Algorithm 1 NS as a breadth-first search algorithm.

```

1 function NS-BFS:
2   given root node (representing the prior volume)
3
4   let Q be a list, sorted by likelihood value
5   add all children of root node to Q
6
7   Z = 0
8   V_remaining = 1
9   posterior_points = []
10  posterior_weights = []
11
12  while Q is not empty:
13    Nlive = length of Q
14    obtain and remove next node from Q
15
16    # optional:
17    node_expanding_agent(node, Q, Z, V_remaining)
18
19    removed_fraction = 1 / Nlive
20    remaining_fraction = 1 - removed_fraction
21    V_dead = V_remaining * removed_fraction
22    weight = V_dead * node.likelihood_value
23
24    add node to posterior_points
25    add weight to posterior_weights
26    Z += weight
27    V_remaining *= remaining_fraction
28
29    add all children of node to Q
30  return Z, posterior_points, posterior_weights

```

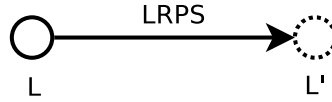


图4. 插入操作: 向树中添加一个子节点 (右) 意味着在父节点 (左) 的似然阈值下进行LRPS采样。

Algorithm 2 Exploration agent corresponding to constant- N nested sampling.

```

1 function classic_node_expanding_agent:
2   given node          # current node
3   given Q             # current list
4   given Z             # current integral estimate
5   given V_remaining   # volume left to be explored
6   Lmax = (last node in Q).likelihood_value
7   frac_remaining = Lmax * V_remaining / Z
8
9   if frac_remaining > 0.001 and node has no children:
10    new_node = LRPS(node.likelihood_value)
11    add new_node as a child of node

```

Algorithm 1 NS as a breadth-first search algorithm.

```

1 function NS-BFS:
2   given root node (representing the prior volume)
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16    # optional:
17    node_expanding_agent(node, Q, Z, V_remaining)
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19    removed_fraction = 1 / Nlive
20    remaining_fraction = 1 - removed_fraction
21    V_dead = V_remaining * removed_fraction
22    weight = V_dead * node.likelihood_value
23
24    add node to posterior_points
25    add weight to posterior_weights
26    Z += weight
27    V_remaining *= remaining_fraction
28
29    add all children of node to Q
30  return Z, posterior_points, posterior_weights

```

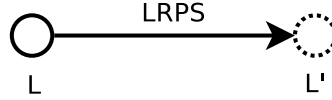


FIGURE 4. Insertion operation: Adding a child node (right) to the tree means LRPS sampling under the likelihood threshold of that parent node (left).

Algorithm 2 Exploration agent corresponding to constant- N nested sampling.

```

1 function classic_node_expanding_agent:
2   given node          # current node
3   given Q              # current list
4   given Z              # current integral estimate
5   given V_remaining    # volume left to be explored
6   Lmax = (last node in Q).likelihood_value
7   frac_remaining = Lmax * V_remaining / Z
8
9   if frac_remaining > 0.001 and node has no children:
10     new_node = LRPS(node.likelihood_value)
11     add new_node as a child of node

```

这种公式意味着，如果保持树形结构，恢复中断的运行非常简单。此外，算法1和2中指定的多个NS运行可以并行独立运行。对于合并，只需要合并根节点，并且仅对最终结果运行算法1的集成即可。该公式还建议了一种将先前采样的点及其阈值转换为树的程序：一个LRPS采样的点可以作为子节点附加到 likelihood 高于或等于所用采样阈值的节点上。

This formulation implies that resuming an interrupted run is trivial, if the tree is kept. Also, several NS runs as specified in Algorithm 1 and 2 can run independently in parallel. For merging, all that is needed is to merge the root nodes, and run only the integration of Algorithm 1 for the final result. The formulation also suggests a procedure for converting previously sampled points with their thresholds into a tree: a LRPS sampled point can be attached as a child to nodes with likelihood above or equal the used sampling threshold.

3.5. 嵌套抽样实现的组成部分

为了以结构化的方式讨论NS，我们识别了嵌套采样实现的组件。这些组件在图5中进行了说明。核心（中心）是一个NS采样器，它保持一组活点，并由最新的死点定义的 likelihood 约束。在每次迭代中，最低的活点可能被一个LRPS替换，该LRPS使用用户提供的应用特定 likelihood 函数和先验空间定义。死点被传递给NS积分器，该积分器对这些死点进行加权以形成后验样本，并估计边际 likelihood Z （参见 § 3 中的公式）。以下几节将讨论各个组件，包括积分器（§ 4）、终止标准（§ 4.3）、LRPS变体（§ 5）和采样器（§ 6）。

4. 积分

4.1. 理论

嵌套抽样（NS）在Skilling (2004)和Skilling等人 (2006)中被引入；收敛性和无偏性在Evans (2007)、Chopin和Robert (2010)、Skilling (2009)以及Keeton (2011)中被讨论和证明。在理论上分析NS的进展一直是少数几篇公开发表的论文（例如，Khanarian和Alvarez, 2013）的关注点。我们建议感兴趣的读者参考Walter (2017)，该文将嵌套抽样与稀有事件模拟的数学理论以及最后粒子算法联系起来，并参考Salomone等人 (2018)，该文将嵌套抽样与序贯蒙特卡洛联系起来，几乎将其视为该框架的一个特例。Birge、Chang和Polson (2012)提出了一种嵌套抽样的推广，并将其与路径抽样和桥抽样联系起来，而Polson和Scott (2014)讨论了其与其他抽样方法（如切片抽样）的联系。Chopin和Robert (2007a)以及Feroz等人 (2013, 附录C)证明，嵌套抽样得到的后验样本分别近似于连续和离散函数的真实后验分布。

4.2. 估测量

NS 基于能够估计体积（收缩量）在每次迭代中，并且 LRPS 样本完全可信。LRPS 问题在 § 5 中讨论，而对于这里的讨论，假设 LRPS 采样完美。这种情况在 Guyader、Hengartner 和 Matzner-Løber (2011) 中被称为理想化算法，在 Higson 等人 (2017) 中被称为完美嵌套采样。

NS 基于能够在每次迭代中估计体积（收缩量）。只有在特殊情况下体积才能被精确知道 (Chopin and Robert, 2008)。通常情况下它们只能被估计。将体积收缩量视为泊松过程可以得到体积收缩量估计器 $\delta V = \frac{1}{N}$ (Huber and Schott, 2010; Guyader, Hengartner and Matzner-Løber, 2011)。基于这种方法，Walter (2015) 推导出了一个无偏估计器 Z 。Chopin and Robert (2007a); Evans (2007) 和 Skilling (2009) 讨论了最终目标是否是获得一个方差最小的对数 Z 或 Z 的无偏估计器 (另见 Keeton, 2011)。Skilling (2009) 认为，因为

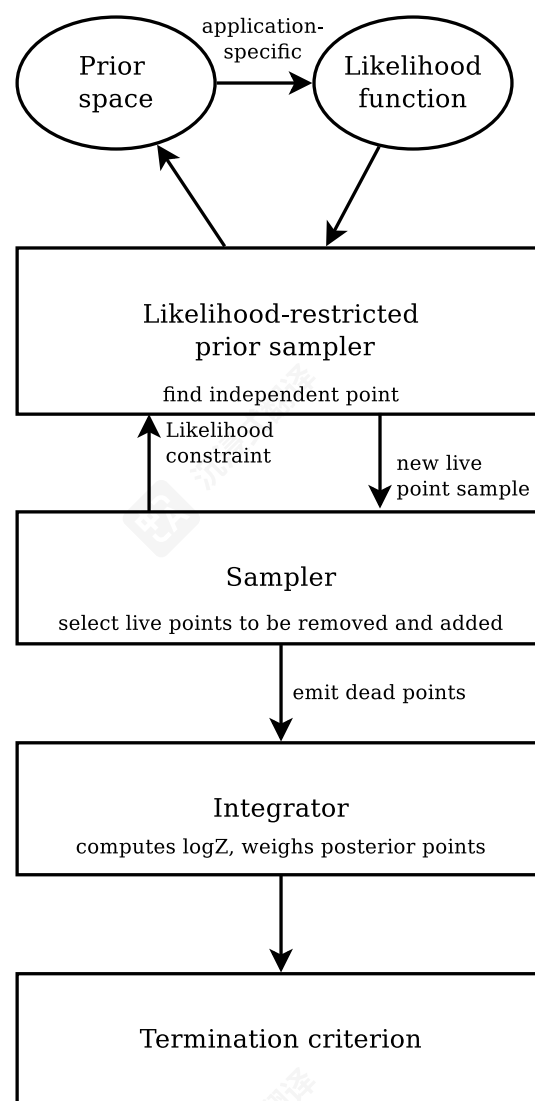


图5. 嵌套抽样实现的组成部分。

3.5. Components of nested sampling implementations

To discuss NS in a structured fashion, we identify components of nested sampling implementations. These are illustrated in Figure 5. The core (centre) is a NS sampler which keeps a set of live points, and the likelihood constraint defined by the most recent dead point. At each iteration, the lowest live point may be replaced by a LRPS, which uses the application-specific likelihood function and prior space definition provided by the user. The dead point is passed to the NS integrator, which weighs these dead points to form a posterior sample, and estimates the marginal likelihood Z (see formulas in §3). The following sections discuss the individual components, including the integrator (§4), the termination criterion (§4.3), the LRPS variants (§5) and samplers (§6).

4. Integration

4.1. Theory

NS was introduced in Skilling (2004); Skilling et al. (2006). Convergence and unbiasedness was discussed and proven in Evans (2007); Chopin and Robert (2010); Skilling (2009); Keeton (2011). Making advances to analyse NS theoretically has been the focus of a few publications (e.g., Khanarian and Alvarez, 2013). We refer the interested reader to Walter (2017), which makes connections to the mathematical theory of rare event simulation and the last particle algorithm, and Salomone et al. (2018), which draws connections to Sequential Monte Carlo, nearly considering NS as a special case of that framework. Birge, Chang and Polson (2012) presents a generalisation of NS and connects it to path and bridge sampling, while Polson and Scott (2014) discusses connections to, among others, slice sampling. Chopin and Robert (2007a) and Feroz et al. (2013, appendix C) prove that posterior samples from nested sampling approximate the true posterior for continuous and discontinuous functions, respectively.

4.2. Estimators

NS rests on being able to estimate the volume (shrinkage) at each iteration, and that the LRPS samples faithfully. LRPS issues are discussed in §5, while for the discussion here, it is assumed that the LRPS is sampling perfectly. This case is called the idealised algorithm in Guyader, Hengartner and Matzner-Löber (2011), and perfect nested sampling in Higson et al. (2017).

NS rests on being able to estimate the volume (shrinkage) at each iteration. Only in special circumstances the volumes are known precisely (Chopin and Robert, 2008). In general they can only be estimated. Casting the volume shrinkage as a Poisson process yields the volume shrinkage estimator $\delta V = \frac{1}{N}$ (Huber and Schott, 2010; Guyader, Hengartner and Matzner-Löber, 2011). Based on this approach, Walter (2015) derived an unbiased estimator of Z . Chopin and Robert (2007a); Evans (2007) and Skilling (2009) discussed whether the ultimate goal is to obtain an unbiased estimator with minimal variance of $\log Z$ or Z (see also Keeton, 2011). Skilling (2009) argued that because

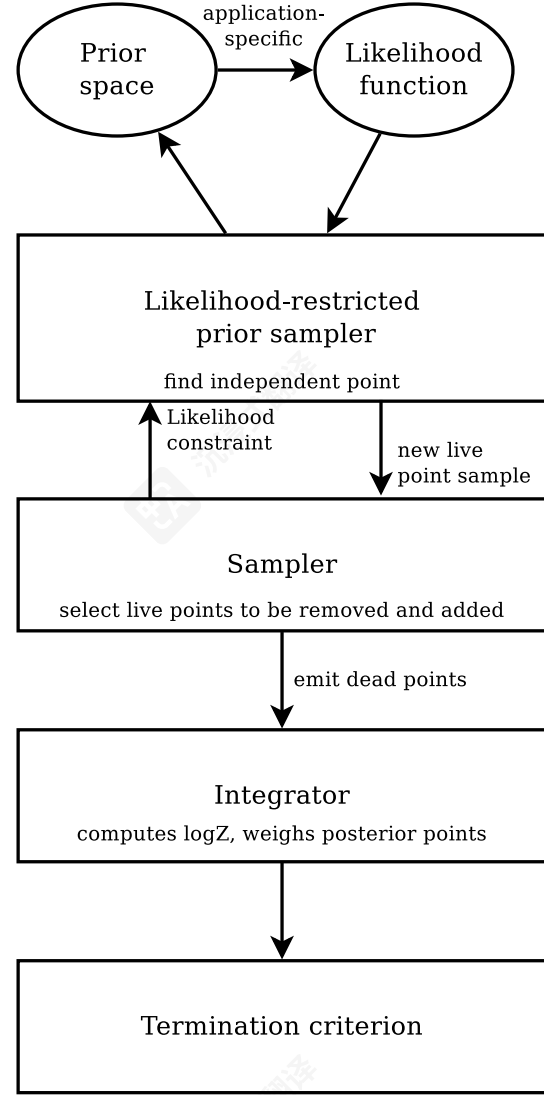


FIGURE 5. Components of a nested sampling implementation.

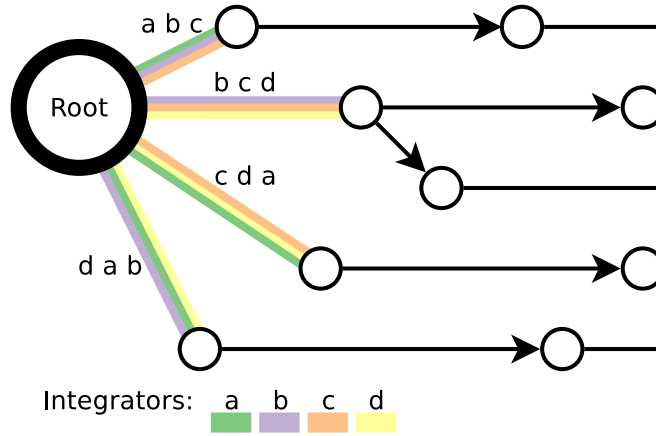


图6. 使用K折交叉验证的NS误差估计：调用四个积分器，每个积分器都对部分根子节点（颜色）保持无知。积分结果的分布提供了误差估计。

贝叶斯因子和对数尺度上的后验概率比是目标，应该寻求对数 Z 的无偏估计器。

积分估计器依赖于在每次迭代中估计压缩率。在Skilling (2004) 的顺序统计方法中，它们获得 $V_{i+1}/V_i \sim \text{Beta}(N, 1)$ ，并估计几何对数体积进展为 $\log V_{i+1}/V_i = -1/N$ 。Beta分布的均值给出 $\delta V = \frac{1}{N+1}$ ，这与上述无偏估计不同。即使不是最优的，如果其偏差相对于其均方根误差可以忽略不计，那么略微有偏的估计器适用于贝叶斯推理。对于上述估计器及其差异，当 $N \gg 1$ ，例如 $N = 100$ (Salomone等人, 2018) 时，这就是情况。对于估计器的理论及在线误差分析，我们建议读者参考上述文献。

迭代过程的终止也可能引入偏差 (Walter, 2015年)。Walter (2015年) 探讨了使用估计器族，每个对应不同的随机终止，以消除这种偏差。

N 活点的有限分辨率会导致噪声探索和似然波动 (Higson等人, 2018年)。这可以通过子采样来实际且优雅地解决 (由Higson等人, 2018年首次提出)。在 § 3.4 的树形表示中，一些根子节点可以以自举或K折的方式解除链接。这在图6中进行了说明。在树搜索表示中，可以有多个积分器，它们只看到一些根子节点，并使用积分结果的分布（后验和 $\log Z$ ）来衡量不确定性。收缩体积可以在每一步随机生成（使用 $\text{Beta}(1, N)$ 分布，Skilling等人, 2006年），这也会导致估计器的分布。结合采样产生的两种方差，即使是单个NS运行也能产生逼真的不确定性 (Higson等人, 2018年)。

4.3. 终止标准

何时终止嵌套采样是一个问题。在树形公式 (§ 3.4) 中，这意味着代理何时应该停止将新的子节点插入树中。

蒙特卡洛积分技术在积分黑盒函数方面存在固有的局限性。尖峰-板条问题说明了这一点。在图1中，我们展示了一个小的高峰（尖峰）和一个更宽的脊（板条）。如果尖峰非常窄，它可能被随机样本忽略，而随机样本会专注于更宽的板条。然而，如果尖峰非常高，它对积分至关重要。正如约翰·斯金林所说：“在大西洋里找一根旗杆是不可能的”。稀疏采样的NS积分可能从未在尖峰中放置活动点，从而通过了将尖峰和板条分开的似然阈值。因此，如果尖峰小于 V_i/N (Pártay, Bartók and Csányi, 2010)，它很可能被检测不到。然而，如果尖峰位于板条的峰顶，NS会找到它，除非它过早终止。

对于不太可能出现的异常情况，可以在运行时确定合理的终止条件。Skilling 等人 (2006) 提出在运行过程中同时估计信息增益 (H ，也用于误差估计) 以及估计通过大部分后验分布所需的迭代次数 HN 。类似地，样本熵也被视为终止条件。或者，Skilling 等人 (2006) 建议将死点积分 $Z_i = \sum_i L_i \delta V_i$ 与最大可能的活点贡献

$Z_{\text{live}} = L_{\text{max}} \times V_i$ 进行比较，并在比率变得非常小 $Z_{\text{live}}/Z_i < \epsilon$ 时终止，例如 $\epsilon \ll 1$ ，例如 $\epsilon = 10^{-3}$ 。这也是代理行为

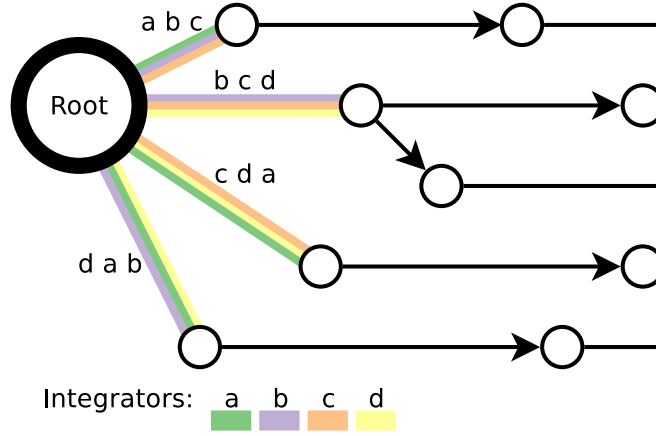


FIGURE 6. NS error estimation with K -fold cross-validation: Four integrators are invoked, which each are made oblivious to some of the root children (colours). The spread of integration results provides an error estimate.

Bayes factors and posterior odds ratios interpreted on log-scales are the goal, unbiased estimators of $\log Z$ should be sought.

Integral estimators rely on estimating the compression ratio at each iteration. In the order statistics approach of Skilling (2004), they obtain $V_{i+1}/V_i \sim \text{Beta}(N, 1)$, and estimate a geometric log-volume progression as $\log V_{i+1}/V_i = -1/N$. The mean of the Beta distribution gives $\delta V = \frac{1}{N+1}$, which differs from the unbiased estimator above. Even if not optimal, a slightly biased estimator is suitable for Bayesian inference if its bias is negligible relative to its root-mean-square error. This is the case for the above estimators and their differences, when $N \gg 1$, e.g., $N = 100$ (Salomone et al., 2018). For theoretical and online error analyses of the estimators, we refer the reader to the above works.

The termination of the iterative procedure can also induce a bias (Walter, 2015). Walter (2015) explores the use of families of estimators, each corresponding to a different random termination, to remove this bias.

The finite resolution of the N live points leads to a noisy exploration and likelihood fluctuations (Higson et al., 2018). This can be practically and elegantly addressed by sub-sampling (first proposed by Higson et al., 2018). In the tree formulation of §3.4, some of the root children can be unlinked in a bootstrapping or K -fold fashion. This is illustrated in Figure 6. In the tree search formulation one could have several integrators which see only some of the root children, and use the spread of integration results (posterior and $\log Z$) to measure the uncertainty. The shrinkage volume can be randomly generated at each step (using a $\text{Beta}(1, N)$ distribution, Skilling et al., 2006), which then also leads to a distribution of estimators. Incorporating the two sources of variance with sampling yields realistic uncertainty also for single NS runs (Higson et al., 2018).

4.3. Termination criteria

The question arises when to terminate nested sampling. In the tree formulation (§3.4), this means when agents should stop inserting new children nodes into the tree.

Monte Carlo integration techniques have intrinsic limitations in integrating black-box functions. The spike-and-slab problem illustrates this. In Figure 1, we illustrate a small, high peak (spike) and a wider ridge (slab). If the spike is very narrow, it can go unseen by random samples, which will focus on the wider slab. However, if the spike is very high, it will be crucial for the integral. As John Skilling puts it: “It is impossible to find a flag pole in the Atlantic ocean”. NS integration with sparse sampling may pass the likelihood threshold that separates the spike and slab without ever placing live points into the spike. Thus, a spike will likely go undetected if it is smaller than V_i/N (Pártay, Bartók and Csányi, 2010). However, if the spike is located on the peak of the slab, then NS will find it, unless it terminates too early.

For less perverse likelihoods, sensible termination can be determined at runtime. Skilling et al. (2006) proposed simultaneously estimating the information gain (H , also employed in an error estimate) during the run and estimating the minimum number of iterations needed to pass the bulk of the posterior as HN . Similarly, the sample entropy was considered as a termination criterion. Alternatively, Skilling et al. (2006) suggests comparing the dead point integral $Z_i = \sum_i L_i \delta V_i$ with the largest possible live point contribution $Z_{\text{live}} = L_{\text{max}} \times V_i$, and terminating when the ratio becomes very small, $Z_{\text{live}}/Z_i < \epsilon$, with $\epsilon \ll 1$, for example $\epsilon = 10^{-3}$. This is also the agent behaviour

在算法 2 中实现。这可以通过 $Z_{\text{live}} = \frac{V_i}{N} \sum_{j=1}^N L_j$ 、梯形规则积分或自举方法进一步优化，但在实践中并不更可靠。

已经考虑了多种其他终止标准。例如，Schöniger 等人 (2014) 提出当 LRPS 变得极其低效时终止。由复杂退化引起的低效可能表明模型可以从重新参数化中受益（参见 Papaspiliopoulos、Roberts 和 Sköld, 2007，对于 MCMC 的类似情况）。Baldock 等人 (2016) 建议监控温度（HMC 动量，见下文 § 5.1.3）。

对于运行期间可用的信息，尚未识别的、隐藏的峰值问题无法以通用和可靠的方式解决。因此，在实践中，运行比看似需要的时间稍长一些是最有效的。这就是 ϵ 剩余分数标准实际所做的工作。

终止也可以通过领域知识和重新参数化来解决。某些似然函数有绝对上限。例如，在具有固定不确定性 σ_i 的测量值 d_i 并用任意模型 m 拟合的高斯似然函数中，加权平方偏差之和 χ^2 是正的：

$$-2 \times \log L = \chi^2 = \sum_i \left(\frac{m(i, \theta) - d_i}{\sigma_i} \right)^2 > 0. \text{ 似然边界直接给出了未来迭代 } Z_{\text{live}} \text{ 的上限。}$$

4.4. 复杂度扩展

现在我们可以考虑 NS 集成的计算复杂度。这取决于 (1) 后验相对于先验的信息增益 (I)，它决定了达到后验主体所需的收缩量，(2) 活动点的数量，它决定了每次迭代的收缩量，以及 (3) LRPSperNS 迭代的计算复杂度以找到可靠的新点，这受到特殊退化 (P)、多模态 (M) 和维度 (D) 问题的制约。

NS 的复杂度随活动点数量 $O(N)$ 线性扩展，这是由于收缩速度较慢 (Skilling, 2004)。除此之外，LRPS 方法的扩展性可能因需要导航的复杂后验形状而任意受阻，直到获得新的独立样本。假设后者随维度 d 线性扩展，Skilling (2009) 给出了 NS 的成本扩展 $O(d^2)$ 。在实践中，Handley、Hobson 和 Lasenby (2015a) 证明了他们的实现扩展为 $O(d^3)$ 。这使得 NS 对于具有数千个参数的非常高维问题（如拟合大型分层贝叶斯模型或神经网络）可能不太吸引人。然而，Javid 等人 (2020) 证明神经网络可以被拟合，并且 NS 具有吸引力，因为全局探索避免了选择过于复杂的网络。

然而，成本与活点之间的关系更为复杂。虽然随着活点数量的增加，收敛速度会变慢，但某些 LRPS 方法在活点数量更多时效率更高，因为它们有助于绘制可能性约束，并可以识别新点可能成功的近似邻域（参见 § 5）。根据 Allison 和 Dunkley (2014) 的研究，这通过椭球拒绝采样技术（将在 § 5.2 中进一步讨论）来说明椭球单峰似然的情况。为此，LRPS 每次迭代的成本经验性地发现（在附录 B 中）按以下方式扩展：

$$O(C_{\text{ell}}) = \exp \left\{ \left(\frac{6.83 \times d^{1.9}}{N} \right)^{3/4} \right\} \quad (3)$$

当活点数量较少时，指数增长变得至关重要。这很可能编码了维度的诅咒，以及拒绝技术的固有局限性。方程 3 表明，椭球采样所需的活点数量不应低于 $7 \times d^2$ 。

总成本由每次迭代的成本和所需的迭代次数获得。从先验体积 V_p 收缩到目标后验体积 V_t 的低分数 ϵ 需要 $i = N \times \ln \frac{V_p}{V_t \times \epsilon}$ 次迭代（参见 § 3.1）。这里的比率 $H = \ln \frac{V_p}{V_t}$ 是从先验到后验的信息增益。结合方程 3 的接受率公式和迭代次数 i ，以及初始活点 N 的采样，Allison 和 Dunkley, 2014 年获得了嵌套采样的总成本：

$$C = N + i \times C_{\text{ell}} = N + N \times \ln \frac{V_p}{V_t \times \epsilon} \times C_{\text{ell}} \quad (4)$$

该因素 V_p/V_t 是特定于问题的。研究其随维度的缩放并不容易，因为改变维度意味着分析不同的问题。在这里，我们考虑两种情况：(A) 如果添加了新的参数，并且它们都使用相同的信息增益进行更新，那么 $V_p/V_t = K^d$ 。这增加了 C 到 $O(N \times C_{\text{ell}} \times d)$ 的成本。(B) 如果添加新参数仅重新分配相同的信息，那么 $V_p/V_t = K$ 保持与维度无关，因此成本为 V_p/V_t 的量级。对于这两种情况，总成本 $V_p/V_t = K^d$ 在图 7 中绘制，并使用 C 。这些归一化是任意的，无法比较。椭球 NS 在 $O(N \times C_{\text{ell}} \times d)$ 维度中效果最佳。这符合我们对 MultiNest 椭球采样实现的实际经验，该实现接近该维度时开始失效。为了比较，还显示了上述 MCMC LRPS 算法的 $V_p/V_t = K$ 缩放，作为红线，并添加了任意选择的基准成本。有趣的是，上述分析表明，对于椭球问题，如果增加活点的数量呈平方级增长，则可以使用椭球采样实现缩放。然而，确实具有椭圆轮廓的乐观情况已被分析，而对于其他情况，成本可能更高。

implemented in Algorithm 2. This can be further refined with $Z_{\text{live}} = \frac{V_t}{N} \sum_{j=1}^N L_j$, trapezoid rule integration, or bootstrapping, but in practice is not more reliable.

A variety of other termination criteria have been considered. For example, Schöniger et al. (2014) proposes terminating when the LRPS becomes extremely inefficient. Low efficiencies caused by complex degeneracies, can indicate that the model could benefit from reparametrisation (see Papaspiliopoulos, Roberts and Sköld, 2007, for a similar situation with MCMC). Baldock et al. (2016) suggests monitoring the temperature (HMC momenta, see below §5.1.3).

The problem of yet unidentified, hidden peaks cannot be addressed in a general and reliable way with information available during the run. Therefore running a few iterations longer than seemingly needed is most effective in practice. This is what the ϵ remainder fraction criterion effectively does.

Termination can also be addressed with domain knowledge and reparametrisation. Some likelihoods have absolute upper bounds. For example, in a Gaussian likelihood with measurements d_i with fixed uncertainties σ_i fitted with an arbitrary model m , the weighted sum of squared deviations, χ^2 , is positive: $-2 \times \log L = \chi^2 = \sum_i \left(\frac{m(i, \theta) - d_i}{\sigma_i} \right)^2 > 0$. The likelihood bound directly gives upper bounds on Z_{live} of future iterations.

4.4. Complexity scaling

We can now consider the computational complexity of NS integration. This depends on (1) the information gain of the posterior compared with the prior (I), which determines the shrinkage necessary to reach the bulk of the posterior, (2) the number of live points, which determines the shrinking per iteration, and (3) the computational complexity of the LRPS per NS iteration to find a reliable new point, which is subject to peculiar degeneracies (P), multi-modality (M) and dimensionality (D) issues.

The NS complexity scales linearly with the number of live points $O(N)$, due to the slower shrinkage (Skilling, 2004). Beyond this, the scaling of the LRPS method can be arbitrarily hampered by complex posterior shapes that need to be navigated until a new independent sample is obtained. Assuming the latter scales linearly with dimensionality d , Skilling (2009) gives the cost scaling of NS as $O(d^2)$. In practice, Handley, Hobson and Lasenby (2015a) demonstrate a scaling of $O(d^3)$ for their implementation. This makes NS perhaps less attractive for very high-dimensional problems with many thousands of parameters such as fitting large hierarchical Bayesian models or neural networks. However, Javid et al. (2020) demonstrate that neural networks can be fitted, and NS is attractive because the global exploration avoids choosing overly complex networks.

However, the relation between cost and live points is more complex. While convergence slows with the number of live points, some LRPS methods work more efficiently the more live points, as they help map out the likelihood constraint and can identify the approximate neighbourhood where new points are likely successful (see §5). Following Allison and Dunkley (2014), this is illustrated with the ellipsoidal rejection sampling technique (discussed further in §5.2) for the case of ellipsoidal, mono-modal likelihoods. For this, the LRPS cost per iteration is empirically found In Appendix B to scales as:

$$O(C_{\text{ell}}) = \exp \left\{ \left(\frac{6.83 \times d^{1.9}}{N} \right)^{3/4} \right\} \quad (3)$$

The exponential increase becomes crucially important when the number of live points is small. This likely encodes the curse of dimensionality, and the inherent limitations of rejection techniques. Equation 3 suggests that the number of live points for ellipsoidal sampling should not be lower than $7 \times d^2$.

The total cost is obtained from the per-iteration cost and the number of iterations needed. Shrinking from the prior volume V_p until a low fraction ϵ of a target posterior volume V_t requires $i = N \times \ln \frac{V_p}{V_t \times \epsilon}$ iterations (see §3.1). The ratio $H = \ln \frac{V_p}{V_t}$ here is the information gain from the prior to the posterior. Combining the acceptance rate formula of eq. 3 and the number of iterations i , and the sampling of the initial N live points, Allison and Dunkley, 2014 obtain the total nested sampling cost as:

$$C = N + i \times C_{\text{ell}} = N + N \times \ln \frac{V_p}{V_t \times \epsilon} \times C_{\text{ell}} \quad (4)$$

The factor V_p/V_t is problem-specific. It is not easy to study the scaling with dimensionality, as varying the dimensionality implies analysing a different problem. Here we consider two cases: (A) If new parameters are added, and they are all updated with the same information gain, then $V_p/V_t = K^d$. This increases the cost of C to $O(N \times C_{\text{ell}} \times d)$. (B) If adding new parameters only redistributes the same information, then $V_p/V_t = K$ remains

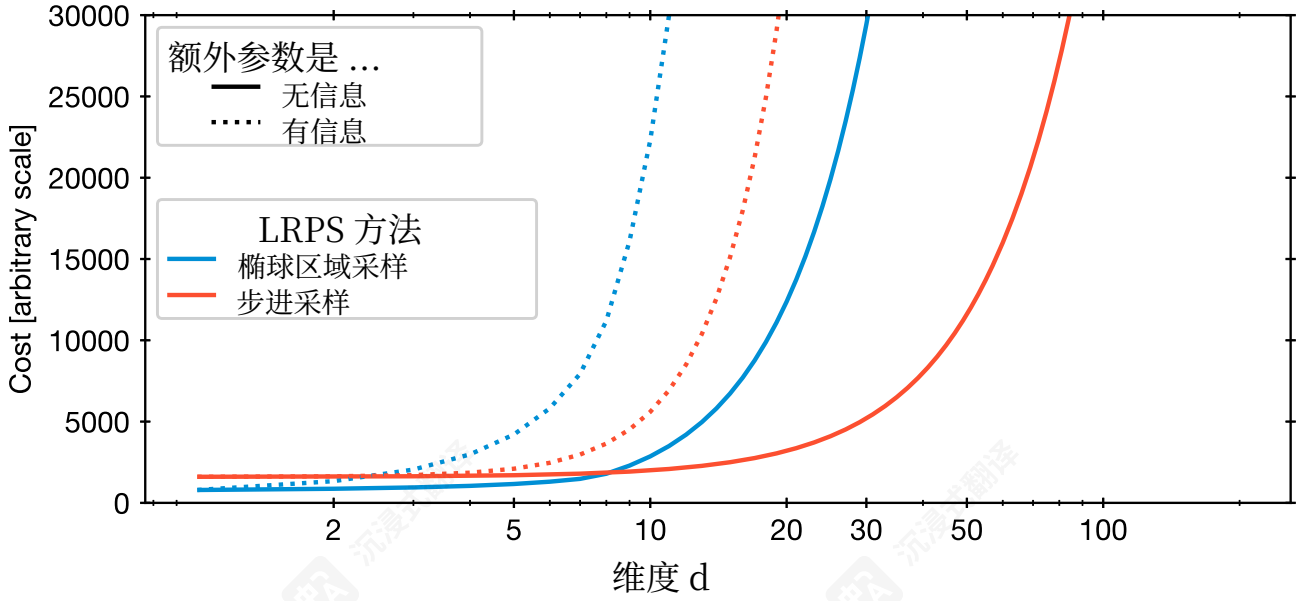


图7. 高斯型似然函数上的似然评估次数。蓝色曲线展示了两个在不同维度下信息增益变化不同的案例。两个案例均表明，在最优条件下（具有椭球等高线的似然），椭球嵌套采样在 $d < 30$ 维度时的成本较低。红色曲线则说明了步进采样更浅的、多项式缩放的特性。

对于椭球问题，如果增加活点的数量呈平方级增长，则可以使用椭球采样实现 d^2 缩放。然而，确实具有椭圆轮廓的乐观情况已被分析，而对于其他情况，成本可能更高。

在实践中，计算成本也受模型结构的影响。因此需要进行数值测试（例如参见Pitkin等人(2017)中的图15和Trassinelli与Ciccodicola(2020)中的图6）。最后，似然峰的发现也受 N 的调节（参见 § 4.4）。正因如此，使用非常小的 N 运行嵌套采样并不一定能得到有用的“快速预览”结果。

4.5. 正确性诊断

如何评估一个完整的嵌套采样实现（包括积分器、LRPS、采样器、终止准则）的正确性？我们可以根据Stokes、Tuyl和Hudson（2016）的分类，将测试分为两类。

4.5.1. 外向式：应用于已知函数

首先，NS实现可用于测试具有已知属性的测试问题。最简单的情况是似然函数，其中积分 Z 是解析已知的（例如，Preuss和von Toussaint，2007年）。这里的主要限制是解析似然函数通常不类似于现实世界的问题。另一个问题是，例如，在选择高斯似然表达式时，积分通常由狭窄范围的似然值主导。因此，LRPS和积分因此可能仅在少数迭代上被测试。这可以通过选择重尾分布来改进。一个极端的例子是 $L = \min(\theta^{-1}, e^{100})$ ，其中 θ 定义在单位区间上，这使得所有死点产生大约相等的权重。

其次，LRPS 可以独立进行测试。Buchner (2014) 提出了一种 LRPS 测试方法，验证 LRPS 引起的收缩行为是否符合预期 $(1 - 1/N)$ 。这是基于似然函数的，其中可以在给定似然 $X(L)$ 处解析地计算所包围的体积。例如，在正态似然中，圆形似然轮廓可以与椭球体对应。然后，从重复的 LRPS 获得的一系列似然中，可以计算体积收缩并与预期进行比较。如果找到点的似然系统地较低或较高，那么 LRPS 就明显不正确。这可以应用于许多后验形状，包括多模态高斯分布。一个特别灵敏的测试是超矩形 $1 - 1/N \max X$ ，因为它的形状远非高斯分布，并在高维中表现出许多角点。重要的是，这个测试与尾部权重无关，因为似然仅在点的 NS 加权中起作用。它也可以应用于非常高维空间来调整 LRPS 参数。

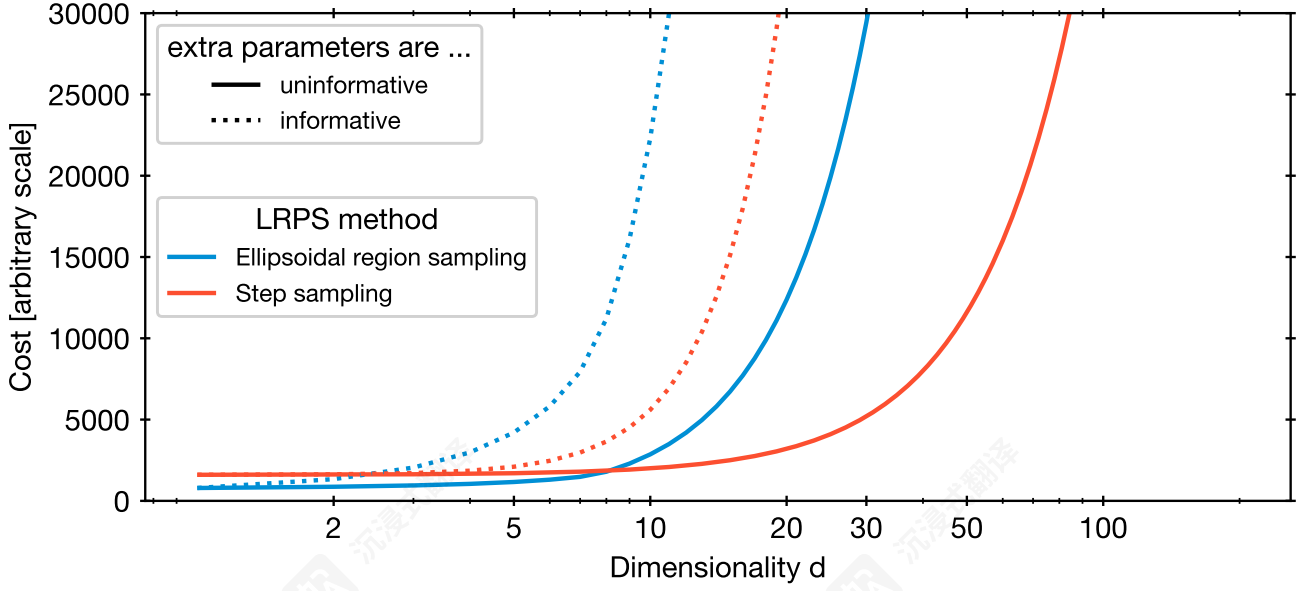


FIGURE 7. Number of likelihood evaluations on Gaussian-like likelihood functions. The blue curves show two cases with different information gain changes with dimensionality. Both cases indicate a low cost at $d < 30$ dimensions for ellipsoidal NS under optimal conditions (likelihood with ellipsoidal contours). Red curves illustrate the shallower, polynomial scaling of step samplers.

constant with dimensionality, and thus the cost is of order $O(N \times C_{\text{ell}})$. For these two cases, the total costs C are plotted in Figure 7, with $N = 400$. The normalisations are arbitrary and cannot be compared. Ellipsoidal NS clearly works best in $d < 30$ dimensions. This agrees with our practical experience with the MultiNest ellipsoidal sampling implementation, which begins to break down close to that dimensionality. For comparison, the aforementioned d^2 scaling of a MCMC LRPS algorithm is also shown as red lines, with an arbitrarily chosen base cost added. Interestingly, the above analysis suggests that for ellipsoidal problems, a d^2 scaling is possible with ellipsoidal sampling if the live points are increased quadratically. However, the optimistic case where the likelihood indeed has elliptical contours was analysed, while for other cases, the cost may be higher.

In practice, the computational cost is specific to the model structure as well. Thus numerical testing is required (see for example Figure 15 in Pitkin et al. (2017) and Figure 6 in Trassinelli and Ciccodicola, 2020). Finally, the discovery of likelihood peaks is also regulated by N (see §4.4). Because of this, running NS with very small N does not necessarily give useful “quick look” results.

4.5. Correctness diagnostics

How can a complete NS implementation (including integrator, LRPS, sampler, termination criterion) be evaluated for correctness? We can divide in two categories of tests, following Stokes, Tuyl and Hudson (2016).

4.5.1. Outside-in: Application to known functions

Firstly, the NS implementation can be applied to test problems with known properties. The simplest case is likelihood functions where the integral Z is analytically known (e.g., Preuss and von Toussaint, 2007). The main limitation here is that analytic likelihood functions often do not resemble real-world problems. A further problem is that often, for example, when choosing a Gaussian likelihood expression, the integral is dominated by a narrow range of likelihood values. Thus, the LRPS and integration are therefore potentially only tested on a low number of iterations. This can be improved by choosing heavy-tailed distributions. An extreme case is $L = \min(\theta^{-1}, e^{100})$ with θ defined over the unit interval, which makes all dead points yield approximately equal weights.

Secondly, the LRPS can be tested in isolation. Buchner (2014) proposed a LRPS test that verifies that the shrinkage caused by the LRPS behaves as expected $(1 - 1/N)$. This is based on likelihood functions where one can analytically compute the volume enclosed at a given likelihood $X(L)$. For example, in a Gaussian likelihood, the circular likelihood contours can be identified with an ellipsoid. Then, from a sequence of likelihoods obtained from

这可以应用于许多后验形状，包括多模态高斯分布。一个特别灵敏的测试是超矩形 $L = \max_i \|\theta_i - \frac{1}{2}\|^{-1}$ ，因为它的形状远非高斯分布，并在高维中表现出许多角点。重要的是，这个测试与尾部权重无关，因为似然仅在点的 NS 加权中起作用。它也可以应用于非常高维空间来调整 LRPS 参数。

4.5.2. 从内向外：运行时或运行后的诊断

第二组测试试图在 NS 运行过程中发现假设何时被破坏。

具有平台值的似然函数会在嵌套采样中引起问题 (Skilling 等人, 2006 年; Schittenhelm 和 Wacker, 2020 年)。这是因为先验空间的顺序不可用，并且等式 2 中的大体积与消失的似然区间相关联。因此，如果两个活点具有完全相同的似然值，这应该引起警报。为了解决这个问题，Fowlie、Handley 和 Su (2021 年) 对嵌套采样算法迭代提出了一种小的修改，在用大小为 N 重新补充活点集之前，先不替换地移除所有具有 $L_{\min}$ 的活点。

一个假设是 LRPS 根据约束先验正确地采样。Stokes、Tuyl 和 Hudson (2016 年) 提出了在二维问题中进行均匀性测试。首先，他们计算分割中的空单元格，并将它们与贝叶斯因子进行比较，与均匀期望进行比较。其次，他们开发了一个等分布测试，该测试通过熵测量样本的集中度，并将其与均匀期望进行比较。最后，他们使用分位数-分位数图可视化均匀性偏差。然而，这些测试似乎仅限于非常低维的问题。

Higson 等人 (2018, 2019) 贡献了关于推断中不确定性的可视化，特别是后验分布的。这是通过子采样 (参见 § 4.2) 完成的嵌套采样运行，并绘制后验分布的分布范围来实现的。相关地，他们提出了体积 ($\log X$) 与参数值 (θ_i) 的图，这可以洞察参数空间的结构，以及 LRPS 如何替换样本。

Higson 等人 (2019) 进一步提出了测试多个独立 NS 运行之间的方差与从单个运行中子采样所预期的方差。在这里，可以检查每个参数 θ_i 、组合 $f(\theta)$ 或 $\log Z$ 的期望值。当 LRPS 采样不完美时，可能会出现超额方差，并且其样本和收缩值是相关的。Higson 等人 (2019) 还考虑了仅两个运行之间的期望值，当计算成本非常高时。

最后，Fowlie、Handley 和 Su (2020) 指出，如果 LRPS 返回的样本是独立的，并且完美地按照约束先验分布，那么它们插入到排序的活点列表中的位置应该是均匀分布的。他们提出了一个插入顺序测试来检查这个条件。每个新样本的插入顺序被收集。在每 N 次迭代和整个运行中，使用 Kolmogorov-Smirnov (KS) 测试来检验顺序分布。Fowlie、Handley 和 Su (2020) 证明了这在玩具问题上实际检测有问题的运行时效果很好。或者，插入顺序测试也可以用作质量指标。当测试触发时，样本收集会被重置。然后，直到测试触发的迭代次数可以类似于随机游走 MCMC 算法中的自相关长度来解释。然而，样本收集可能仍然需要偶尔截断，以便最近添加的差样本不会被前面的大量好样本稀释。这里的限制是，通常实现的 KS 测试仅适用于连续变量。Fowlie、Handley 和 Su (2020) 表明，四舍五入到整数使分布变得非均匀，并且测试的敏感性低于其应有的水平。

我们提出一种改进的检验方法，使用适合于检验离散数是否均匀分布的统计量。我们首先从威尔科克森-曼-惠特尼 U 检验开始，曼和惠特尼 (1947 年) 提出的检验方法，该方法检验两个长度为 n_1 和 n_2 的观测序列。对于第一个序列中的每个观测值，记录其在第二个序列中较小或相等的观测值的数量 (“胜利”， W)，以及相等的观测值的数量 (“平局”， T)。然后计算 U 统计量：

$$U = W + \frac{1}{2}T$$

然后 U 服从均值为

$$m_U = \frac{n_1 \times n_2}{2}$$

和标准差

$$\sigma = \sqrt{\frac{n_1 n_2}{12} \left(n + 1 - \sum_{i=1}^k \frac{t_i^3 - t_i}{n(n-1)} \right)}$$

repeated LRPS, volume shrinkages can be computed and compared to expectations. If the likelihood of the found point is systematically lower or higher, then the LRPS is noticeably incorrect. This can be applied to many posterior shapes, including multi-modal Gaussians. A particularly sensitive test is the hyper-rectangle $L = \max_i \|\theta_i - \frac{1}{2}\|^{-1}$, because its shape is far from Gaussian and exhibits many corners in high dimensions. Importantly, this test is independent of the tail weight, as the likelihood only enters in NS weighting of points. It can also be applied in very high dimensions to tune LRPS parameters.

4.5.2. Inside-out: Diagnostics at or after run-time

The second group of tests tries to notice during the NS run when assumptions are broken.

Likelihood functions with plateaus can cause problems in nested sampling (Skilling et al., 2006; Schittenhelm and Wacker, 2020). This is because the ordering of the prior space is not available, and a large volume is associated with a vanishing likelihood interval in eq. 2. Therefore, if two live points have the exact same likelihood, this should cause alarm. To address this, Fowlie, Handley and Su (2021) proposed a small modification to the nested sampling algorithm iterations to remove all live points with $L_{\min}$ without replacement before replenishing the live point set to size N .

One assumption is that the LRPS samples correctly according to the constrained prior. Stokes, Tuyl and Hudson (2016) proposed tests for uniformity in 2-dimensional problems. Firstly, they count the empty cells in a segmentation, and compare them with a Bayes factor to uniform expectation. Secondly, they develop an equi-distribution test that measures the concentration of samples with an entropy, and compares that to a uniform expectation. Finally, they visualise deviations from uniformity with quantile-quantile plots. These tests however appear limited to very low-dimensional problems.

Higson et al. (2018, 2019) contributed visualisations of the uncertainty in the inference, in particular of the posterior distributions. This is achieved by sub-sampling (see §4.2) completed nested sampling runs, and plotting the spread of posteriors. Relatedly, they propose diagrams of volume ($\log X$) vs. parameter value (θ_i), which allows insight into the structure of the parameter space, and how the LRPS replaces samples.

Higson et al. (2019) further proposes testing the variance between multiple independent NS runs to the variance expected from sub-sampling a single run. Here, one can check the expectation for each parameter θ_i , a combination $f(\theta)$, or $\log Z$. Excess variance can occur when the LRPS is sampling imperfectly, and its samples and shrinkages are correlated. Higson et al. (2019) also considered the expectations between only two runs, when the computation is very costly.

Finally, Fowlie, Handley and Su (2020) pointed out that if the samples returned from the LRPS are independent and perfectly distributed according to the constrained prior, then where they are inserted into the sorted live points list should be uniformly distributed. They proposed an insertion order test to check this condition. The insertion order of each new sample is collected. The order distribution is tested with a Kolmogorov-Smirnov (KS) test every N iterations, and for the full run. Fowlie, Handley and Su (2020) demonstrates that this works well to detect problematic runs in practice on toy problems. Alternatively, the insertion order test could also be used as a quality indicator. When the test triggers, the sample collection is reset. Then the number of iterations until the test triggers can be interpreted similar to an auto-correlation length in random walk MCMC algorithms. However, the samples collection likely still needs to be truncated occasionally, so that a recent addition of poor samples is not diluted by a preceding high number of good samples. A limitation here is that the KS test as typically implemented is only valid for continuous variables. Fowlie, Handley and Su (2020) show that the rounding to integers makes the distribution non-uniform, and the test is less sensitive than it could be.

We propose an improvement of the power of the test with a statistic suitable for testing whether discrete numbers are uniformly distributed. We begin with the Wilcoxon-Mann-Whitney U test Mann and Whitney (1947), which tests two sequences of observations, of length n_1 and n_2 . For each observation in the first sequence, the number of smaller and equal observations in the second sequence is recorded (“wins”, W), as well as the number of equal observations (“ties”, T). Then the U statistic is:

$$U = W + \frac{1}{2}T$$

Then U is normal distributed with mean

$$m_U = \frac{n_1 \times n_2}{2}$$

and standard deviation

$$\sigma = \sqrt{\frac{n_1 n_2}{12} \left(n + 1 - \sum_{i=1}^k \frac{t_i^3 - t_i}{n(n-1)} \right)}$$

N	覆盖率	KS检验	U检验	N	倾斜度	KS检验	U测试
1000	0.9	1.00000	0.99723	1000	0.9	0.31010	0.44932
1000	0.96	0.13472	0.19984	1000	0.96	0.01864	0.02873
1000	0.98	0.01416	0.02627	1000	0.98	0.00505	0.00728
400	0.9	1.00000	0.69506	400	0.9	0.06385	0.11478
400	0.96	0.02344	0.04745	400	0.96	0.00635	0.01084
400	0.98	0.00589	0.00926	400	0.98	0.00343	0.00425
100	0.9	0.07364	0.07745	100	0.9	0.01205	0.01653
100	0.96	0.00988	0.00771	100	0.96	0.00419	0.00363
100	0.98	0.00593	0.00383	100	0.98	0.00335	0.00281

表 1

KS 测试和 U 测试的性能比较。在左侧表格中，整数从 0 到 $N \times$ 覆盖率分布，在右侧表格中则按概率分布。右侧报告了报告 3σ 显著性的测试比例。在几乎所有情况下， U 测试的比例都更高。
($p < 0.0027$)

与 t_i 共享顺序 i 和 $n = n_1 + n_2$ 的观测数量。换句话说， $z = \frac{U - m_U}{\sigma_U}$ 近似于标准正态分布，并且很少出现，例如 $|z| > 3$ ，这种 3σ 的偏离。

现在让我们想象第二个序列是一个非常大的参考点样本 ($n_2 = N \times M$ 个 $M \gg 1$)。它们均匀分布在 0 到 $N - 1$ 之间，因此每个顺序由 M 个样本表示。第一个观察序列是收集的插入顺序， O_i 。这些是从 0 到 $N - 1$ 的整数。在这种情况下，一个观察到的插入顺序 O 将有 M 个平局和 $O \times M$ 个胜利。检验统计量变为：

$$U = M \times \sum_{i=1}^{n_1} \left(O_i + \frac{1}{2} \right)$$

均值变为 $m_U = \frac{1}{2} \times n_1 \times N \times M$ ，而标准差简化为 $M \gg 1$ 和 $n_2 \gg n_1$ ，变为 $\sigma = \sqrt{\frac{n_1}{12}} \times N \times M$ 。消去在 U 、 m_U 和 σ 中出现的 M ，我们发现

$$z = \frac{\left(\sum_{i=1}^{n_1} \frac{2O_i + 1}{N} \right) - n_1}{\sqrt{n_1/3}}$$

服从标准正态分布。这可以通过对 n_1 个在 $[0, N]$ 中均匀采样的整数进行采样并绘制 z 值来容易地数值确认。我们做出两点重要说明：首先，在这种形式下， U 检验允许 N 在每次迭代中变化。其次， z 统计量的符号指示偏差的方向。

现在可以比较KS检验和 U 检验的灵敏度（均为双尾检验）。对于合理的活点数，表1考虑了两种情况：在第一个模拟（顶部），生成的插入顺序被截断，仅覆盖0到 $\lceil N \times C \rceil$ 的范围，并使用 $C < 1$ 。在第二个模拟（底部），生成的插入顺序模拟自一个轻微倾斜的幂律分布，低端的顺序 i 更可能。我们模拟了100,000个大小为 $N = 1000$ 的样本，400、100，并应用KS和 U 检验。表1比较了报告 3σ 偏差的检验比例。通常， U 检验具有更高的检测率。然而，当检测率已经很高 ($> 50\%$) 或非常接近预期的假阳性率时，KS检验有时表现相似或更好。但这是较不有趣的边缘情况。除了更灵敏之外， U 检验实现起来稍微简单一些，因为只需要累积 $\sum_{i=1}^{n_1} \frac{2O_i + 1}{N}$ 和样本数 n_1 ，而不是整个直方图。

该测试可以以三种不同的方式应用：(1) 像Fowlie、Handley和Su (2020)那样在完整运行中应用，(2) 每 N 次迭代应用 (Fowlie、Handley和Su, 2020)，可能需要使用Bonferroni校正，以及(3) 累积直到 $|z|$ 超过预定义的阈值。例如，在模拟 10^7 次迭代且插入顺序均匀的情况下，当 $|z| > 4$ 导致片段长度不短于 $10^{5.5}$ 时重置累积统计量。如果经常产生较短的片段，或者更具体地说，如果NS运行中的片段数量大于迭代次数除以105.5，这表明该运行存在偏差。此选项避免了选择一个chunksize来应用测试。

5. 基于似然的先验采样 (LRPS)

LRPS是NS的核心。关于NS能做什么或不能做什么的经验性陈述取决于LRPS的实现方式。本节试图传达的是，多种解决方案是可能的，并且已经被考虑过。

LRPS 应该从似然受限先验中独立地采样一个新的点。但这可能很难完美实现。然而，在实际中，正如Salomone 等人 (2018) 通过理论论证所阐述的，即使点之间存在轻微相关性，也能获得良好的 goodNS 结果。

N	Coverage	KS test	U test	N	Slant	KS test	U test
1000	0.9	1.00000	0.99723	1000	0.9	0.31010	0.44932
1000	0.96	0.13472	0.19984	1000	0.96	0.01864	0.02873
1000	0.98	0.01416	0.02627	1000	0.98	0.00505	0.00728
400	0.9	1.00000	0.69506	400	0.9	0.06385	0.11478
400	0.96	0.02344	0.04745	400	0.96	0.00635	0.01084
400	0.98	0.00589	0.00926	400	0.98	0.00343	0.00425
100	0.9	0.07364	0.07745	100	0.9	0.01205	0.01653
100	0.96	0.00988	0.00771	100	0.96	0.00419	0.00363
100	0.98	0.00593	0.00383	100	0.98	0.00335	0.00281

TABLE 1

Performance comparison of KS test and U test. In the left table, the integers are distributed from 0 to $N \times \text{Coverage}$, in the right table they are distributed with probability $p_i \propto i^{\text{Slant}}$. The fraction of tests reporting a 3σ significance ($p < 0.0027$) is reported on the right. The U test fractions are higher in almost all cases.

with t_i the number of observations that share order i and $n = n_1 + n_2$. In other words, $z = \frac{U - m_U}{\sigma_U}$ is approximately standard normal and should rarely exhibit, for example, $|z| > 3$, a 3σ excursion.

Lets now imagine that the second sequence is a very large sample of reference points ($n_2 = N \times M$ with $M \gg 1$). They are uniformly spread from 0 to $N - 1$, thus each order is represented by M samples. The first sequence of observations are the collected insertion orders, O_i . These are integers ranging from 0 to $N - 1$. In that case, a observed insertion order O will have M ties and $O \times M$ wins. The test statistic becomes:

$$U = M \times \sum_{i=1}^{n_1} \left(O_i + \frac{1}{2} \right)$$

The mean becomes $m_U = \frac{1}{2} \times n_1 \times N \times M$ while the standard deviation simplifies because $M \gg 1$ and $n_2 \gg n_1$, to $\sigma = \sqrt{\frac{n_1}{12}} \times N \times M$. Cancelling out M which occurs in U , m_U and σ , we find that

$$z = \frac{\left(\sum_{i=1}^{n_1} \frac{2O_i + 1}{N} \right) - n_1}{\sqrt{n_1/3}}$$

is standard normal distributed. This is easy to numerically confirm by sampling n_1 integers sampled uniformly in $[0, N)$ and plotting the z values. We make two important remarks: Firstly, in this form, the U test allows N to vary from iteration to iteration. Secondly, the sign of the z statistic indicates the direction of the bias.

Now the sensitivity of the KS test and the U test (both two-sided) can be compared. For reasonable numbers of live points, two scenarios are considered in Table 1: In the first simulation (top), generated insertion orders are truncated to only cover the range 0 to $\lceil N \times C \rceil$, with $C < 1$. In the second simulation (bottom), generated insertion orders are simulated from a mildly slanted power law distribution, with order i more likely at the low end. We simulated 100,000 samples of size $N = 1000, 400, 100$ and apply the KS and U tests. The fraction of tests reporting a 3σ deviation are compared in Table 1. In general, the U test has a higher detection rate. However, when the detection rate is already high ($> 50\%$), or very close to the expected false positive rate, the KS test sometimes performs similarly or better. However, these are less interesting edge cases. Additional to being more sensitive, the U test is slightly simpler to implement, as only $\frac{2O+1}{N}$ and the number of samples n_1 need to be accumulated, instead of entire histograms.

The test can be applied in three different ways: (1) on the full run as in Fowlie, Handley and Su (2020), (2) every N iterations (Fowlie, Handley and Su, 2020), probably with a Bonferroni correction, and (3) accumulate until $|z|$ exceeds a predefined threshold. For example, when simulating 10^7 iterations with uniform insertion order, resetting the accumulation statistic when $|z| > 4$ leads to segment lengths no shorter than $10^{5.5}$. If shorter segments are regularly produced, or more specifically, if the number of segments of a NS run is larger than the number of iterations divided by $10^{5.5}$, this is an indication that the run is biased. This option avoids selecting a chunk size to apply the test.

5. Likelihood-restricted prior sampling (LRPS)

LRPS is the crux of NS. Empirical statements about what NS can or cannot do are at the mercy of the LRPS implementation. This section tries to convey that a large diversity of solutions are possible and have been considered.

LRPS is supposed to deliver an independent, new point sampled from the likelihood-restricted prior. This can be difficult to achieve perfectly. However, in practice, and as elaborated in Salomone et al. (2018) with theoretical arguments, good NS results can be achieved also with mildly correlated points.

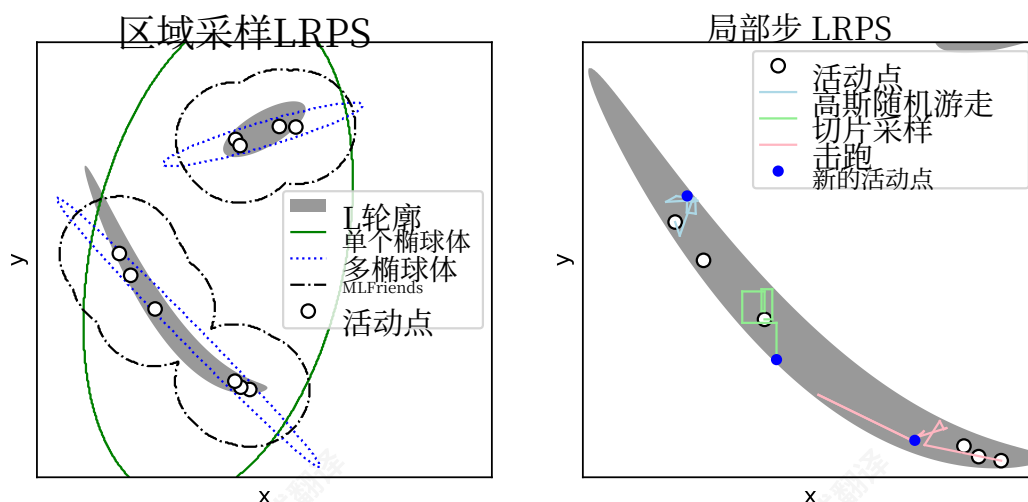


图8. LRPS方法。在图1示例的某次迭代中，展示了寻找新活点的不同方法。左图：区域采样方法基于活点（白色圆圈）构建区域，以近似未知似然轮廓（灰色）。如果轮廓过大，基于这些轮廓的拒绝采样会变得低效，并且如果它遗漏了某个区域（例如左上角尾部），则对NS积分造成问题。这里的放大图特意选得太小。右图：局部步长方法从活点（白色圆圈）开始Metropolis随机游走（彩色线）。似然轮廓外的步长被拒绝。如果步长提议良好且步数足够多，就会达到一个新点（蓝色圆圈），该点与起点无关。

5.1. 局部步长算法（基于MCMC）

可以使用MCMC (Skilling, 2004) 从先验中采样一个新的点。通常，要么选择最近被移除的点，要么选择一个随机点作为随机游走的起点，或者更一般地，相对于受限先验的不变移动步长。提出一个新的点，如果它超过当前的似然阈值，则被接受。在平坦先验中，这使得随机游走成为一种纯粹的几何探索。这种随机游走如图8的右图所示。随机游走进行若干步后，最终点被返回给NS采样器，作为一个（几乎）独立的样本。

任何用于LRPS的随机游走MCMC解决方案都需要(1)步长提议，(2)一个适应NS进行持续缩小时的提议调整方案，以及(3)步数。不幸的是，文献中常常缺乏这些实现细节，这严重限制了数值比较的实用性。

5.1.1. 利用实时点知识

要制定一个好的步骤提案，可以利用两个属性：首先，当前的实时点已经至少大致按照当前轮廓分布，并且可以追踪相关空间及其几何形状，如果它们从先验到当前迭代已经被忠实地采样。这个假设对于直接从先验中抽取的初始实时点当然是合理的。其次，轮廓的行为在迭代之间变化不大，因为体积只缩小了很小一部分，因此必须依次解决非常相似的问题。

第二个属性通过在迭代之间重用优化的提案来利用，例如，如果前一次迭代的接受率较低，提案在使用下一次迭代之前会被缩小（例如，Sivia和Skilling, 2006年）。然而，这种调整不会保持详细平衡。为了解决这个问题，Salomone等人（2018年）建议进行一个带有自适应提案的预热NS运行，并将每个迭代的提案存储起来，然后进行一次最终运行，使用这些提案，但现在不再进行调整。

第一个特性被许多作者利用，通过从实时点估计样本协方差 (Veitch和Vecchio, 2010; Schuet, Timucin和Wheeler, 2011) 或确定主方向 (Nikolic, 2009) 来理解当前空间的大小和方向。

处理多模态数据需要更复杂的流程。Martiniani等人（2014）预计算了一个极小值数据库，并执行交换移动，使得MCMC可以在这些极小值之间进行交换。Handley、Hobson和Lasenby（2015b）利用活点，执行迭代Jarvis-Patrick聚类 (Jarvis和Patrick, 1973)，然后基于每个聚类的成员点估计协方差，从而获得

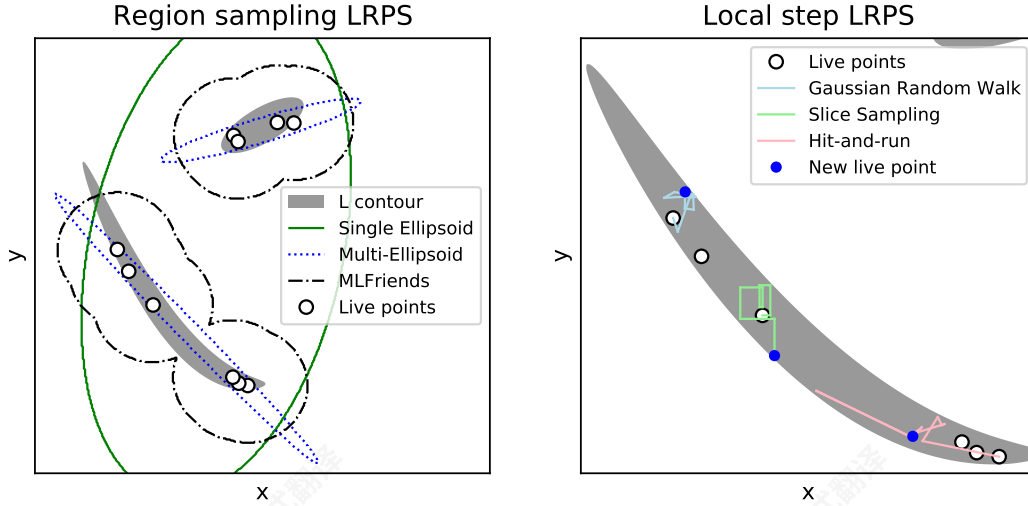


FIGURE 8. LRPS methods. At an iteration for the example from Figure 1, different methods for finding a new live point are illustrated. Left panel: Region sampling methods construct regions based on the live points (white circles) to approximate the unknown likelihood contour (grey). Rejection sampling based on these contours becomes inefficient if the contours are too large, and problematic for NS integration if it misses a region (e.g., the top left tail). The enlargement is intentionally chosen too small here. Right panel: Local step methods start Metropolis random walks (coloured lines) from a live point (white circle). Steps outside the likelihood contour are rejected. If the step proposal is good and the number of steps large, a new point (blue circle) is reached that is independent of the starting point.

5.1. Local step algorithms (MCMC-based)

A new point can be sampled from the prior with MCMC (Skilling, 2004). Typically, either the recently deceased point, or a random point are chosen as the start of a random walk, or more generally, invariant move steps with respect to the constrained prior. A new point is proposed, and accepted if it exceeds the current likelihood threshold. In flat priors, this makes the random walk a purely geometric exploration. Such random walks are illustrated in the right panel of Figure 8. The random walk proceeds for a number of steps, after which the final point is returned to the NS sampler as a (nearly) independent sample.

Any random walk MCMC solution to LRPS needs (1) a step proposal, (2) a recipe for adapting the proposal to the continuous shrinking as NS progresses and (3) the number of steps. Unfortunately, the literature often lacks these implementation details which severely limits the usefulness of numerical comparisons.

5.1.1. Leveraging live point knowledge

To craft a good step proposal, two properties can be leveraged: Firstly, the current live points are already distributed at least approximately according to the current contour, and trace out the relevant space and its geometry, if they have been sampled faithfully from the prior up to the current iteration. This assumption is certainly justified for the initial live points, which are drawn directly from the prior. Secondly, the behaviour of the contour changes little from iteration to iteration, as the volume shrinks by a small fraction, so that very similar problems have to be solved in sequence.

The second property is leveraged by reusing optimised proposals between iterations, for example if the previous iteration's acceptance rate was low, the proposal is shrunk before use in the next iteration (e.g., Sivia and Skilling, 2006). However, such adaptations do not maintain detailed balance. To address this, Salomone et al. (2018) suggest a warm-up NS run with adaptive proposals turned on and stored for each iteration, and followed by a final run that uses these proposals, now without adaptations.

The first property is leveraged by many authors by estimating the sample covariance from the live points (Veitch and Vecchio, 2010; Schuet, Timucin and Wheeler, 2011) or determining the principal directions (Nikolic, 2009) to understand size and orientation of the current space.

More sophisticated procedures are necessary to handle multi-modality. Martiniani et al. (2014) pre-computes a database of minima and performs an exchange move, whereby the MCMC can swap between them. Taking advantage of the live points, Handley, Hobson and Lasenby (2015b) performs iterative Jarvis-Patrick clustering (Jarvis and Patrick, 1973) and then estimates covariances based on the member points of each cluster, obtaining

一个局部协方差，用于从该聚类的成员点开始进行随机游走。原本为单模态分布设计的 MCMC 方案可以通过对活点应用聚类算法来处理多个模态，通过从成员点中减去聚类均值来叠加聚类点，并从偏移点（例如它们的样本协方差）中推导出提议。然后 MCMC 算法从随机选定的一个活点开始其链的初始点。

5.1.2. 临近采样

首个完全指定的建议由 Sivia 和 Skilling (2006) 提出。使用多维高斯分布，从死点开始。如果接受数 a 占据主导地位，拒绝数 r ，则标准差增加 $\exp(1/a)$ 倍，否则减少 $\exp(1/r)$ 倍。Brewer、Pártay 和 Csányi (2011) 更倾向于使用重尾、高度多尺度的建议，以避免在当前迭代中进行自适应。许多其他 MCMC 建议分布已被应用（例如，Liu 等人，2016 年；Beaton 和 Xiang，2017 年；Polido、Jablonski 和 Lépine，2013 年）。高斯随机游走以蓝色线条显示在图 8 的右图中。

另一种解决方案来自 Veitch 和 Vecchio (2010)，他们使用具有 2 个自由度的学生-t 分布建议，缩放到活点协方差的 10%，每 10 次迭代估计一次。然而，10% 的步骤使用受差分进化启发的建议：选择两个额外的活点，并将差向量加到当前点。这些尝试解决多模态分布；然而，他们观察到对于特定问题，即使有 1000 个 MCMC 链步，LRPS 艺术品仍然存在。为了在中间或高维问题中获得更高的效率，Gibbs (Murray 等人，2006 年) 和沿主成分的逐分量建议 Nikolic (2009 年) 被提出。Trassinelli (2016 年) 主要使用具有均匀分布的每个参数 MCMC 建议，但也包括一个交叉步骤来组合活点。这项工作没有验证其实现的正确性。同一作者后来的出版物包括不保持详细平衡的 MCMC 建议 (Trassinelli, 2019 年，他们的第一个高失败恢复程序)。

为结合问题参数空间几何的知识，Javid (2019) 提出了沿球形弧线和包裹圆形参数的方案。

Moss (2020) 将方案空间变形以平滑特殊形状，并将多模态分布聚集在一起。通过优化神经网络将信息损失最少的活点转换为标准高斯分布。从更简单的表面上采样可以更高效，如果方案考虑了雅可比非线性流对空间的压缩，那么可以从（受限）先验中获取样本。这种方法看起来非常有前景且通用，但在实践中实现和正确训练也相当复杂。

5.1.3. 按方向采样

Betancourt (2011) 推导了约束哈密顿蒙特卡洛 (CHMC)，即将哈密顿蒙特卡洛直接应用于似然约束先验采样。为简化起见，我们描述其在适当参数化下的行为，其中先验是平坦的。然后 CHMC 在选定方向上以长度 ξ 的直步前进，直到步长违反约束。在那里，它从边界反射，即动量向量反转垂直于似然（约束）梯度，并继续。因此，该技术需要步长和可计算的似然梯度。伽利略蒙特卡洛 (GMC; Skilling, 2012) 在拒绝反射时为 CHMC 添加一个反向步，从而使链更频繁地成功。该技术与 Neal (2003) 中讨论的反射有关。恶魔嵌套采样 (Habeck, 2015) 通过在恶魔中存储超额似然（能量）来软化似然轮廓。这意味着 MCMC 过程倾向于避开边界，而无需使用梯度。当梯度可用时，恶魔嵌套采样可以存储 CHMC 速度向量，并解决组合动量-位置空间。Baldock 等人 (2017) 还提出了一种基于 HMC 的嵌套采样扩展，并将 GMC 与存储每个活点动量的 CHMC 版本进行了比较，发现它们优于更简单的随机游走 MCMC 算法（另见 Nielsen, 2013）。

另一种方法是沿一条线从现有点进行采样。这里，介绍了三种此类算法，它们都可在 UltraNest 中使用：

1) 在切片采样 (Neal, 2003) 中，参数空间轴按提议方向进行迭代。均匀采样通过选择较远的边界实现，在它们之间均匀采样，并在每次拒绝时将边界向该侧收缩，直到接受一个样本 (Handley, Hobson and Lasenby, 2015b)。切片采样随机游走用绿色线条在图 8 的右图中表示。对于非均匀先验分布，需要额外的一步，即使用辅助变量采样先验分布的高度。对于平坦先验分布，或通过重新参数化空间将先验分布变为平坦的切片，则使用似然函数作为预言者（在似然阈值之上，或不在）。NS 文献通常对切片采样的具体变体保持模糊，

a local covariance used for walks started from members of that cluster. MCMC schemes originally intended for unimodal distributions can be extended to handle multiple modes by applying a clustering algorithm to the live points, overlaying the cluster points by subtracting cluster means from member points, and to derive the proposal from the shifted points (e.g., their sample covariance). Then the MCMC algorithm begins its chain from a random live point as its starting point.

5.1.2. Sampling by vicinity

The first fully specified proposal was presented in [Sivia and Skilling \(2006\)](#). A multi-dimensional Gaussian is used, starting from the dead point. If the number of accepts a dominates the number of rejects r , the standard deviation is increased by a factor of $\exp(1/a)$, otherwise decreased by a factor of $\exp(1/r)$. [Brewer, Pártay and Csányi \(2011\)](#) prefers a heavy-tailed, highly multi-scale proposal to avoid adaptation at the current iteration. Numerous other MCMC proposal distributions have been applied (e.g., [Liu et al., 2016](#); [Beaton and Xiang, 2017](#); [Polido, Jablonski and Lépine, 2013](#)). A Gaussian random walk is illustrated with blue lines in the right panel of Figure 8.

Another solution comes from [Veitch and Vecchio \(2010\)](#), who uses a Student-t distribution with 2 degrees of freedom proposal scaled to 10% of the covariance of live points, estimated every 10 iterations. However, 10% of steps use a proposal inspired by differential evolution: two additional live points are selected, and the difference vector added to the current point. These attempt to address multi-modal distributions; however, they observe that for a specific problem even with 1000 MCMC chain steps, LRPS artefacts remain. For higher efficiency in intermediate or high dimensional problems, Gibbs ([Murray et al., 2006](#)) and component-wise proposals along the principal components [Nikolic \(2009\)](#) have been proposed. [Trassinelli \(2016\)](#) primarily uses per-parameter MCMC proposals with a uniform distribution, but also includes a crossover step to combine live points. This work does not verify the correctness of their implementation. A later publication by the same author includes MCMC proposals that do not preserve detailed balance ([Trassinelli, 2019](#), their first high-failure recovery procedure).

To incorporate knowledge of the problem parameter space geometry, [Javid \(2019\)](#) introduced proposals along spherical arcs and wrapping of circular parameters.

[Moss \(2020\)](#) deforms the proposal space to flatten peculiar shapes and bring multi-modal distributions together. A neural network is optimised to transform the live points with least information loss to a standard Gaussian distribution. Sampling from that simpler surface can be more efficient, and if proposals consider the space compression by the non-linear flow via the Jacobian, samples from the (restricted) prior can be obtained. This method appears to be very promising and general, but also complex to implement and train correctly in practice.

5.1.3. Sampling by direction

[Betancourt \(2011\)](#) derived Constrained Hamiltonian Monte Carlo (CHMC), the straightforward application of Hamiltonian Monte Carlo to likelihood constrained prior sampling. For simplicity, we describe its behaviour in a suitable parametrisation where the prior is flat. Then CHMC makes straight steps of length ξ in a chosen direction until the step violates the constraint. There, it reflects off the boundary, i.e. the momentum vector reverses normal to likelihood (constraint) gradient, and continues. This technique thus requires a step size and a computable likelihood gradient. Galilean Monte Carlo (GMC; [Skilling, 2012](#)) adds a reverse step to CHMC upon rejection of the reflection, so that the chain succeeds more often. This technique is related to reflections discussed in [Neal \(2003\)](#). Demonic Nested Sampling ([Habeck, 2015](#)) softens the likelihood contours by storing excess likelihood (energy) in a demon. This means that a MCMC procedure then tends to turn away from the border without requiring the use of gradients. When gradients are available, Demonic Nested Sampling can store the CHMC velocity vector, and solve a combined momentum-position space. [Baldock et al. \(2017\)](#) also presents a HMC-based nested sampling extension, and compares GMC with a CHMC version that stores the momentum of each live point, finding them to outperform simpler random walk MCMC algorithms (see also [Nielsen, 2013](#)).

Another procedure is the sampling from an existing point along a line. Here, three families of such algorithms are presented, all of which are available in *UltraNest*:

- 1) In slice sampling ([Neal, 2003](#)), parameter space axes are iterated through for the proposal direction. Uniform sampling is achieved by choosing distant bounds, sampling uniformly between them and shrinking the bound towards that side for every reject until a sample is accepted ([Handley, Hobson and Lasenby, 2015b](#)). The slice sampling random walk is illustrated with green lines in the right panel of Figure 8. For non-uniform priors, an additional step is needed, by sampling the height of the prior distribution with an auxiliary variable. For flat priors, or placing slices through a reparameterised space which make the prior flat, the likelihood function is used as an oracle (above the likelihood threshold, or not). The NS literature often remains vague which exact variant of slice sampling is

一个正面的例子是PolyChord，它在下一个主轴上提出（Handley、Hobson和Lasenby，2015b）。主轴是从起点（见§ 5.1.1）的活点簇的样本协方差矩阵中获得的。

2) 在碰撞逃逸蒙特卡洛（HARM, Turchin, 1971; Smith, 1984）中，每一步会随机选择一个方向。具有边界收缩的算法变体由Kiatsupaibul、Smith和Zabinsky（2011）明确提出，它与切片采样关系最密切。这种游走过程在图8右侧面板中以粉色线条表示。即使在复杂的几何形状中，HARM在混合方面非常有效，并且随着维度的增加也能很好地扩展（参见Collins等人，2013; Kiatsupaibul、Smith和Zabinsky，2011及其参考文献），Collins等人（2013）还表明它在切片采样方面表现更优。Stokes、Tuyl和Hudson（2017）测试了这种技术用于具有非凸表面的NS，尽管是在低维度下。

3) 通过选择另一个随机活动点来绘制方向。例如，Stokes、Tuyl和Hudson（2017）使用受单纯形启发的游走来利用活动点的分布。Pitkin等人（2017）结合了多种提议，包括均匀局部步长提议、差分进化和仿射不变集合采样MCMC（Goodman和Weare，2010）。仿射不变集合采样器是MCMC提议的常用选择，因为它不需要调参。对于NS来说，这似乎是一个自然的选择，因为它已经维护了一个点集。然而，在实践中，这种提议主要在高斯形状中表现良好，Huijser、Goodman和Brewer（2015）证明在高维度中，采样器点集会坍缩到低维平面。

还有更多几何随机游走算法存在，并且它们在NS中的LRPS应用中似乎具有先验的适用性。例如，可以参考Vempala（2005）的综述。

5.1.4. 独立性的步数

所有上述方法都需要计算找到一个新的、假设为独立的点的步数。如何选择步数？这显然至少取决于P/D问题。一种简单的方法是观察在一系列步数不断增加的NS运行中 Z 的变化（例如，Higson等人，2019）。

或者，活点已经均匀分布的假设表明了一个简单的调谐标准。如果局部MCMC链没有进展到两个活点之间的典型距离，那么它很可能没有走够远（Salomone等人，2018）。因此，一种简单的自动调谐方法是在达到或不达到该标准时，为下一次迭代增加/减少步数。对于典型的活点距离，UltraNest对HARM自动调谐方法（“adapt=move-distance”）的实现使用所有活点对之间的平均马氏距离。Salomone等人（2018）强调，为了保持详细平衡，必须使用预定义的步数（即至少与自动调谐确定的步数一样多）执行没有适应的新运行。

在CHMC和Galilean Monte Carlo中，反射可能导致随着步数增加，无法探索更多参数空间，从而形成循环。为了解决HMC中类似的问题，Hoffman和Gelman（2014）提出构建仅向前和向后进行HMC轨迹，直到它们再次转向（无回转采样器，NUTS）。通过随机选择向前或向后移动，并迭代加倍步数，可以保持详细平衡。当终点向量的点积为正时，可以识别出U转。然后从轨迹中采样一个点，其接受概率由总能量（目标概率和步长动量）决定。NUTS的变体可能是Demonic Nested Sampling扩展的一个有趣的研究方向。

为了将这种方法应用于NS，Griffiths和Wales（2019）开发了无Galilean U-Turn采样器（NoGUTS）。在平坦先验下，CHMC和Galilean Monte Carlo中，动量保持不变，目标概率要么是常数，要么为零。因此，轨迹直到边界反射前都是直线。这简化了与NUTS相比的问题。然而，NS中采样空间的持续收缩，如果采用简单的步长自适应方法（Griffiths和Wales，2019），会导致偏差。

为了使采样空间具有良好分辨率且不频繁超出边界，Skilling（2012）建议调整步长，以便大多数步长被接受。但这意味着大多数步长构成了一条直线的进展，包含很少的额外信息。目前，关于NoGUTS、HARM和切片采样在不同问题中的混合质量比较研究尚未进行。

5.2. 区域采样算法（非MCMC）

从先验分布中在似然约束下进行采样最安全的方法是随机地从先验分布中采样，如果似然约束不满足则拒绝该点。这种拒绝采样需要一个好的提议函数才能高效。在LRPS的情况下，提议函数的支持集对应于参数空间的一个区域。为了正确性，它必须完全包含（未知的）似然等高线，即似然受限

implemented. A positive example is *PolyChord*, which proposes along principle axis after the next (Handley, Hobson and Lasenby, 2015b). The principle axis are obtained from the sample covariance matrix of live point cluster where the walk has started (see §5.1.1).

2) In hit-and-run Monte Carlo (HARM, Turchin, 1971; Smith, 1984), a random direction is instead chosen in each step. The algorithm variant with bound shrinkage was presented explicitly by Kiatsupaibul, Smith and Zabinsky (2011) and is most closely related to slice sampling. Such a walk is illustrated with pink lines in the right panel of Figure 8. Even in complicated geometries, HARM is highly effective in mixing and scales well with dimensionality (see Collins et al., 2013; Kiatsupaibul, Smith and Zabinsky, 2011, and references therein), Collins et al. (2013) also show that it outperforms slice sampling. Stokes, Tuyl and Hudson (2017) tested this technique for NS with non-convex surfaces, albeit in low dimensions.

3) Drawing the direction by choosing another random live point. For example, Stokes, Tuyl and Hudson (2017) uses a simplex-inspired walk to take advantage of the distributions of live points. Pitkin et al. (2017) combines of several proposals including uniform local step proposals, differential evolution, and affine-invariant ensemble sampling MCMC (Goodman and Weare, 2010). The affine-invariant ensemble sampler is a popular choice for a MCMC proposal, because it does not need tuning. This appears as a natural choice for NS, which already maintains a population of points. However, in practice, this proposal performs well primarily in Gaussian shapes, and Huijser, Goodman and Brewer (2015) demonstrate that in high dimensions, the sampler population collapses into a lower-dimensional plane.

Many more geometric random walk algorithms exist and appear a-priori suitable for LRPS in NS. For example, see the survey by Vempala (2005).

5.1.4. Number of steps for independence

All aforementioned methods require the number of steps until a new, supposedly independent point is found. How to chose the number of steps? This is surely dependent on at least P/D issues. A simple technique is to observe the change of Z in a series of NS runs with increasing number of steps (e.g., Higson et al., 2019).

Alternatively, that the live points are already uniformly distributed suggests a simple tuning criterion. If the local MCMC chain has not progressed further than the typical distance between two live points, it has likely not stepped far enough (Salomone et al., 2018). A simple auto-tuning method is thus to increase/decrease the number of steps for the next iteration when that criterion is reached/not reached. For the typical live point distance, *UltraNest*'s implementation of HARM auto-tune method ("adapt=move-distance") uses a mean Mahalanobis distance of all pairs of live points. Salomone et al. (2018) emphasises that for conserving detailed balance, a fresh run with no adaptation has to be performed with a predefined number of steps, i.e., at least as many as auto-tuning determined.

In CHMC and Galilean Monte Carlo, the reflections can lead to cycles that do not explore more of the parameter space with increasing number of steps. To address a similar problem in HMC, Hoffman and Gelman (2014) proposed to construct forward and backward HMC trajectories only until they turn back (No U-turn sampler, NUTS). Detailed balance is preserved by randomly considering going forward or backwards while iteratively doubling the number of steps. A U-turn can be identified when the end point vectors show a positive dot product. A point is then sampled from the trajectory, with acceptance probabilities determined by the total energy (target probability and step momentum). Variants of NUTS may be an interesting research direction for Demonic Nested Sampling extensions.

To transfer this approach to NS, Griffiths and Wales (2019) developed the No Galilean U-Turn Sampler (NoGUTS). In the case of CHMC and Galilean Monte Carlo under a flat prior, the momentum remains constant and the target probability is either a constant or zero. Therefore, the trajectories are straight until boundary reflections. This simplifies the problem compared with NUTS. However, the constant shrinkage of the sampling space in NS leads to biases if simple step size adaptations are employed (Griffiths and Wales, 2019).

To allow good resolution of the sampling space and not stepping out of the boundary too often, Skilling (2012) recommends tuning the step size so that most steps are accepted. This however means that most of the steps constitute the progression of a straight line and contain little additional information. A comparative study of the mixing quality of NoGUTS, HARM and slice sampling in different problems remains unstudied as of yet.

5.2. Region sampling algorithms (non-MCMC)

The safest way to sample from the prior under a likelihood constraint is to sample randomly from the prior and reject the point if the likelihood constraint is not fulfilled. This rejection sampling requires a good proposal function to be efficient. In the case of LRPS, the support of the proposal function corresponds to a parameter space region. For correctness, it must fully contain the (unknown) likelihood contour, i.e., the support of the likelihood-restricted

先前。如果一个参数空间的部分被遗漏，如图8所示，体积比 V_{i+1}/V_i 将会被高估。然而， Z 估计可能被高估或低估，这取决于被遗漏区域相对于构建区域的概率。本节提出的方法试图重建这个概率轮廓本身，或者至少是其超集。

Mukherjee、Parkinson和Liddle (2006) 计算了包含所有活点的最小边界椭球，并将其扩展了 ~ 1.7 倍。图8的左面板说明了这样一个包裹活点的椭球。从椭球中迭代地抽取样本，并评估其似然性（拒绝采样）。因此，该方法有两个参数：活点数量，它有助于描绘参数空间，以及扩大因子。如果椭球扩展不足，参数空间的一些区域将永远不会被采样；如果扩展过多，拒绝采样将效率低下。扩大因子的选择取决于模型和维度。除了用缩放的协方差度量近似最小体积边界椭球之外，更高级的约束优化算法可以产生更小的体积（Rollins, 2015）。为什么椭球会比，比如说，盒子（用于Obrezanova等人，2007年；Möller等人，2013年）更受欢迎？在高数据情况下，似然函数往往成为椭圆分布（如多元高斯分布），它们具有椭圆轮廓（例如，参见Wilks, 1938年）。

如果问题呈现多模态性，模式之间的空间使得拒绝采样非常低效。因此，Shaw、Bridges和Hobson (2007) 使用递归k-means对活点进行聚类，并对每个聚类采用椭球采样。相比之下，Theisen和Jülich (2013) 在采样效率低于阈值时增加聚类数量。Feroz和Hobson (2008) 进一步考虑了x-means、g-means和pg-means进行聚类，最终使用x-means。该算法选择包围活点的椭球主轴的端点，并尝试使用两个聚类进行k-means聚类。活点被分配到两个聚类中的一个，并用于构建包围椭球。如果两个椭球比围绕所有活点的单个椭球更好地描述活点，则聚类被接受。该过程递归重复，直到收敛。需要定义接受拆分时的条件，Shaw、Bridges和Hobson (2007)、Feroz和Hobson (2008)、Feroz、Hobson和Bridges (2009) 测试了多种标准，包括信息标准。最简单的方法是考虑体积是否至少减少了某个因子，以及椭球是否足够分离。MultiNest算法将这些标准作为椭球扩大因子的附加参数。图8的左面板说明了多椭球聚类。

Feroz和Hobson (2008) 也考虑了随迭代次数和椭球体积减小的自适应扩大因子，但这些最终没有在Feroz、Hobson和Bridges (2009) 提出的MultiNest算法中使用。使用多个椭球对于近似特殊形状也很有用。效率使得MultiNest成为了一种流行的算法，拥有多种实现和接口，包括用于Python的PyMultiNest (Buchner等人, 2014)、用于R的RMultiNest (Buchner, 2015)、一个未命名的Mathematica实现 (Gervino、Mana和Palmisano, 2016)、用于C的DIAMONDS++ (Corsaro和De Ridder, 2014)、用于jax GPU编程语言的JAXNS (Albert, 2020)、nestle (Barbary, 2016)，以及从后者派生出的用于Python的dynesty Speagle (2020)。

为了避免为每个问题选择一个扩大因子，Buchner (2014) 通过交叉验证从活动点估计它。如果某些随机活动点子集未知，我们会构建一个足够大的区域来采样它们吗？如果没有，扩大就不充分，必须增加。经过几次交叉验证轮次，找到了一个足够大的扩大。作为一个具体示例，Buchner (2014) 的RadFriends算法将每个活动点转换为椭球的中心，其形状由距离度量的选择决定。虽然原始的RadFriends使用球体，但Buchner (2019) 的MLFriends使用活动点的协方差来定义椭球，从而带来了显著的性能提升。图8的左面板说明了MLFriends构建的区域。通过检查哪些活动点包含在其他活动点的椭球中，可以自然地定义聚类。从每个活动点减去聚类均值，并取结果的协方差，可以进一步改进几何结构，而无需专门的聚类算法。

5.3. 混合方法

混合方法结合了区域重建与MCMC采样。这些方法通过排除极有可能位于轮廓之外的区域来减少似然评估次数，同时保留了MCMC算法的维度缩放特性。

Feroz和Hobson (2008) 将MultiNest与Sivia和Skilling (2006) 的MCMC提议（20步）结合，并测试了高达100维的问题。有趣的是，他们发现使用与局部协方差调优的提议效果较差。这种混合组合在dynesty (Speagle, 2020) 中可用。

prior. If a portion of the parameter space is left out, as illustrated in Figure 8, the volume ratio V_{i+1}/V_i will be overestimated. However, the Z estimate can be over or underestimated, depending on the likelihood in the left-out region relative to the constructed region. The methods presented in this section try to reconstruct this likelihood contour itself, or at least a super-set of it.

Mukherjee, Parkinson and Liddle (2006) compute the smallest bounding ellipsoid that contains all live points, and expands this by a factor (~ 1.7). The left panel of Figure 8 illustrates such an ellipsoid wrapping the live points. Iteratively samples are drawn from the ellipsoid and their likelihood is evaluated (rejection sampling). This method thus has two parameters: The number of live points, which helps trace out the parameter space, and the enlargement factor. If the ellipsoid is not expanded enough, regions of the parameter space are never sampled, if it is expanded too much, the rejection sampling is inefficient. The choice of the enlargement is model and dimension-dependent. Beyond approximating the minimum-volume bounding ellipsoid with a scaled covariance metric, more advanced constrained optimisation algorithms yield smaller volumes (Rollins, 2015). Why would ellipsoids be preferred over, say, a box (used in Obrezanova et al., 2007; Möller et al., 2013)? In the high-data regime, likelihood functions tend to become elliptical distributions (such as a multi-variate Gaussian), which have ellipsoidal contours (see e.g., Wilks, 1938).

If the problem presents multi-modality, the space between modes makes the rejection sampling very inefficient. Therefore, Shaw, Bridges and Hobson (2007) cluster the live points with recursive k-means and employs ellipsoid sampling for each cluster. In contrast, Theisen and Jülich (2013) increases the number of clusters when the sampling efficiency drops below a threshold. Feroz and Hobson (2008) further consider x-means, g-means and pg-means for the clustering and uses x-means in the end. The algorithm chooses the end points of the major axis of the ellipsoid enclosing the live points, and attempts a k-means clustering with two clusters. Live points are assigned to one of the two clusters, and used to construct enclosing ellipsoids. If the two ellipsoids describe the live points better than a single ellipsoid around all live points, the clustering is accepted. This procedure is recursively repeated, until convergence. The condition when a split is accepted needs to be defined, and a variety of criteria are tested in Shaw, Bridges and Hobson (2007); Feroz and Hobson (2008); Feroz, Hobson and Bridges (2009), including information criteria. The perhaps simplest is to consider whether the volume is decreased by at least a certain factor, and whether the ellipsoids are sufficiently apart. The MultiNest algorithm has these criteria as parameters additional to the ellipsoid enlargement factor. The left panel of Figure 8 illustrates the multiple-ellipsoids clustering.

Feroz and Hobson (2008) also considers adaptive enlargement factors that decrease with iteration and ellipsoid volume, but these are ultimately not used in the MultiNest algorithm presented in Feroz, Hobson and Bridges (2009). The use of multiple ellipses is also useful for approximating peculiar shapes. The efficiency has made MultiNest a popular algorithm, with multiple implementations and interfaces, including PyMultiNest (Buchner et al., 2014) for Python, RMultiNest (Buchner, 2015) for R, an unnamed Mathematica implementation (Gervino, Mana and Palmisano, 2016), DIAMONDS for C++ (Corsaro and De Ridder, 2014), JAXNS for the jax GPU programming language (Albert, 2020), nestle (Barbary, 2016) and, derived from the latter, dynesty Speagle (2020), also for Python.

To avoid choosing an enlargement factor for every problem, Buchner (2014) estimates it from the live points by cross-validation. If some random subset of live points were unknown, would we construct a region large enough sample them? If not, the enlargement is insufficient and must be increased. After several cross-validation rounds, a large-enough enlargement is found. As a specific example, the RadFriends algorithm Buchner (2014) turns every live point into the centre of an ellipsoid, whose shape determined by the choice of distance metric. While the original RadFriends used spheres, Buchner (2019)’s MLFriends uses the covariance of live points to define the ellipsoids, leading to substantial speed improvements. The left panel of Figure 8 illustrates the region constructed by MLFriends. Clusters are naturally defined by checking which live points are contained in other live points’ ellipsoids. Subtracting the cluster means from each live point, and taking the resulting covariance improves the geometry further, without requiring a dedicated clustering algorithm.

5.3. Hybrid methods

Hybrid methods combine the region reconstruction with MCMC sampling. These reduce the number of likelihood evaluations by excluding space that is very likely outside the contour, while retaining the dimensionality scaling of MCMC algorithms.

Feroz and Hobson (2008) combined MultiNest with the MCMC proposal of Sivia and Skilling (2006) of 20 steps and tested problems in up to 100 dimensions. Interestingly, they find that using a proposal tuned with local covariances is inferior. Such hybrid combinations are available in dynesty (Speagle, 2020).

与MultiNest的x-means不同, Trassinelli (2019) 探索了高斯混合方法和均值漂移聚类方法与MCMC结合, 但缺乏对提议和聚类方法的数值比较, 以判断其是否优于先前发表的研究。

UltraNest 包件可以将两种区域构造方法 (MLFriends 和单个椭圆) 结合起来。前者在低维和多个簇的情况下表现良好, 而后者适用于高斯类似似然。在这两种情况下, 都使用自助法估计放大因子。只有当新点同时落在两种区域构造内时才会考虑。然后步进采样方法可以利用这两个区域来预过滤建议点, 以避免模型评估。

6. Nested sampling 变体

在标准 NS 中, 采样器维护一个固定的活点数。这涉及识别最低似然点并用 LRPS 替换它, 从而单调增加似然阈值。以下小节将探讨该方案的变体, 包括软似然约束 (§ 6.1)、活点数的变化 (§ 6.2) 以及算法并行化的可能性 (§ 6.3)。

6.1. 软化硬似然约束

LRPS 方法处理硬似然阈值, 因此只能测试点是否可接受。结果低于轮廓的点会被丢弃, 而且很难知道随机游走轨迹何时接近轮廓。同时, 当 LRPS 方法错误地排除参数空间时, 收缩会加速, 而缓慢的 LRPS 采样 (例如随机游走太短) 会导致点移动不足, 从而减缓收缩。为了解决这些问题, 已经开发出了一些通过避免硬轮廓来放宽问题的方法。

重要性抽样从解析形状中抽取样本, 以近似目标未知概率分布, 并对样本进行重新加权。重要性嵌套抽样 (INS, Chopin和Robert, 2007b; Chopin和Robert, 2008; Chopin和Robert, 2010) 通过这种方式推广嵌套抽样的概念来应用相同的思想。Feroz等人 (2013) 采用多椭圆抽样应用这一概念, 并证明使用原本被丢弃的样本可以显著提高 $\log Z$ 估计的精度。实际上, INS估计器可以在某些困难问题中部分纠正MultiNest不正确的LRPS抽样 (Buchner, 2014; Feroz等人, 2013)。然而, Nelson等人 (2020) 在一个系外行星的应用中表明, 即使在正确模型下, MultiNest中的标准NS和INS估计器在多次运行之间也存在超出其不确定性的显著散差 (参见其附录A9, 图7和8)。实际上, 运行之间的散差可能是LRPS不可靠的指标 (参见 § 4.5.2)。

在扩散嵌套采样 (Brewer、Pártay和Csányi, 2011) 中, 由收缩创建的似然轮廓被可逆地探索。粒子不仅被允许在当前的似然约束内移动, 还可以向上 (向下) 移动到更 (少) 约束的先验似然水平。所使用的水平并非基于所有找到的似然值。相反, 会保持少数 (几十个) 水平。在这些水平内, 采样的点被用于估计体积内的平均似然。这意味着相对粗糙的MCMC提议可以作为LRPS程序使用, 因为仅需确保水平平均值最终收敛到真实值。

扩散嵌套采样的上下移动通过Metropolis提议随机实现。因此, 扩散嵌套采样将整个嵌套采样探索转换为具有任意精度的MCMC过程。将NS作为MCMC过程使其在理论 (收敛性) 分析中具有吸引力。然而, 这也需要使用MCMC收敛性诊断来确定运行终点, 而与标准嵌套采样相比, 这一点并不那么明确。

恶魔嵌套采样 (Habeck, 2015) 通过将多余的似然 (能量) 存储在恶魔变量中来平滑似然轮廓。这可以与哈密顿蒙特卡洛结合以存储动量, 同时通过温度演化 (另见Baldock等人, 2017) 提供有关运行状态的诊断。

除了提升 LRPS 之外, 似然约束可能需要更宽松, 因为精确的似然无法计算。对于似然无信息推理, 即仅能获得随机但无偏的似然估计的情况, Mikelson 和 Khammash (2020) 提出了一种 NS 变体。

Different to MultiNest’s x-means, [Trassinelli \(2019\)](#) explores Gaussian mixture method and the mean-shift clustering method with MCMC, but lacks numerical comparisons to judge any improvements over previous publications, both for the proposal and clustering method.

The *UltraNest* package can combine two region construction methods, MLFriends and a single ellipsoid. The former works well in low dimensions and with multiple clusters, while the latter works well for Gaussian-like likelihoods. The enlargement factors are estimated in both cases with bootstrapping. New points are only considered when they fall within both region constructions. Step sampling methods can then take advantage of both regions to pre-filter proposed points to avoid model evaluations.

6. Nested sampling variations

In standard NS, the sampler maintains a fixed number of live points. This involves identifying the lowest likelihood point and replacing it with LRPS, thereby increasing the likelihood threshold monotonically. The following subsections look at variations of this scheme, including soft likelihood constraints (§6.1), varying the number of live points (§6.2) and possibilities for parallelisation (§6.3) of the algorithm.

6.1. Softening the hard likelihood constraint

LRPS methods deal with a hard likelihood threshold, and thus can only test whether a point is acceptable or not. Points which turn out to be below the contour are discarded, and it is difficult to know when a random walk trajectory approaches the contour. At the same time, when LRPS methods mistakenly exclude parameter space the shrinking is accelerated, while slow LRPS sampling (such as too short random walks) can cause points not to move enough, leading to slowed shrinkage. To address these problems, methods have been developed which relax the problem by avoiding a hard contour.

Importance sampling draws from an analytic shape to approximate the unknown probability distribution of interest and reweighs the samples. Importance Nested Sampling (INS, [Chopin and Robert, 2007b](#); [Chopin and Robert, 2008](#); [Chopin and Robert, 2010](#)) applies the same concept by generalising nested sampling in this fashion. [Feroz et al. \(2013\)](#) employed this concept with multi-ellipsoidal sampling, and demonstrates that the use of otherwise discarded samples leads to a substantially more precise $\log Z$ estimate. Indeed, the INS estimator can somewhat correct for the incorrect LRPS sampling of MultiNest in some difficult problems ([Buchner, 2014](#); [Feroz et al., 2013](#)). However, [Nelson et al. \(2020\)](#) demonstrate in an application to exoplanets that both the standard NS and INS estimators in MultiNest can show substantial scatter between runs beyond their uncertainties even in the correct model (see their appendix A9, Figure 7 and 8). Indeed, the scatter between runs can be an indicator that the LRPS is unreliable (see §4.5.2).

In Diffusive Nested Sampling ([Brewer, Pártay and Csányi, 2011](#)) the likelihood contours created by shrinkages are reversibly explored. Particles are not only allowed to traverse within the current likelihood constraint, but to also to move up (down) in likelihood levels to a more (less) constrained prior. The levels used are not based on all likelihoods found. Instead a low number (dozens) of levels are maintained. Within these levels, the sampled points are used to estimate the average likelihood across the volume. This implies that relatively crude MCMC proposals can be employed as LRPS procedures, as it is only required that the level averages ultimately converge to the true value.

The up/down move of Diffusive Nested Sampling is achieved stochastically with a Metropolis proposal. Thus Diffusive Nested Sampling turns the entire nested sampling exploration into a MCMC process with arbitrary precision. Having NS as a MCMC process makes it appealing for theoretical (convergence) analysis. However, it also requires the use of MCMC convergence diagnostics to determine the end point of a run, which is not as well-defined as with standard nested sampling.

Demonic Nested Sampling ([Habeck, 2015](#)) softens the likelihood contours by storing excess likelihood (energy) in a demon variable. This can be combined with Hamiltonian Monte Carlo to store momenta, and also delivers diagnostics about the state of the run through the temperature evolution (see also [Baldock et al., 2017](#)).

Beyond improving LRPS, the likelihood constraint may need to be softer because the exact likelihood cannot be computed. For likelihood-free inference, where only a stochastic but unbiased estimate of the likelihood is available, [Mikelson and Khammash \(2020\)](#) present an NS variant.

6.2. 改变活点数量

原始的 NS 公式考虑了一个固定大小的活点种群。在发现不重叠的活点簇时，Feroz 和 Hobson (2008) 提出将嵌套采样运行拆分为独立的多个运行。该程序被采用以防止丢失峰，并将每个（可能大小不等）峰中的活点数量增加到 N 。每个子运行相关的体积需要被估计，这在 MultiNest 中通过数值方法完成。然而，尽管后续论文采用了相同的方法，但该问题和防止峰丢失的问题并未得到明确说明，并且已知文献中没有测试问题容易出现此问题。

NS 的主要缺点在于它需要系统地整个先验分布推进到可能非常小的后验分布，这需要许多迭代。这与 MCMC 等方法形成对比，后者可以快速接近后验质量集中区域，甚至可以从有利的初始位置开始。在某些问题中，特别是如果只确定存在一个后验模态时，因此可能有效尝试加速 NS 中的“查找”阶段。

为解决这个问题，Higson 等人 (2017 年) 提出了动态 Nested Sampling。首先，使用固定且较低数量的活点运行 NS。然后，构建后验权重的经验 CDF，以及一个指示每个死点上方剩余先验体积分数的经验 CDF。这两个 CDF 按 1:3 的比例线性组合。然后，寻找包含大部分 CDF 的 10% 和 90% 分位数，即 $L_{10\%}$ 和 $L_{90\%}$ ，并从 $L_{10\%}$ 开始启动新的 NS 运行，通过在阈值处使用 LRPS 创建 N' 个活点进行采样。然后继续运行该 NS 直到 $L_{90\%}$ 。将新的和原始的 NS 运行合并，并重复 CDF 过程，直到满足某些标准，例如 $\log Z$ 不确定性。Higson 等人 (2017 年) 在玩具问题和实际问题展示了效率提升。这实际上采用了一种临时的线性缩放方法来优化多目标计划。

考虑到第 3.4 节中提出的树形结构，我们也可以将动态嵌套采样解释为反复运行树搜索，并添加子节点。更普遍的看法是，多个智能体可以在树上操作，直到它们各自的收敛标准得到满足。我们将这种方案称为反应式嵌套采样，因为智能体分析树并对其状态做出反应。例如，动态 NS 智能体选择位于 $L_{10\%}$ 上方的节点，为它们添加子节点，并继续这些新分支，直到达到某个标准 ($L > L_{90\%}$)。在 UltraNest 的实现中，三个智能体分析树并通过添加子节点来做出反应：首先，通过根据后验权重随机采样节点来改进有效样本量标准 (Higson 等人, 2017 年)。其次，通过根据忽略该节点所造成的损失信息来随机采样节点来改进采样不确定性 (参见 Speagle, 2020 年)。第三， Z 不确定性的改进主要受制于活点较少的阶段，因为 $\log Z$ 上的不确定性是 $\sigma = \sqrt{\sum_i \frac{1}{N_i^2}}$ (Higson 等人, 2017 年)。因此，该策略确定了达到目标 $\log Z$ 不确定性所需的最小 N ，并在整个运行过程中强制执行这一要求。Speagle (2020 年) 分析了不同活点添加程序的不确定性估计。

据我们所知，添加新子节点没有错误的方法，因此代理可以积极优化以符合其标准。因此，可以考虑智能代理和博弈论 (例如，极小极大算法) 等计算机科学概念。但话说回来，最初使用，例如 $N = 1$ 运行 NS 并不可取。部分原因在于不会探索多种模式，还因为某些 LRPS 程序可以从少量活动点中显著受益，以便它们可以构建，例如，协方差矩阵来估计当前几何形状或至少其粗略尺度。

6.3. 并行化

利用多个处理单元可以在多个层次上实现：

在似然函数内并行化 概念上最简单的并行化是在似然函数中使用多个计算核心，例如处理大型数据集或评估复杂模型。这不需要对基本 NS 算法进行任何更改。与此相关的是 Graff 等人 (2012) 的方法，即在 NS 运行期间训练一个神经网络来模拟似然函数，以避免当网络精度足够时评估昂贵的似然函数。

当 LRPS 效率低时对其进行并行化 不高效的搜索新活点可以通过让多个核心通过工作进程独立执行 LRPS 来实现 (例如 Feroz、Hobson 和 Bridges, 2009 年)。第一个成功的抽取被接受并返回给主进程。然后使用后续阈值重新启动并行化。这种方法易于实现，因为它需要很少的通信，并且不需要修改其他 NS 组件。

6.2. Varying the number of live points

The original formulation of NS considered a population of live points of fixed size. Upon finding disjoint clusters of live points, [Feroz and Hobson \(2008\)](#) proposed to split the nested sampling run into separate, independent runs. This procedure is adopted to prevent the loss of modes and increases the number of live points to N in each (potentially unequally sized) mode. The volume associated with each sub-run needs to be estimated, which is done numerically in MultiNest. However, although following papers adopt the same procedure, the problem and prevention loss of modes is not clearly demonstrated, and no test problem examples in the literature are known to be prone to this issue.

A major slowdown of NS is that it needs to systematically progress from the entire prior to the potentially extremely small posterior, and this can take many iterations. This is in contrast to, for example, MCMC, which can head quickly towards the posterior mass concentration, or even be started in favourable locations. In some problems, especially if one is confident only one posterior mode is present, it may therefore be efficient to try to accelerate this “finding” phase in NS.

To address this, Dynamic Nested Sampling was proposed by [Higson et al. \(2017\)](#). First, NS is run with a fixed, low number of live points. Then, an empirical CDF of the posterior weights is built, and an empirical CDF indicating the fraction of the prior volume remaining above each dead point. These two CDFs are linearly combined with ratios 1:3. Then, the 10% and 90% quantiles, $L_{10\%}$ and $L_{90\%}$, are sought which contain most of the CDF. A new NS run is started from $L_{10\%}$, by creating N' live points sampled by the LRPS at that threshold. This NS run is then continued until $L_{90\%}$. The new and original NS runs are merged, and the CDF procedure repeated until some criterion is met, such as the log Z uncertainty. [Higson et al. \(2017\)](#) demonstrate efficiency gains on toy and real-world problems. This effectively employs an ad-hoc linear scalization to optimize a multi-objective plan.

Considering the tree formulation presented in Section 3.4, we can also interpret Dynamic Nested Sampling as repeatedly running a tree search, and adding child nodes. The more general view is that multiple agents could operate on the tree and until each of their convergence criteria are met. We term this scheme Reactive Nested Sampling, because an agent analyses the tree and reacts to its state. For example, the Dynamic NS agent selects nodes just above $L_{10\%}$, adds children to them and continues these new branches until the some criterion is reached ($L > L_{90\%}$). In *UltraNest*’s implementation, three agents analyse the tree and react by adding children: Firstly, the effective sample size criterion is improved by randomly sampling nodes based on their posterior weights ([Higson et al., 2017](#)). Secondly, the sampling uncertainty is improved by randomly sampling nodes based on the information loss of leaving it out (see [Speagle, 2020](#)). Thirdly, improvements to the Z uncertainty are primarily limited by phases with few live points, as the uncertainty on log Z is $\sigma = \sqrt{\sum_i \frac{1}{N_i^2}}$ ([Higson et al., 2017](#)). Therefore, the strategy identifies the minimum N required to reach the targeted log Z uncertainty, and enforces that throughout the run. [Speagle \(2020\)](#) analysed the uncertainty estimations for different live point addition procedures.

As far as we can tell, there is no wrong way to insert new children, so agents can aggressively optimize towards their criteria. Therefore, computer science concepts such as intelligent agents and game theory (e.g., minimax algorithms) can be considered. That said, running NS initially with, to give an extreme example, $N = 1$, is not advisable. This is in part because multiple modes will not be explored, but also because some LRPS procedures can substantially benefit from having a few live points, so that they can build, for example, a covariance matrix to estimate the current geometry or at least its rough scale.

6.3. Parallelisation

Taking advantage of multiple processing units can be achieved in several levels:

Parallelisation within the likelihood function The conceptionally simplest parallelisation is to use multiple computing cores within the likelihood function, for example to process large data sets or evaluate complex models. This requires no changes to the basic NS algorithm. Related here is [Graff et al. \(2012\)](#)’s approach of training a neural network to emulate the likelihood function during a NS run to avoid evaluating the costly likelihood function when the network accuracy is sufficient.

Parallelisation of the LRPS when its efficiency is low Inefficient searches for new live points can be parallelised by letting multiple cores perform LRPS independently by worker processes (e.g., [Feroz, Hobson and Bridges, 2009](#)). The first successful draw is accepted and returned to the main process. Then the parallelisation is restarted with the subsequent threshold. This method is easy to implement, as it requires little communication and no modification of other NS components.

如果一个迭代同时从多个工作进程产生成功的抽取，如果它们超过连续阈值，则可以按顺序接受。

UltraNest允许每个处理器并行推进其MCMC链。当一个链达到所需步数时，样本被接受，并且似然阈值被提高。然后工作进程找到它们链中满足新似然阈值的最后一个点，并从那里继续。这避免了丢弃整个链。

通过一次添加和移除多个活跃点进行并行化 这允许并行搜索新的样本（Burkoff 等人，2012；Henderson 和 Goggans，2014；Martiniani 等人，2014）。除了 LRPS 并行化之外，Baldock 等人（2017）对此进行了定量测试。Nielsen（2013）提出了一种 NS 的修改版本，其中移除的活跃点数量是根据活跃点的似然分布来选择，以最优地划分空间。

通过使用相同的 PRNG 种子进行多个独立的运行进行并行化 当尝试绘制更高点时，从约束先验（通过区域采样）中抽取的样本数量与进程数相同（Rollins，2015）。然而，每个进程仅评估第 p 个样本的似然，该样本对应其 ID p 。在似然评估后，每个进程将结果分发给其他进程，并就每个样本是否接受或拒绝做出本地决策。因此，每个进程都有相同的嵌套采样运行，但似然评估能够很好地扩展到进程数。

使用多个独立运行进行并行化 多个独立运行可以在后期合并（Skilling，2009；Henderson、Goggans和Cao，2017）。与之前的方法相比，这种方法具有更多的有效活点，因此精度更高。独立运行的缺点是它们无法共享参数空间几何信息。独立运行的好处是它们可以探索系统误差（Higson等人，2019）。Griffiths和Wales（2019）将这种方法推向了令人尴尬的并行化极限，进行了20,000个独立运行，每个运行只有一个活点。然后他们使用专门的算法识别落入相同模式的运行，并将这些运行统一，运行的数量与模式概率成正比。扩散嵌套采样使用类似的并行化方法，让它的游走者主要独立探索（Brewer、Pártay和Csányi，2011，B. Brewer，私人通信）。

同时分析相似数据集时，Buchner (2019) 注意到，当分析相似数据集时，它们在给定迭代中的似然等值线是相似的。通过绘制联合似然等值线可以减少计算量。这在分析大量相似数据或使用相同输入参数的蒙特卡洛模拟数据时最为有用，因为模型评估成本很高。它也可以用来轻微地改变先验和似然假设（T. Enßlin，私人通信），或数据准备过程。

类似地，现有的NS运行可以通过重用成功提出的活点，直到活点顺序显著发散，来使用不同的似然函数进行更新。通过使用标准化肯德尔tau距离来衡量活点顺序之间的相似性，UltraNest实现了这种热启动。

7. 数值实验

我们简要地通过数值演示了讨论的某些方面。为此，我们选择了一个具有解析积分的玩具问题，该问题展示了高维、高信息量问题的某些方面。

$$\begin{aligned}
 r_1 &= 10^{-11} \\
 w_1 &= 0.4 \times r_1 \\
 r_2 &= r_1/40 \\
 w_2 &= w/40 \\
 d_1 &= \sqrt{x^2 + y^2} \\
 d_2 &= \sqrt{(x + r_1)^2 + y^2} \\
 N_i &= \frac{1}{\sqrt{2\pi w_i}} \exp \left[-\frac{1}{2} \left(\frac{d_i - r_i}{w_i} \right)^2 \right] \\
 L &= N_1 + 100 \times N_2
 \end{aligned}$$

If an iteration simultaneously yields successful draws from multiple workers, they can be accepted in order if they exceed the consecutive thresholds. *UltraNest* allows each processor to advance its MCMC chain in parallel. When the required number of steps is reached in one of the chains, the sample is accepted and the likelihood threshold raised. The workers then find the last point in their chain which fulfils the new likelihood threshold, and resume from there. This avoids discarding the entire chain.

Parallelisation by adding and removing several live points at once This allows parallel searches for new samples (Burkoff et al., 2012; Henderson and Goggans, 2014; Martiniani et al., 2014). This is quantitatively tested for example in Baldock et al. (2017), in addition to LRPS parallelisation. Nielsen (2013) presents a modification of NS, where the number of removed live points is selected to optimally divide the space based on the likelihood distribution of live points.

Parallelisation by multiple independent runs with the same PRNG seed When attempting to draw a higher point, as many samples are drawn from the constrained prior (via region sampling) as the number of processes (Rollins, 2015). Each process however only evaluates the likelihood of the p -th sample, corresponding to its ID p . Upon likelihood evaluation, each process distributes the result to the others and decides locally for each sample whether to accept or reject. Therefore, each process has the same nested sampling run but the likelihood evaluations scale well with the number of processes.

Parallelisation with multiple independent runs Multiple, independent runs can be merged later (Skilling, 2009; Henderson, Goggans and Cao, 2017). Compared with the previous method, this has a higher number of effective live points, and thus a higher accuracy. The drawback of independent runs is that they cannot share information about the parameter space geometry. The benefit of independent runs is that systematic errors can be explored (Higson et al., 2019). Griffiths and Wales (2019) take this approach to the embarrassingly parallel extreme, with 20,000 independent runs with only one live point each. They then identify runs that landed in the same mode using a specialised algorithm, and unify those runs, with the number of runs proportional to the mode probability. The Diffusive Nested Sampling uses a similar parallelisation approach by letting its walkers explore mostly independently (Brewer, Pártay and Csányi, 2011, B. Brewer, priv. comm.).

Analysing similar datasets simultaneously Buchner (2019) noticed that when similar datasets are analysed, their likelihood contours at a given iteration are similar. Computation can be reduced by drawing in the joint likelihood contour. This is most useful in large surveys of similar data or with Monte Carlo simulated data under the same input parameters, when model evaluations are costly. It could also be used to vary mildly the prior and likelihood assumptions (T. Enßlin, priv. comm.), or the data preparation.

Similarly, existing NS runs can be updated with a different likelihood function, by reusing the successfully proposed live points until the live point order diverges significantly. With similarity between live point ordering measured by the normalized Kendall tau distance, such a warm start is implemented in *UltraNest*.

7. Numerical experiment

We briefly demonstrate numerically some aspects discussed. To that end, we choose a toy problem with analytic integral, which demonstrates some aspects of high-dimensional, highly informative problems.

$$\begin{aligned}
 r_1 &= 10^{-11} \\
 w_1 &= 0.4 \times r_1 \\
 r_2 &= r_1/40 \\
 w_2 &= w/40 \\
 d_1 &= \sqrt{x^2 + y^2} \\
 d_2 &= \sqrt{(x + r_1)^2 + y^2} \\
 N_i &= \frac{1}{\sqrt{2\pi w_i}} \exp \left[-\frac{1}{2} \left(\frac{d_i - r_i}{w_i} \right)^2 \right] \\
 L &= N_1 + 100 \times N_2
 \end{aligned}$$

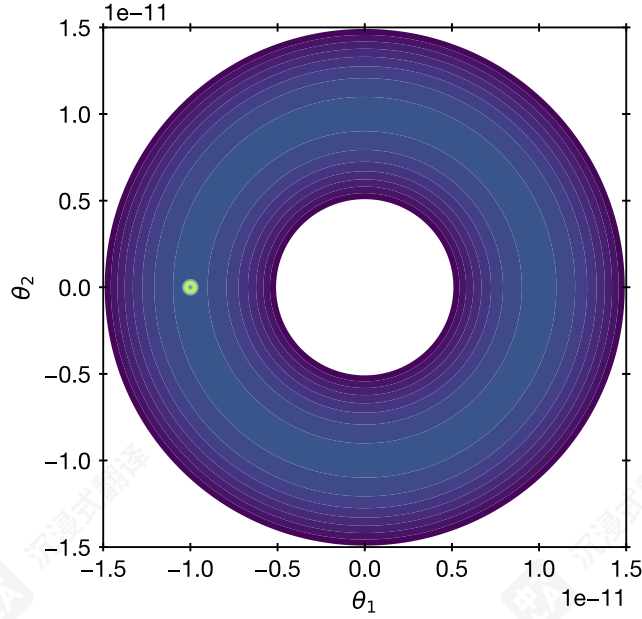


图9. 钻石环玩具问题的似然等高线。一个高斯壳（浅绿色）放置在更大的高斯壳上。这是一个小尖峰在板上的例子。注意与先验界限 $[-1, 1]$ 相比，小尺度 (10^{-11})。

参数空间是二维的，其中 x 和 y 在 -1 和 $+1$ 之间先验均匀分布。似然函数在图9中进行了可视化。将环半径 r 与距离 d 进行比较，使用高斯似然函数来比较宽度 w 。高斯壳使得MCMC提议变得困难，因为它们必须缓慢地沿着壳游走。添加了两个高斯壳，其比例为1:100，这使得该问题成为一个尖峰-板条问题。与先验范围相比，后验分布非常窄，需要许多NS迭代才能找到后验分布。

我们使用固定的、相对较低数量的活动点 ($N = 100$) 运行NS。首先，测试LRPS方法中的MCMC步数。我们选择HARM，每个NS迭代中的步数从1到64。我们还添加了第5.1.3节中介绍的自调谐方法。图10显示了运行结果。在顶部面板中，自调谐方法上升两次，即当遇到两个壳并必须导航时。之前，步数可以调到低值，使其比在整个过程中使用安全步数效率高几个数量级。图10的底部面板显示了累积积分估计。NS运行之间的最终 $\log Z$ 估计在最终估计的尺度上随机变化，与不确定性相当。误差条相对较大，因为 N 较小，但与解析值（虚线水平线）在可接受的 ($< 2\sigma$) 一致范围内。累积 $\log Z$ 在迭代5500附近显示平台。这是相变阶段，当小“钻石”尖峰被发现位于较大的环（板）顶部时。如果运行在之前停止，基于固定的计算预算或因为活动点看起来相似，则不会发现额外的概率。

与此同时，参数空间也变得难以沿着细环导航。穿越的小欧几里得距离导致自动调谐方法增加其步数（顶部面板中灰色曲线上升）。U检验展示在中间面板。曲线主要呈随机变化，即使对于最短的MCMC步数也未能检测到强偏差。这些曲线根据所选的累积带宽（此处：1000次迭代）差异很大，表明插入顺序在采样上没有问题。这与相互一致的 $\log Z$ 估计（底部面板中的误差线）一致，后者通常对不正确的LRPS采样非常敏感。

综合这些结果，人们可能希望至少以4步运行HARM，直到对应于第5000次迭代的似然值，然后以200步继续。

图11比较了各种方法提取有效样本的效率。为了避免给基于区域的方法带来强烈的几何优势，故意选择了较低的 N 。MultiNest和MLFriends在三个因子范围内表现相似（蓝色、橙色叉号）。如上所述，HARM自动调谐（黄色叉号）比每次迭代具有大量固定步数的HARM（绿色叉号）效率高得多。

通过动态嵌套采样，以 $N = 50$ 开始，并迭代添加50个活点，dynesty can 可以高效地提取越来越多有效的样本（图11中的蓝色曲线）。这里开发的反应性NS方法是

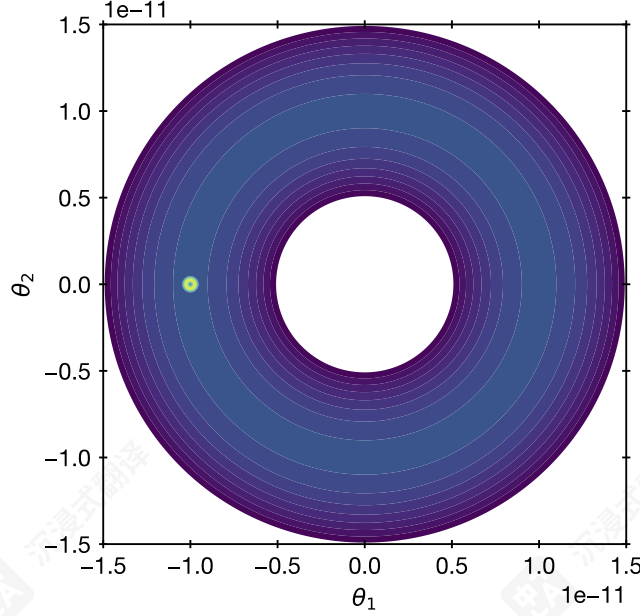


FIGURE 9. Likelihood contours of the diamond ring toy problem. A Gaussian shell (light green) is placed on a larger Gaussian shell. This is an example of a small spike on a slab. Note the small scales (10^{-11}) compared with the prior bounds $[-1, 1]$.

The parameter space is two-dimensional, with x and y a priori uniform between -1 and $+1$. The likelihood is visualised in Figure 9. The distance d from the ring radius r is compared with the width w using a Gaussian likelihood. The Gaussian shells make MCMC proposals difficult, as they have to slowly wander along the shell. Two Gaussian shells are added, with ratios $1 : 100$, making this a spike-and-slab problem. The posterior is very narrow compared with the prior range, requiring many NS iterations until the posterior is found.

We run NS with a fixed, relatively low number of live points ($N = 100$). First, the number of MCMC steps in a LRPS method is tested. We choose HARM, with the number of steps per NS iteration ranging from 1 to 64. We also add the auto-tune method presented in Section 5.1.3. Figure 10 shows the run. In the top panel, the auto-tuning method rises twice, namely when the two shells are encountered and have to be navigated. Before, the number of steps could be tuned to low values, allowing it to be orders of magnitudes more efficient than using a safe number of steps throughout. The bottom panel of Figure 10 presents the accumulated integral estimate. The final $\log Z$ estimate between the NS runs vary stochastically in their final estimate, on a scale comparable to the uncertainty. The error bars are relatively large because of the small N , but are in acceptable ($< 2\sigma$) agreement with the analytic value (dashed horizontal line). The accumulated $\log Z$ shows a plateau near iteration 5500. This is the phase transition when the small “diamond” spike is being discovered on top of the larger ring (slab). If runs had stopped before, based on a fixed computing budget or because the live points appear similar, the additional probability would not have been discovered.

At the same time, the parameter space also becomes difficult to navigate along the thin ring. The small euclidean distances traversed cause the auto-tune method to increase its number of steps (grey curve rises in the top panel). The U test is presented in the middle panel. The curves vary mostly stochastically, not detecting strong deviations even for the shortest MCMC step numbers. These curves look substantially different, depending on what accumulation bandwidth is chosen (here: 1000 iterations), indicating that the insertion order does not see problems with the sampling. This agrees with the mutually consistent $\log Z$ estimates (error bars in bottom panel), which are usually very sensitive to incorrect LRPS sampling.

Taking these results together, one may want to run HARM with at least 4 steps until the likelihood corresponding to iteration 5000, and subsequently with 200 steps.

The efficiency in extracting effective samples of various methods is compared in Figure 11. To avoid giving a strong geometric advantages to region-based methods, low N are deliberately chosen. MultiNest and MLFriends perform similarly within a factor of three (blue, orange cross). As remarked above, HARM auto-tune (yellow cross) is substantially more efficient than HARM with a large, fixed number of steps per iterations (green cross).

With Dynamic Nested Sampling, started with $N = 50$, and iteratively adding 50 live points, dynesty can efficiently extract more and more effective samples (blue curve in Figure 11). The Reactive NS methods developed here are

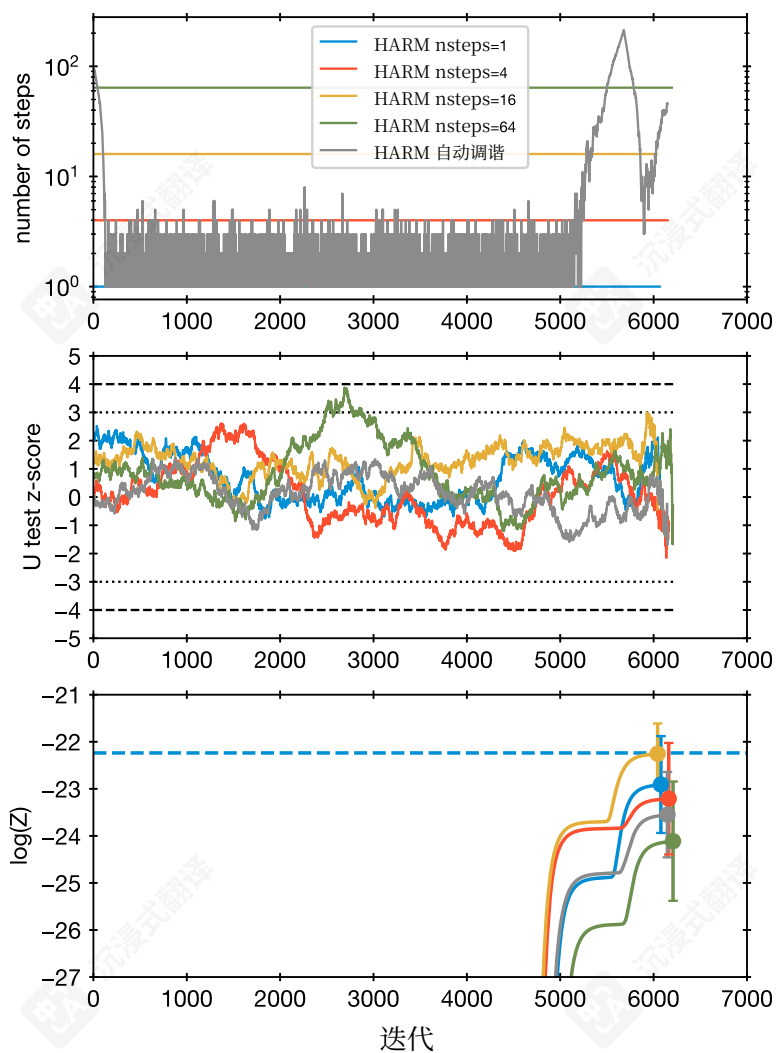


图10. MCMC步数。使用HARM (Hit-and-Run蒙特卡洛) 方法对玩具问题进行的五次运行结果。上面板：每个新样本的步数。这是在自动调优方法中变化的。中面板：在1000次迭代的滚动窗口中计算的U检验z分数。下面板：积分估计。在迭代5000到6000之间，可以观察到相变，其中初始时积分估计趋于平稳，然后迅速再次增加。虚线是真值。

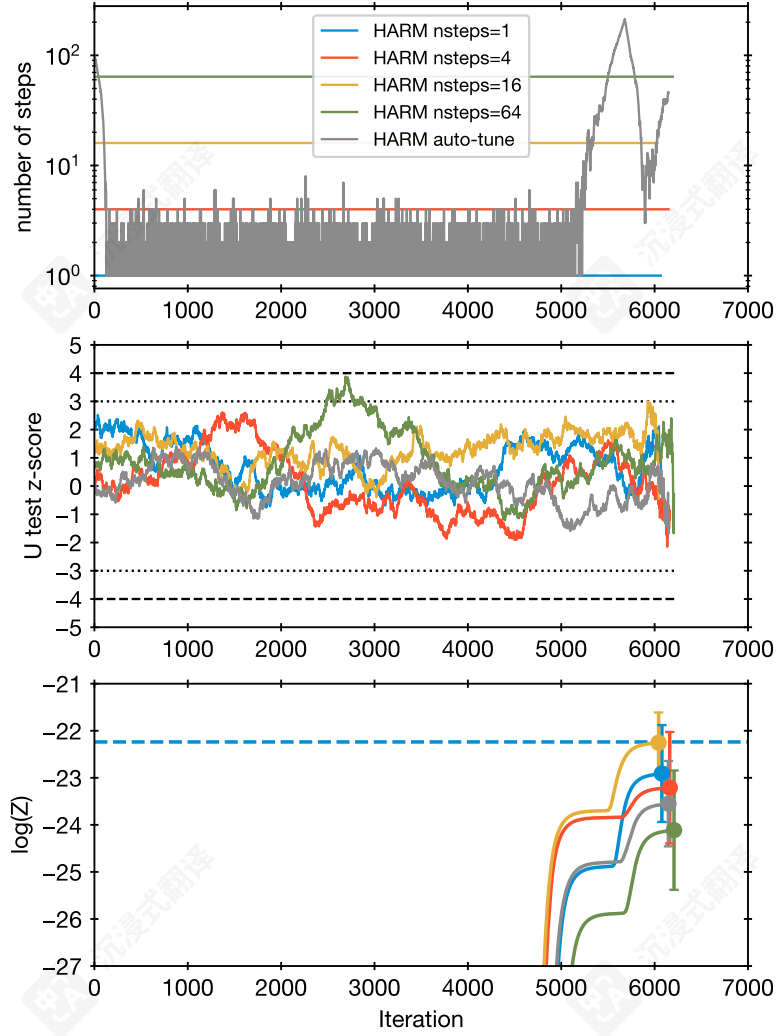


FIGURE 10. Number of MCMC steps. Five runs on the toy problem are shown using HARM (Hit-and-Run Monte Carlo) method. Top panel: Number of steps per new sample. This is varied in the auto-tuning method. Middle panel: z-score of the U test computed in a rolling window of 1000 iterations. Bottom panel: Integral estimates. Between iteration 5000 and 6000, a phase transition can be observed where initially the integral estimate plateaus and then rapidly increases again. The dashed line is the true value.

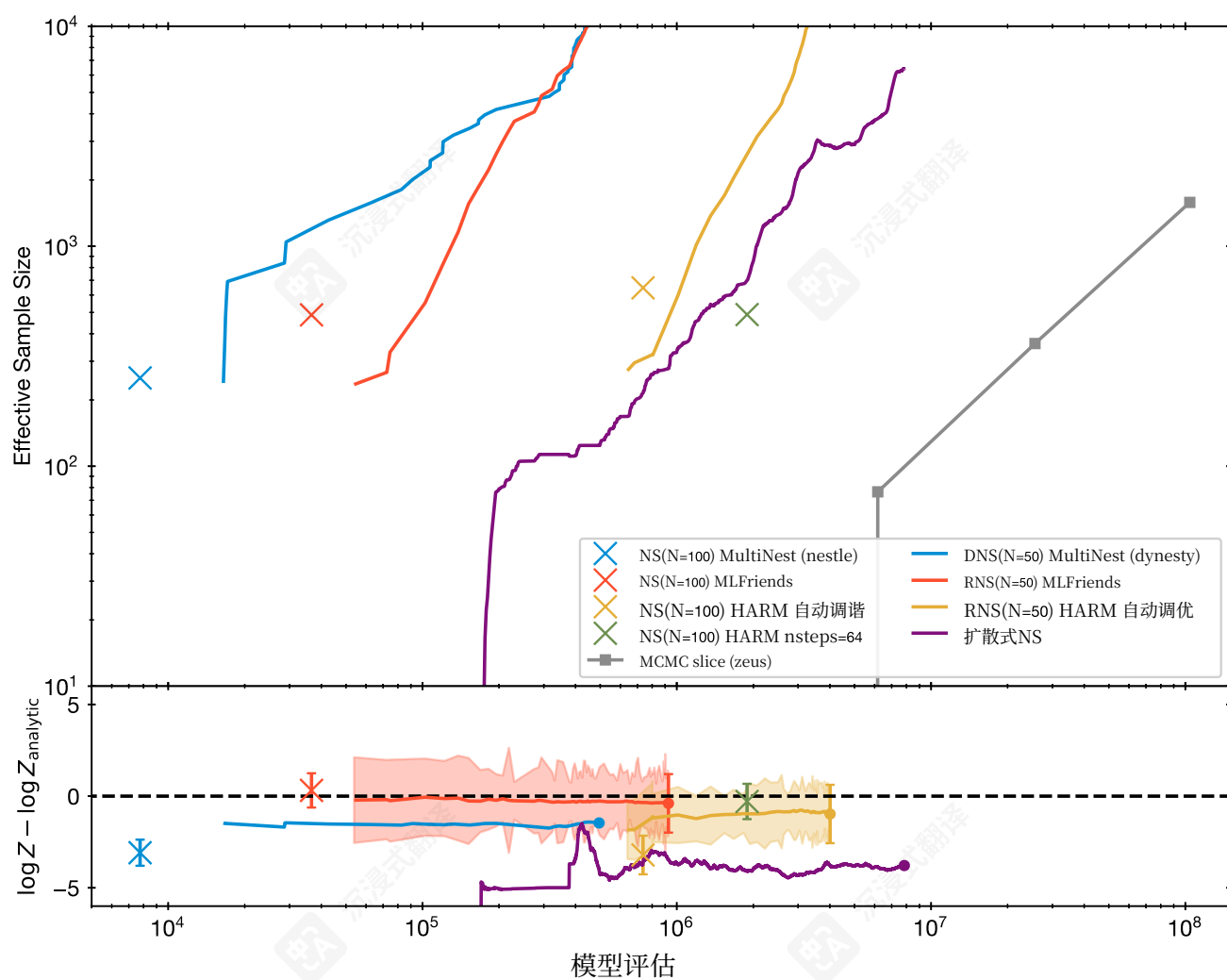


图11. 不同LRPS方法的效率比较，其中一些方法使用动态NS或反应式NS。顶部面板：有效样本提取数量（左上优于右下）。MCMC在识别后验分布的初始阶段后，近似线性扩展。MLFriends（橙色）和自动调谐HARM（绿色）在每单位计算成本下显示出更陡峭的样本增加。底部面板：积分估计值，相对于解析值。误差线和阴影区域显示了 1σ 的不确定性估计，当可用时。

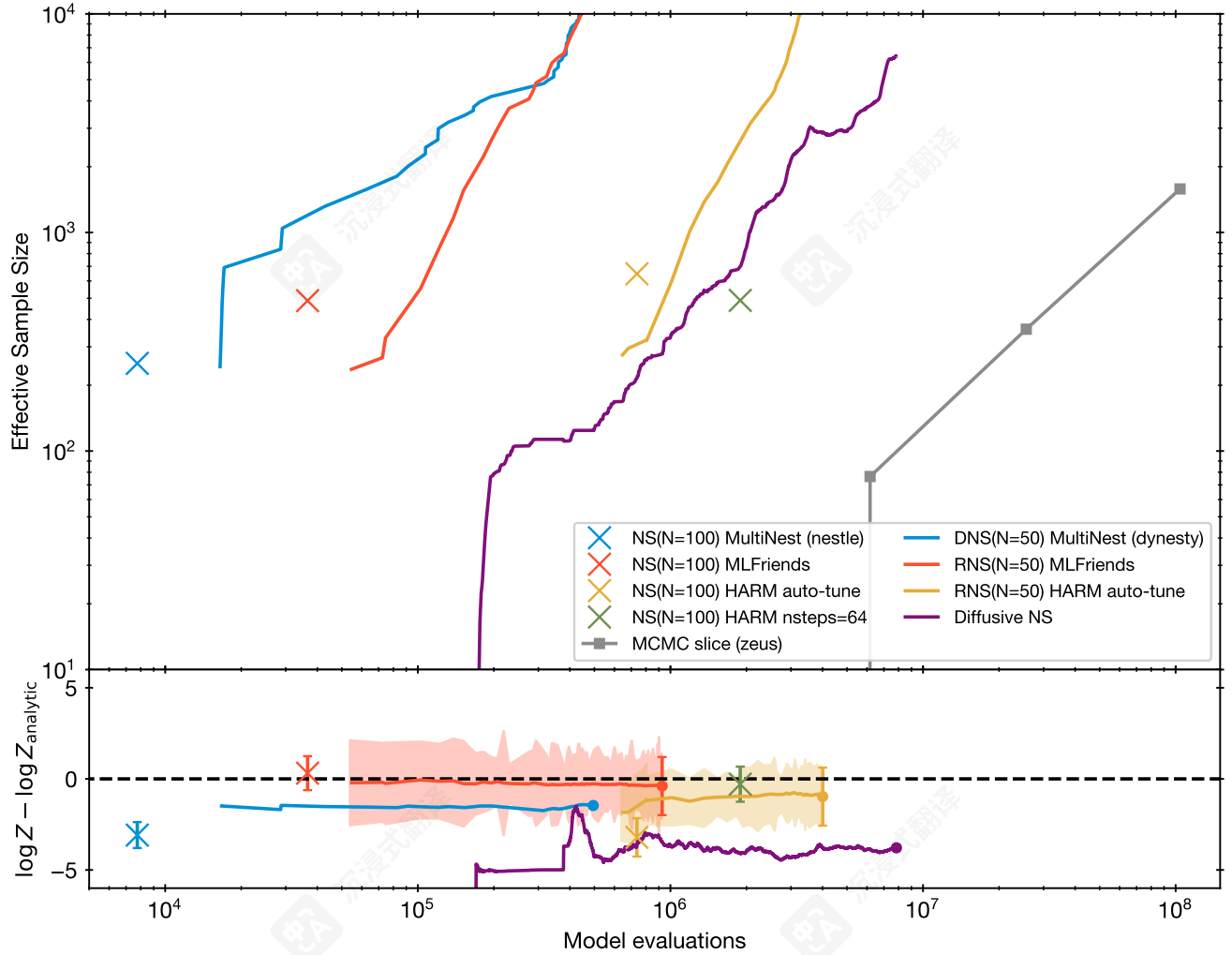


FIGURE 11. Efficiency comparison of different LRPS methods, some using Dynamic NS or Reactive NS. Top panel: Effective number of samples extracted (upper left is better than lower right). MCMC, after an initial phase of identifying the posterior, scales approximately linearly. MLFriends (orange) and auto-tuning HARM (green) show the steeper increase in samples per computation cost. Bottom panel: Integral estimate, relative to analytic value. The error bar and shaded areas show 1σ uncertainty estimates, when available.

与一个根据后验权重选择点并扩展其父节点的代理一起使用。这应用于MLFriends和HARM自动调优（分别为红色和黄色曲线）。有趣的是，MLFriends的曲线随似然成本呈超线性增长。在这个问题中，MLFriends与活点数量成反比。这是因为采样区域可以更好地近似，从而导致拒绝率更低。当然，这只有在问题维度较低的情况下才可能实现。在这里，MLFriends比MultiNest表现更好，因为椭圆聚类对高斯壳层的描述很差。当然，也存在相反的情况，例如L形似然轮廓，其中MultiNest明显优于MLFriends。有趣的是，HARM自动调优也随似然评估急剧增长。更多的活点意味着它们之间的平均距离更小，因此到达新点所需的步数更少。

扩散嵌套采样器DNest4显示出近似线性扩展（图11中的紫色曲线）。所需的似然评估次数高于其他方法。为了比较，我们还包含了一种MCMC方法，使用切片采样器zeus（Karamanis和Beutler，2020）。这表明在识别后验分布的初始阶段后，呈现线性扩展（灰色曲线）。在这种情况下，NS方法提取后验样本的效率要高得多，因为MCMC自相关非常差。这个问题的计算成本比较不能代表高维问题，在高维问题中，方法的排序可能不同，甚至可能相反。

图11的底部面板比较了积分估计。在这里，MultiNest估计偏移，因为它排除了其非常小的误差范围内的真实值。MLFriends和HARM方法在 2σ 内与解析值一致。动态和反应式嵌套采样运行即使有大量似然评估仍然保留相当大的不确定性。这是因为新的活点不会添加到运行的初始阶段，导致收缩估计保持不确定。DNest4最初显示出收敛到真实值，但后来这种趋势没有改善。DNest4不报告不确定性。

MLFriends（用于低维情况）和HARM自动调优（用于低维和高维情况）方法看起来很有前景。它们已实现于UltraNest Python包中，并支持在集群上进行大规模并行计算。

8. 总结

本次综述涵盖了跨多个子领域的嵌套抽样文献，并整理了提出的想法和概念。我们描述了嵌套抽样适合的问题类型，以及实现需要解决的实践困难。总结如下：

1. 嵌套抽样是一种实用的贝叶斯模型比较和参数推断算法。它能全局探索参数空间。这在具有潜在复杂、多模态似然的问题中很重要。探索可以在很大程度上无监督且无需特定问题调优，直到达到一个明确的收敛点。嵌套抽样在扩展到非常高维和高度信息量的问题时存在一些局限性。
2. 本次综述强调了已开发的嵌套抽样（NS）变体的多样性。这些包括：无任何MCMC的NS（如MultiNest和MLFriends）、在NS中使用MCMC，以及将NS作为MCMC链运行（扩散式嵌套抽样）。一个相对较新的发展是更深入地集成HMC，并调整如hit-and-run和切片抽样的几何随机游走算法。
3. 近年来，对运行质量和正确性的诊断与可视化有了显著改进。

9. 未来研究

大量文献涉及将嵌套采样应用于特定问题，评估其质量，并提出新的NS变体和实现。除此之外，我们未来还需要系统性的理论和实践评估：

1. 使用扩展的序贯蒙特卡洛框架研究理论基础似乎特别有前景。鉴于NS只变异一个粒子，应该研究哪些定理可以转移到这种扩展框架中。
2. 需要在PMDIT空间中对各种问题进行系统的数值比较，以判断NS变体的能力和局限性，包括LRPS方法和它们的调优参数。

used with an agent that selects points based on posterior weights, and expands their parent nodes. This is applied to MLFriends and HARM auto-tune (red and yellow curves, respectively). Interestingly, the curve for MLFriends rises super-linearly with likelihood cost. MLFriends scales in this problem inversely with the number of live points. This is because the sampling region can be approximated better, leading to smaller rejection rates. Of course, this is only possible because of the low dimension of this problem. MLFriends scales better than MultiNest here, because the ellipsoid clustering is a poor description of the Gaussian shells. There are of course problems where the converse is true, such as L-shaped likelihood contours, where MultiNest substantially outperforms MLFriends. Interestingly, also HARM auto-tune scales steeply with likelihood evaluations. More live points mean the mean distance between them is smaller, therefore the number of steps needed to reach a new point is smaller.

The Diffusive Nested Sampler DNest4 shows approximately a linear scaling (purple curve in Figure 11). The number of likelihood evaluations needed is higher than for the other methods. For comparison, we also include a MCMC method, using the slice sampler zeus (Karamanis and Beutler, 2020). This shows a linear scaling (grey curve) after an initial phase where the posterior needs to be identified. The efficiency of extracting posterior samples in this case is substantially higher in NS methods, because the MCMC auto-correlation is exceptionally poor. The computational cost comparison in this problem is not representative of high-dimensional problems, where the ranking of methods may be different and even reversed.

The bottom panel of Figure 11 compares the integral estimates. Here, the MultiNest estimate is off, as it excludes the true value within its very small error bars. MLFriends and HARM methods agree with the analytic value within 2σ . The dynamic and reactive nested sampling runs retain substantial uncertainties even with many likelihood evaluations. This is because new live points are not added to the initial phase of the run, causing the shrinkage estimates to remain uncertain. DNest4 initially shows convergence to the true value, but this trend is not improving later. DNest4 does not report uncertainties.

The MLFriends (for low-d) and HARM auto-tune (for low and high-d) methods appear promising. They are implemented in the *UltraNest* Python package, with support for massively parallel computing on clusters.

8. Summary

This review has surveyed the nested sampling literature across many subfields, and compiled proposed ideas and concepts. We have described the problem types nested sampling is suitable for, and laid out the practical difficulties implementations need to solve. To summarise:

1. Nested sampling is a practically useful algorithm for Bayesian model comparison and parameter inference. It globally explores the parameter space. This is important in problems with potentially complex, multi-modal likelihoods. The exploration can largely proceed unsupervised and without problem-specific tuning until a well-defined convergence point. Nested sampling has some limitations to scale to very high dimensional and highly informative problems.
2. This review highlighted the diversity of NS variants have been developed. These include: NS without any MCMC (such as MultiNest and MLFriends), using MCMC within NS, and running NS as a MCMC chain (Diffusive Nested Sampling). A relatively recent development is to integrate HMC more deeply, and adapting geometric random walk algorithms such as hit-and-run and slice sampling.
3. Diagnostics and visualisations of the quality and correctness of runs have improved substantially in the last few years.

9. Future research

Much of the literature is involved with the application of nested sampling to specific problems, evaluating its quality, and also proposing new NS variants and implementations. Beyond this, we see the need for systematic theoretical and practical evaluations in the future:

1. Investigating the theoretical foundations with an extended Sequential Monte Carlo framework seems particularly promising. Given that NS mutates only one particle, it should be investigated what theorems can be transferred to such an extended framework.
2. Systematic numerical comparisons of a wide range of problems across the PMDIT space are needed to judge the capabilities and limitations of NS variants, including the LRPS methods and their tuning parameters.

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附录A：关于编码先验分布

单位超立方体变换， $u \rightarrow \theta$ ，是一种通过以自然概率单位参数化来编码先验分布的方法， $u_i \in [0, 1]$ 。在分解先验分布中，该变换通过逆累积分布函数， $\theta_i = F^{-1}(u_i)$ ，实现。它很方便，因为恒定能量的轨迹是直的，并且可以从几何形状中采样而不需要额外的拒绝。因此，一些（但并非所有）流行的NS实现允许用户通过单位超立方体变换指定先验分布，并且在许多示例中仅展示分解先验分布。这导致一些人认为NS需要超立方体和分解先验分布。在这里，我们澄清情况并非如此。

首先，只要定义了LRPS方法和似然函数，NS就可以继续进行。如果先验空间不是 $\mathbb{R}^d$ ，而是任意的“对象”，一个简单的例子包括Metropolis采样器，它根据先验密度扰动样本。实际上，DNest4（Brewer和Foreman-Mackey，2018）中可以实现这种通用的处理方式。这包括允许维度变化，如Brewer（2014）所示。

其次，我们展示了两个实际案例，其中依赖先验可以通过NS和单位超立方体变换进行编码。考虑一个由均值 μ 和协方差 Σ 定义的相关高斯先验。从单位区间 $u \in [0, 1]$ 到不相关、标准正态先验的变换可以通过逆累积分布函数 $z_i = F^{-1}(u_i)$ 进行编码。其次，从标准正态先验到一般高斯的变换使用仿射变换 $\theta = Az + \mu$ ，其中 $\Sigma = AA^T$ 。因此，得到了一个相关的样本。这种方法不仅限于多元高斯分布。例如，Student-t分布也会以相同的方式工作，并且可以接收自由度参数。这种方法的推广是Copula模型（例如，Nelsen，2007）。

通常来说，首先转换一个变量，然后迭代地考虑下一个变量的条件累积分布，根据前一个变量。这种迭代条件方法也可以应用，如果非解析先验仅以多维直方图或样本 s 的形式提供。

通常，随机数生成的程序可以加以适应。例如，考虑将分数编码到参数化中的问题。这发生在尝试拟合某些物理对象的元素丰度时，其中相对分数必须总和为1。在某些情况下，如果更接近可观测值，可能值得拟合绝对参数，例如该元素的总质量，并从后验中获取分数。然而，这需要为质量设置先验，这可能很困难。因此，我们假设我们希望每个分数都接收均匀先验，但必须同时总和为1。这种情况下的适当分布是具有 $\alpha = 1$ 的平坦Dirichlet分布。使用超立方体变换获取先验样本，我们获得独立的伽马变量 $z_i \sim \text{Gamma}(\alpha, 1)$ ，即 $z_i = -\log u_i$ ，并获取分数变量为 $\theta_i = z_i / \sum_i z_i$ 。可以选择更复杂的变换来提高LRPS效率（Betancourt，2012）。

我们顺便指出，最先进的HMC框架在内部使用类似的变换来避免轨迹超出先验支持范围（Carpenter等人，2017年）。

附录B：椭球似然函数的椭球采样效率

在椭球似然轮廓的情况下，数值研究了椭球嵌套采样的计算成本（Mukherjee、Parkinson和Liddle，2006年），遵循Allison和Dunkley（2014年）的方法。与我们固定的放大处理不同，这里我们确定了用于拒绝采样的放大量：我们从维度为 d 的超球体中生成 N 个点。在一个自助采样方案中，我们随机剔除一些点（Buchner，2016年），并从剩余部分计算样本协方差矩阵。这用于构建一个椭球，通过所需的因子放大以恢复所有剔除的点。重复50次，并存储最大的放大因子。

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Appendix A: On encoding prior distributions

The unit hypercube transformation, $u \rightarrow \theta$, is a way to encode priors by parametrising them in natural probability units, $u_i \in [0, 1]$. In factorised priors, the transformation is achieved with inverse cumulative distribution functions, $\theta_i = F^{-1}(u_i)$. It is convenient as constant-energy trajectories are straight, and sampling from geometrical shapes can be achieved without an additional rejection. For this reason, some (but not all) popular NS implementations let users specify priors via unit hypercube transformations, and in many examples demonstrate only factorised priors. This has lead some to believe that NS requires hypercubes and factorised priors. Here we clarify that neither is the case.

Firstly, NS can proceed as long as a LRPS method and a likelihood are defined. If the prior space is not $\mathbb{R}^d$, but arbitrary “objects”, a simple example includes Metropolis samplers that perturb the sample following the prior density. Indeed, such generic treatments are possible in DNest4 (Brewer and Foreman-Mackey, 2018). This includes allowing the dimensionality to vary, as demonstrated in Brewer (2014).

Secondly, we demonstrate two real-world cases where dependent priors can be encoded in NS with unit hypercube transformations. Consider a correlated Gaussian prior, defined by mean μ and covariance Σ . The transform from a unit interval $u \in [0, 1]$ to an uncorrelated, standard normal prior can be encoded through inverse cumulative distribution functions, $z_i = F^{-1}(u_i)$. Secondly, the transform from a standard normal prior to a general Gaussian is performed with an affine transform, $\theta = Az + \mu$, where $\Sigma = AA^T$. Therefore, a correlated sample is obtained. This approach is not limited to multi-variate Gaussians. For example, a Student-t would work the same and can receive a degrees of freedom parameter. The generalisation of this approach are copula models (e.g., Nelsen, 2007).

As a general approach, it can be useful to first transform one variable, and then iteratively consider the conditional cumulative distribution of the next given the previous. This iterative conditional approach can also be applied if a non-analytic prior is only available as a multi-dimensional histogram or samples s .

Often, procedures from random number generation can be adapted. As an example, consider the problem of encoding fractions into the parametrisation. This occurs for example when trying to fit for the elemental abundance of some physical object, where the relative fractions must sum to one. In some cases, it may be worthwhile to fit for an absolute parameter, such as the total mass of that element, if that is closer to the observable, and obtain fractions from the posterior. However, this requires placing priors on the masses, which may be difficult. Lets therefore assume we want to assume that the fractions each receive uniform priors, but simultaneously must sum to unity. The appropriate distribution for this scenario is the flat Dirichlet distribution with $\alpha = 1$. To obtain prior samples using a hypercube transformation, we obtain independent gamma variables $z_i \sim \text{Gamma}(\alpha, 1)$, i.e., $z_i = -\log u_i$, and obtain the fraction variables as $\theta_i = z_i / \sum_i z_i$. More sophisticated transformations may be chosen to improve LRPS efficiency (Betancourt, 2012).

We comment in passing that state-of-the-art HMC frameworks use similar transformations internally to avoid trajectories exceeding the prior support (Carpenter et al., 2017).

Appendix B: Ellipsoidal sampling efficiency for ellipsoidal likelihoods

The computational cost of ellipsoidal nested sampling (Mukherjee, Parkinson and Liddle, 2006) in the case of ellipsoidal likelihood contours is numerically explored, following Allison and Dunkley (2014). Different to their fixed enlargement treatment, here we determine the enlargement needed for rejection sampling: We generate N points from a hypersphere of dimension d . In a bootstrapping scheme, we randomly leave points out (Buchner, 2016) and compute from the remainder a sample covariance matrix. This is used to construct an ellipsoid, enlarged by the factor needed to recover all left-out points. This is repeated 50 times, and the largest enlargement factor stored. The

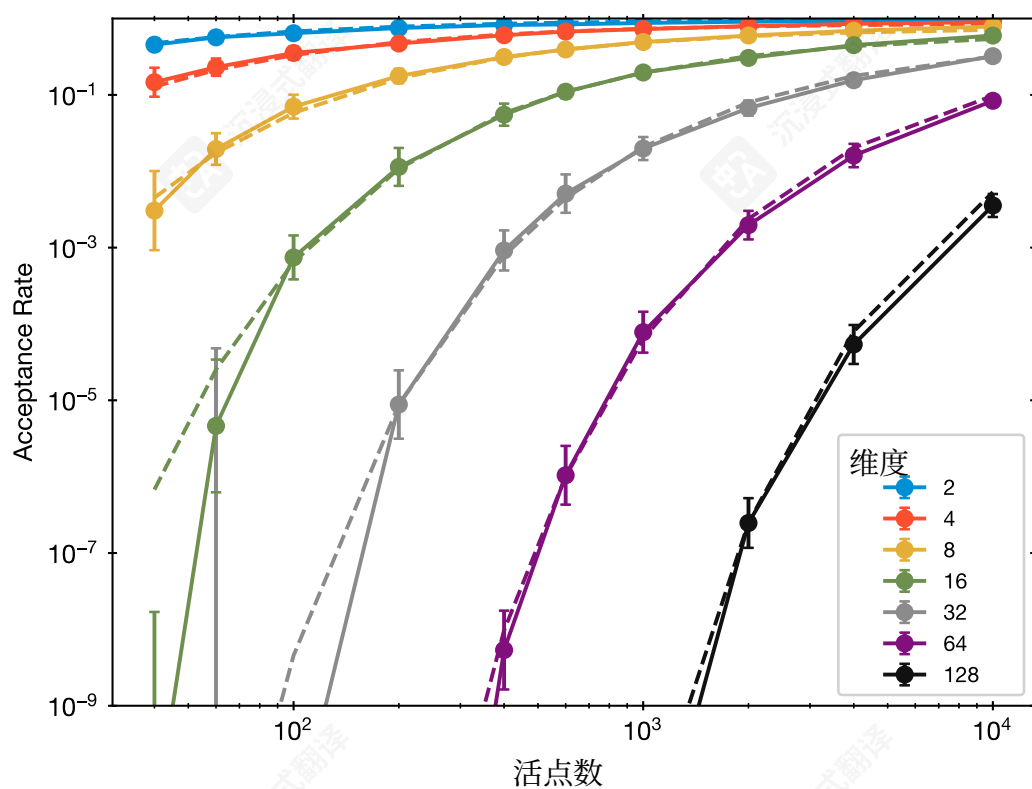


图12。椭圆拒绝采样的接受率。这些测量构建的椭圆体的逆超额体积，以可靠地采样后验分布。更多的活点会导致更紧的椭圆体和更高的接受率。增加维度数量会降低接受率。虚线显示了方程3的经验近似，颜色与每个维度的数据点相同。

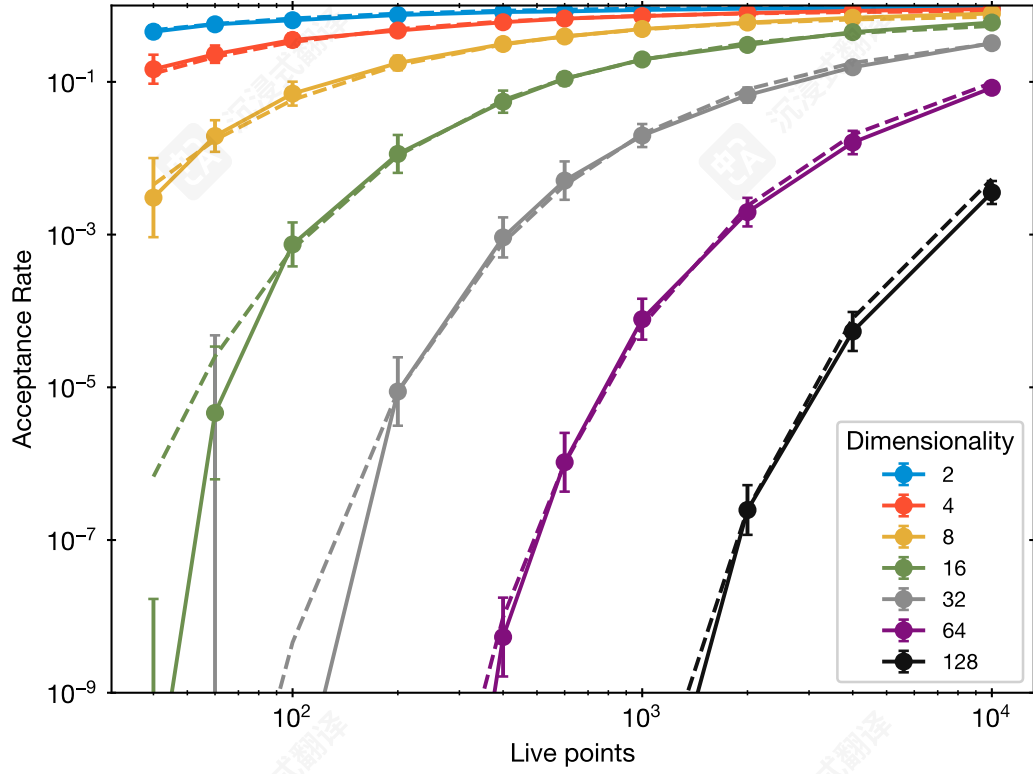


FIGURE 12. Acceptance rate for ellipsoidal rejection sampling. These measure the inverse excess volume of the constructed ellipsoids to reliably sample the posterior. More live points lead to tighter ellipsoids and higher acceptance rates. Increasing the number of dimensions decreases the acceptance rate. The dashed curve shows the empirical approximation of eq. 3, with the same colours as the data points for each dimensionality.

原始球体与构造椭球体的椭球体积比给出了椭球拒绝采样的接受率。

图12展示了接受率 α 作为 N 和 d 的函数。误差线表示40次独立计算的 α 的标准差。图12显示接受率随活点数 N 的增加而增加，但随维度的增加而减少。图12表明，当 d 翻倍时，如果 N 增加五倍，可以保持50%的接受率。当 N 远小于维度时，这种方法会急剧失效。数据点可以通过以下公式（虚线）经验性地描述：

$$\alpha = \left(1.07 - \log d^{1/3}\right) \times \exp \left\{ - \left(\frac{6.83 \times d^{1.9}}{N} \right)^{3/4} \right\} \quad (5)$$

第一项随维度增加对接受率的影响很小，因此在成本缩放计算中被忽略。平均成本，即所需的模型评估次数，是 $C_{\text{ell}} = 1/\alpha_0$ 。

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ellipsoid volume ratio of the original sphere and the constructed ellipsoid gives the acceptance rate of ellipsoidal rejection sampling.

Figure 12 presents the acceptance rate α as a function of N and d . Error bars indicate the standard deviation across 40 independently computed α . Figure 12 shows that the acceptance rate increases with the number of live points N , but decreases with dimensionality. Figure 12 suggests that acceptance rates of 50% can be maintained when d doubles if N increases five-fold. When N is much smaller than the dimensionality, this method breaks down dramatically. The data points can be empirically described (dashed curves) by the following formula:

$$\alpha = \left(1.07 - \log d^{1/3}\right) \times \exp \left\{ - \left(\frac{6.83 \times d^{1.9}}{N} \right)^{3/4} \right\} \quad (5)$$

The first term decreases the acceptance rate only mildly with dimensionality, and is therefore neglected in the cost scaling computations. The average cost, i.e., the number of model evaluations needed, is $C_{\text{ell}} = 1/\alpha$.

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